AI Financial Advice: Supply, Demand, and Life Cycle Implications*

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Abstract

We ask a representative sample to write prompts seeking spending and investing advice from LLMs, then simulate the lifetime effects of following the advice under realistic asset and labor market conditions. Applying this method to `GPT-5.2` and `Gemini 3 Flash`, we find the advice would move respondents toward life cycle theory: broader participation in diversified equity, age-declining equity shares, and larger savings buffers. Recommendations vary systematically by gender, AI experience, and financial literacy. For gender, two-thirds of equity-share differences arise from men and women writing different prompts (demand), while one-third arises from gender labels attached to otherwise identical prompts (supply).

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Millions of people now turn to large language models (LLMs) for personal financial advice. Within a few years of these tools becoming widely available, over half of adults in the U.S. and the U.K. report using them for financial guidance, a share that in some surveys rivals or exceeds the proportion who consult a human financial advisor (Lloyds Banking Group 2025; J.D. Power 2025; Gallup 2025). This rapid adoption raises the prospect that affordable, widely accessible, high-quality financial advice may be closer than ever. However, whether that promise is realized depends on two distinct forces: the content of the advice these models give, and how people change their behavior in response to it. Generative AI may differ from existing sources of advice on both margins. A natural first step, and the focus of this paper, is to characterize the advice itself: what LLMs recommend, and how those recommendations vary across individuals.

We develop and implement a method to describe quantitatively the personal financial advice that LLMs give. Studying this advice systematically is challenging because it depends on how individual demand for AI financial advice manifests in prompts, on features of the models that supply the AI advice, and on dynamic, uncertain consequences that accumulate over the life cycle. We overcome these three challenges by (i) surveying a demographically representative sample to elicit realistic prompts seeking financial advice from an LLM, (ii) simulating the quantitative implications of the advice received in response to these prompts over the life cycle under realistic asset and labor market conditions, and (iii) evaluating why the advice differs across heterogeneous groups by randomizing prompts to isolate the role of supply (i.e., how models respond to similar prompts) from that of demand (i.e., differences in prompts across individuals).

Applying our method to `GPT-5.2` and `Gemini 3 Flash`, the LLMs used in our analysis, we document three facts about LLM-generated spending, saving, and investing advice. We focus on these domains both because they involve some of the most consequential financial decisions households make and because our survey shows they are the topics people most commonly seek AI financial guidance on. First, following LLM advice would move most survey respondents closer to the broad recommendations of standard life cycle theory relative to their observed behavior, including greater participation in diversified equity funds, equity shares that decline with age, and sizable savings buffers. Second, LLM advice departs from life cycle theory in systematic ways: it implies unusually high patience, relies on simple heuristics, smooths consumption imperfectly after income shocks, and allows portfolios to drift passively with realized returns. Third, LLM advice varies systematically with individual characteristics, such as gender, financial literacy, and prior AI experience. These differences accumulate over the life cycle into wealth differences at age 60 of approximately 4–6% between groups and reflect both demand and supply. For example, by randomizing gender labels, we show that two-thirds of the gender difference in investment advice is accounted for by differences in the advice men and women ask for, while one-third stems from the model supplying different advice to the same questions when labeled as coming from men versus women.

Taken together, our results highlight the potential of generative AI to improve financial decision-making, but suggest that its impact is likely to vary across individuals. The finding that LLM advice broadly aligns with life cycle theory should not be taken for granted. Unlike other forms of automated financial advice, such as robo-advisors, these models are not optimized to improve household financial decisions (D'Acunto et al. 2019; Rossi and Utkus 2020). One might therefore worry that, in pursuit of engagement, LLMs would instead reinforce existing behaviors, validate biases, or simply tell people what they want to hear (Fedyk et al. 2026). Our results suggest a more nuanced conclusion: current general-purpose LLMs often provide advice that aligns with standard financial guidance, but this advice is sensitive to demand and, therefore, differs systematically across users.

Naturally, the advice provided by `GPT-5.2` and `Gemini 3 Flash` may differ from that of future models. Our contribution is therefore not only to characterize the advice of a particular generation of models, but also to provide a framework for studying LLM financial advice. As these models continue to evolve rapidly, three features of our analysis are likely to remain relevant. First, differences in advice arise not only from model capabilities, but also from demand: what users ask, what information they provide, and how they frame their financial problems. These demand-side constraints will likely be more persistent than the fast pace at which models are changing. Second, our survey-and-simulation approach provides a systematic way to track changes in model and user behavior as both evolve over time. Third, progress in applying generative AI to household finance will depend in part on the availability of benchmarks and diagnostic tests against which models can be evaluated and improved. While many such benchmarks exist for other tasks, we provide a set of diagnostics grounded in life cycle theory (e.g., consumption smoothing, diversification, portfolio rebalancing).

Methodology. Our method for studying LLM financial advice proceeds in three steps. First, we survey a demographically representative sample of 1,000 U.S. adults recruited through Prolific. We ask respondents to write three prompts to an LLM. The first prompt is a description of their financial situation, the second is for advice on spending versus saving, and the last is for investing advice. This is the main innovation of our approach: using prompts written by real users introduces meaningful heterogeneity in both the topics people raise and how they express themselves. To our knowledge, our survey provides the first systematic evidence on how prompts vary across individuals seeking financial advice from LLMs. Alongside the prompts, we collect data on respondents' demographics, financial situation, financial literacy, and prior experience seeking financial guidance from LLMs.

In the second step, we build a quantitative life cycle simulation model in the tradition of Gourinchas and Parker (2002) and Cocco et al. (2005) (and most closely related to Choukhmane and de Silva 2026). The model is calibrated to match historical U.S. data and incorporates stochastic income, stochastic asset returns, job-to-job and unemployment transitions, mortality risk, and a tax and social insurance system. We simulate households' consumption-saving decisions and their allocation of savings across four asset classes: fixed income, diversified equity funds, individual stock holdings, and a residual risky asset capturing holdings in crypto, gold, commodities, and collectibles.

The final step in our method combines the prompts written by our survey respondents with the life cycle simulation model. Rather than deriving policy functions from optimal choices, we derive them from LLM recommendations. Specifically, at each age, we randomly match each agent in our simulation with a prompt written by a survey respondent with a similar age, income, and employment status. We then replace the state variables in that prompt with those of the simulated individual and submit two successive LLM queries: first, we submit the prompt and receive textual financial advice from an LLM; then, we use a second LLM to translate this textual advice into quantitative recommendations for saving, consumption, and asset allocations. We repeat this process at each age over the individual's life cycle and across many simulated individuals. Importantly, each query to the LLM is independent, with no memory of past responses; the only link across periods is the evolution of state variables that depend on prior choices.

This method can be reapplied to other populations, advice topics, or models. A key design choice is the trade-off between specificity and external validity. Asking people to write prompts about defined topics, in our case spending, saving, and investing, allows us to compare responses across individuals and integrate them into a parsimonious life cycle framework, but at the cost of some of the open-ended quality of how people actually interact with these tools. We view this as a feature of the approach that can be adjusted depending on the research question. Greater specificity would allow studying decisions that we abstract from (e.g., insurance, housing, credit management), while more open-ended prompts would better reflect real-world usage at the cost of being harder to compare across individuals and incorporate into a simulation framework.

Main findings. Our main results come from applying this method to `GPT-5.2`, which we refer to as "the LLM" throughout. We chose this LLM because ChatGPT is the most commonly used model for financial advice in our survey (and others, e.g., Lloyds Banking Group 2025; J.D. Power 2025) and `GPT-5.2` was the frontier model at the time we ran our analysis (and we find quantitatively similar results using `Gemini 3 Flash`).

We document three facts about LLM advice over the life cycle. First, following LLM recommendations would move most individuals closer to the broad principles of life cycle theory than their observed behavior. On saving, many individuals across all age groups report having accumulated little financial wealth; by contrast, following the advice would lead virtually all simulated individuals to build savings buffers above $10,000 by age 30. On portfolio choice, it would induce near-universal participation in the stock market, with equity shares that decline with age after 45, and risky holdings concentrated in diversified equity funds rather than individual stocks, crypto, gold, commodities, or collectibles. In addition to these broad patterns, the LLM's advice responds to the topics individuals raise in ways that are generally consistent with standard theory. For example, prompts that mention macroeconomic uncertainty lead the LLM to increase saving and reduce equity holdings, consistent with a stronger precautionary motive.

Second, we find that the LLM's advice departs from the more subtle prescriptions of life cycle theory in systematic ways. The advice implies an unusually high discount factor in a standard life cycle model. Instead of tailoring recommendations closely to individuals' evolving economic circumstances, the advice often relies on simple saving and withdrawal heuristics, including bunching at round numbers and recommending fixed withdrawal rules. Consistent with this reliance on heuristics, consumption is insufficiently smoothed: the LLM recommends too little decumulation in retirement and sharp consumption declines after job loss, even when simulated individuals have sufficient liquid wealth. Asset allocation advice exhibits a similar lack of active adjustment as portfolios drift passively with realized returns rather than being actively rebalanced. These departures appear to reflect, in part, the prompts that people write. A more structured "academic" prompt, which provides details on each agent's financial situation and explicitly references life cycle planning and portfolio theory, improves consumption smoothing and reduces reliance on heuristics, though it does not increase active portfolio rebalancing.

Third, the LLM's advice varies systematically with individual characteristics, such as gender, financial literacy, and prior AI experience. For each of these characteristics, we split survey respondents into two groups and repeat our simulation, sampling only prompts written by individuals in each group (e.g., simulating life cycle paths using only prompts written by men versus women). The advice varies meaningfully along each dimension: average wealth at age 60 is around 5% higher when sampling prompts written by men, individuals with high financial literacy, or those with past AI experience. These differences arise because the LLM consistently recommends lower equity shares for prompts written by women or individuals with low financial literacy, and lower saving rates for prompts written by individuals with no prior experience using AI for financial advice.

To understand why LLM advice differs across individuals, we develop an approach to decompose these differences into demand and supply components. In the context of gender, we focus on survey prompts that do not explicitly identify the respondent's gender, and we randomly add gender labels, such as "I am a man." This separates demand-side differences (i.e., men and women write different prompts) from supply-side differences (i.e., the LLM gives different advice to the same question when the stated gender changes). We find that approximately two-thirds of the gender gap in recommended equity allocations is demand-driven: it reflects differences in the content of men's and women's prompts that remain even when both are assigned the same gender label. The remaining one-third is supply-driven: the same prompts receive higher equity recommendations when labeled as coming from a man rather than a woman. These findings echo evidence from human financial advisors, where gender differences in advice also reflect both demand and supply (Bhattacharya et al. 2024; Bucher-Koenen et al. 2025). In contrast, we find that gender differences in recommendations for investing in non-diversified assets, such as individual stocks or crypto, are entirely demand-driven.

Related literature. We contribute to two broad strands of the literature. First, we contribute to the literature on financial advice and financial innovation by introducing a framework to quantitatively study LLM-generated financial advice. This literature is motivated by the evidence that individuals do not, on their own, always make optimal financial decisions.[1] While this creates a role for professional advice, substantial evidence shows that existing financial advice sources are limited by both demand- and supply-side constraints (see Reuter and Schoar 2024 for a recent review). Professional human advisors are often costly, and their advice can be low quality (Linnainmaa et al. 2021; Andries et al. 2025), subject to conflicts of interest (Mullainathan et al. 2012), and affected by misconduct (Egan et al. 2019). Popular personal finance books are cheaper and more accessible, but their guidance often contains fallacies and departs from the prescriptions of normative economic

[1]This includes evidence that individuals do not take full advantage of available 401(k) employer matches (Choi et al. 2011; Choukhmane et al. 2025), do not efficiently allocate assets across accounts (Bergstresser and Poterba 2004), or fail to refinance fixed-rate mortgages when beneficial (Andersen et al. 2020). For reviews, see Beshears et al. (2018), Gomes et al. (2021), and Campbell and Ramadorai (2026).

models (Choi 2022). Automated advice, such as robo-advisors, can improve financial decisions (D'Acunto et al. 2019; Rossi and Utkus 2020; Reher and Sokolinski 2021), but relative to LLMs, robo-advisors may be more limited in their ability to provide personalized conversational advice, as well as the soft skills and emotional support that clients value in human advisors (Greig et al. 2025). Given their rapid growth, a natural question is whether LLMs can provide an affordable, widely accessible source of financial advice that could help overcome some of these issues with traditional forms of advice.

Our contribution is to provide evidence on whether LLM-generated advice can fulfill this promise, taking into account that the advice depends on the questions people ask, the characteristics of the models, and the accumulated impact over the life cycle. In doing so, we extend a longstanding tradition of using quantitative life cycle models to evaluate financial products and policies, with recent applications to adjustable-rate mortgages (Guren et al. 2021; Campbell et al. 2021), target-date funds (Duarte et al. 2025), saving nudges (Choukhmane 2025; Choukhmane et al. 2025), and income-contingent student loans (de Silva 2025). We apply this approach to what is perhaps the fastest-growing area of innovation in personal finance: generative AI.

Second, we contribute to the growing literature on how LLMs are changing economics and finance (see Mo and Ouyang 2025 for a review), and in particular to the emerging literature that studies LLMs as models of human behavior. This literature shows that LLMs often replicate human behavior and biases elicited in surveys and classic economic experiments (Horton et al. 2023; Ross et al. 2024; Bini et al. 2026).[2] A related strand studies LLM behavior in financial settings. Here too, LLMs both resemble and depart from human behavior: Ouyang et al. (2025) find that LLM risk-taking is stable across a variety of behavioral tasks; Fedyk et al. (2026) show that LLM preferences over asset allocations resemble those of younger, high-income individuals, but are less likely to violate transitivity; and Rumpf et al. (2026) compare recommendations from individuals, professional advisors, and LLMs, and find that LLMs are far more sensitive to risk preferences but also exhibit substantial stochasticity. A related set of recent papers studies LLMs directly as providers of financial advice using researcher-designed personas or hypothetical scenarios, examining whether LLMs can infer investor preferences, tailor portfolio recommendations, and provide useful guidance while also documenting limitations such as stochasticity, investment biases, and sensitivity to demographic attributes (for example, Fieberg et al. 2025; Takayanagi

[2]Also related is the strand of this literature that studies agent-based economic models populated by LLM agents, including Hao and Xie (2025), who study LLMs' solutions to a two-period consumption-saving problem, and Verstyuk and Douglas (2025), who study a Krusell and Smith (1998) economy populated with LLM agents.

et al. 2025; Foltyn and Olsson 2026).

Relative to this literature, our approach makes two contributions. First, instead of using LLMs to learn about, simulate, or replace human survey responses, we use surveys of humans to learn about LLMs: prompts written by a survey sample allow us to characterize both the supply *and* demand sides of human-LLM interactions. In this sense, our approach uses humans to learn about AI. Second, and more specific to our financial advice application, we embed LLM advice in a dynamic, stochastic life cycle model. This allows us to characterize the effects of LLM advice over an entire lifetime and compare the resulting behavior with both observed household choices and quantitative economic models. This life cycle perspective distinguishes our analysis from studies that evaluate LLM advice primarily in static survey, experimental, or portfolio-choice environments.

1 Methodology for Simulating LLM Advice

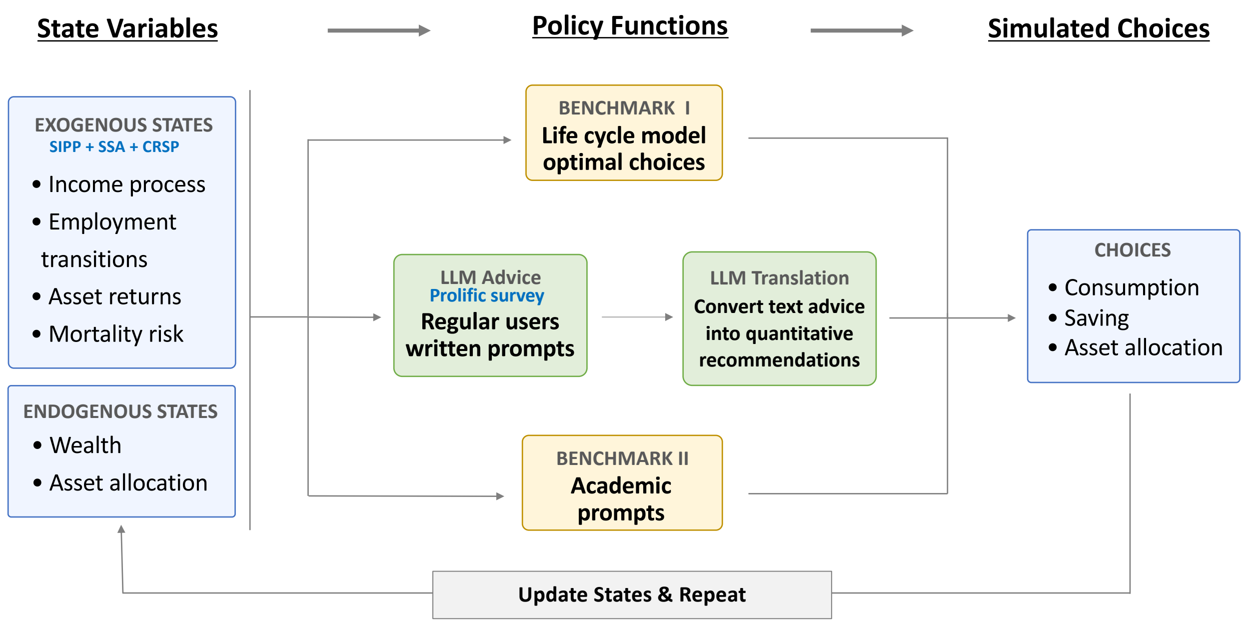
We begin by describing our methodology, which is summarized in Figure 1 and consists of three steps. First, we conduct a survey and ask a demographically representative sample to write prompts seeking saving and investing advice from an LLM (Section 1.1). Second, we calibrate a quantitative life cycle model to U.S. income dynamics, employment transitions, asset returns, and tax rules; this model provides both the economic environment for our simulations and a normative benchmark grounded in economic theory (Section 1.2). Finally, we combine the survey with the model to simulate the life cycle paths of individuals who would follow an LLM's advice each year (Section 1.3). After describing the methodology and benchmarks, which apply in general to any LLM, we describe our model selection.

1.1 Survey

The first step in our methodology is to conduct a survey that elicits prompts in which individuals seek financial guidance from an LLM. We design the survey in Qualtrics and administer it to a demographically balanced U.S. sample recruited through the Prolific online platform. Our initial sample consists of 1,000 complete responses, of which 952 pass both Prolific's authenticity check to screen out AI-generated responses and a minimum specificity requirement. Appendix A.2 provides more details on this sample selection.

The survey asks each respondent to write three free-text prompts to an LLM financial advisor (see Figure E1 for the exact wording). The first prompt asks respondents to describe their financial situation, including whatever details they think would help an LLM provide useful advice. Respondents are told to assume the LLM has no prior information about them. The second prompt asks respondents to seek advice on how much to spend over the

Figure 1. Overview of Methodology for Simulating LLM Advice



Notes: This figure summarizes the methodology. State variables (left) feed into policy functions (center), which produce simulated choices (right). The life cycle model (Benchmark I) provides optimal policy functions given calibrated state variables. LLM advice uses survey prompts to generate textual recommendations, which are then translated into quantitative choices. Academic prompts (Benchmark II) replace survey prompts with researcher-designed prompts. Endogenous state variables are updated each period and the process repeats over the life cycle. Section 3.1 also defines a one-shot advice benchmark that is not shown in the schematic.

coming year, while the third asks for advice on how to invest their savings between stocks and safer options. For the second and third prompts, respondents are told that the LLM already has the information from their first response. We focus on spending and investing advice because saving, budgeting, and investing are the domains in which people most commonly seek AI financial guidance, both in our survey and in prior evidence (J.D. Power 2025).[3]

In addition to the prompts, we survey respondents about the state variables that will be used in our life cycle simulation (including age, income, employment status, and financial wealth), as well as demographic characteristics such as gender, race, financial literacy, and prior experience with AI. Table E1 compares the demographics of our Prolific sample with those of the Current Population Survey (CPS). The sample broadly matches the CPS across age, gender, and income, though it over-represents the unemployed, under-represents the lowest-income earners, and under-represents respondents older than 70.

Summary statistics for the prompts written by our survey respondents—including word

[3] Among the 48% of respondents to our survey who report having used an AI tool for financial advice or information in the past three months, the most common topics are saving (49%) and investing (49%), followed by budgeting (42%) and financial education (35%) (Figure E2).

counts, character counts, and the prevalence of numerical and dollar-amount mentions—are reported in Table E2–Table E4. The financial situation prompt is the longest of the three (mean of 43 words), with 65% of respondents mentioning specific numbers and 34% mentioning dollar amounts. The spending and investment advice prompts are shorter (mean of 27 words each), with lower rates of quantitative detail.

1.2 Life Cycle Model

In the second step, we build a quantitative life cycle model of consumption, saving, and portfolio choices based on Choukhmane and de Silva (2026). The full specification and calibration of the model are presented in Appendix B.

Each model period corresponds to one year. Individuals enter working life at age 22, retire deterministically at age 65, and face age-dependent mortality risk based on the Social Security Actuarial Life Tables until a maximum age of 90. They have time-separable CRRA utility over consumption with annualized discount factor β and relative risk aversion γ.

Individuals face employment risk and move across four labor-market states: employed, transitioning from job-to-job, unemployed, and retired. While employed, they earn stochastic labor income that combines a deterministic age profile, a persistent AR(1) component, and a transitory shock. Job-to-job transitions are associated, in expectation, with persistent wage gains, whereas unemployment spells lead to persistent wage losses upon reemployment. These parameters of the income process and the employment transition probabilities are estimated using Survey of Income and Program Participation (SIPP) data. A government provides unemployment benefits to the unemployed and Social Security retirement benefits tied to average lifetime earnings to the retired. It also taxes income according to the 2025 U.S. federal schedule and taxes capital returns at a flat rate.

Investors can allocate their savings across four assets: a risk-free bond, a diversified stock market index, an individual stock, and an alternative risky asset representing holdings such as crypto, gold, commodities, and collectibles. The returns of the diversified index are calibrated using historical CRSP total market returns. We refer to the individual stock and the alternative risky asset as non-diversified assets, since they capture risky positions outside the diversified market index that an LLM can recommend. Both non-diversified assets have the same return process, but independent shock realizations, so they represent distinct risky positions. We calibrate their return process using evidence on individual stock returns from Bessembinder (2018). We construct these returns such that neither can improve the Sharpe ratio of a portfolio that combines the bond and the diversified stock market index.

In each period, investors choose consumption, saving, and portfolio shares across the four assets subject to no borrowing and leverage constraints. The model has seven state variables: age, labor productivity, transitory income shock, employment status, tenure, average lifetime income, and savings.

1.3 Simulating Advice over the Life Cycle

The final step in our methodology combines the survey responses with the life cycle model to simulate the life cycle paths of individuals who follow the advice provided by an LLM, as illustrated in Figure 1. Each simulated life path proceeds by iterating, every year from age 22 to 89, through five steps: states, prompt, advice, choices, and shocks. Table 1 illustrates these steps with a concrete example.

Step 1: States. At the beginning of each period, each simulated individual is characterized by the state variables of our life cycle model (Table 1, Panel A). In our example, the individual is 22 years old, employed, earning \$29,480, and has no financial assets.

Step 2: Prompt selection and preparation. We assign each simulated individual to one of 12 buckets defined by employment status, age, and income. We then randomly draw a prompt written by a survey respondent in the same bucket, so that the LLM advice assigned to the simulated individual is based on a prompt from someone with similar observable characteristics (Figure A1).[4] Within each prompt bucket, the characteristics of the simulated individual may still differ from those of the sampled survey respondent. We therefore identify any references to state variables in the prompt and replace them with the corresponding values from the simulation. For example, Table 1, Panel B shows the raw prompt as originally written by a 23-year-old survey respondent earning approximately \$30,000 per year. Panel C shows the same prompt after variable insertion: every mention of a state variable—including age, income, wealth levels, and portfolio allocations—is replaced with the simulated individual's current values. Our insertion rules, described in Appendix A, preserve the author's writing style and concerns while ensuring that the quantitative details match the simulated individual's situation. The three survey prompts (financial situation, spending advice, and investment advice) are then concatenated into a single message and sent to an LLM with an instruction to respond in 200 words or fewer.

Step 3: LLM advice. The LLM responds to the prepared prompt (Table 1, Panel D) with textual advice. The qualitative properties of this advice, including which topics the LLM emphasizes and how they compare with what individuals discuss in their prompts, are described in Section 2.1.

[4] In choosing which characteristics to match on, we face a trade-off between matching simulated individuals more closely to survey respondents and maintaining sufficient sample size within each bucket.

Step 4: Translating advice into quantitative choices. Next, the LLM's textual advice is converted into the quantitative recommendations for the choice variables of our model: consumption and savings held in the four asset classes. We do this by passing the LLM's advice back into another LLM along with deterministic instructions on how to extract numerical recommendations and output them in a structured JSON format. This second LLM call maps qualitative advice (e.g., "save 15–30% of income" and "invest in a total-market index fund") into dollar amounts for consumption and contributions to each asset class (Table 1, Panel E). The extraction follows a priority ordering: annual dollar amounts are used when stated; monthly amounts are annualized; ranges are converted to midpoints; percentages of income are converted to levels. The full translation prompt, including the asset class definitions, extraction priorities, budget constraints, and deterministic rules that ensure all outputs satisfy accounting identities, is in Appendix A.7.

Step 5: Shocks and state transitions. After extracting the quantitative recommendations, we draw realizations of the exogenous shocks in our life cycle model: labor market transitions, persistent and transitory income shocks, and asset returns for each of the four asset classes. The realized shocks, combined with the LLM's choices, determine the state variables at the beginning of the next period (Table 1, Panel F). These updated states are then used as the input for the next iteration.

Simulation design. Each query to the LLM is independent, with no memory of past responses; the only link across periods is the evolution of state variables that depend on prior choices. In each run of the model, we simulate the same 1,000 individuals, each starting life at age 22 with no wealth, retiring at age 65, and dying by age 90. A new prompt is drawn and prepared in every period for every individual, and the distribution from which prompts are drawn evolves as an individual's state variables change.

1.4 Benchmarks to Compare with LLM Advice

We compare our simulated LLM advice to three benchmarks. The first is respondents' status-quo self-reported behavior, which captures what people do on their own. The second is optimal behavior in the life cycle model, which provides a normative benchmark. The third is LLM advice generated from a structured researcher-designed prompt, which tests whether providing more complete and standardized information moves advice closer to the model benchmark. The latter two benchmarks are illustrated in Figure 1.

Benchmark I: Observed behavior. The observed-behavior benchmark uses respondents' self-reported savings and portfolio choices from our survey. The behavior of our survey respondents is broadly consistent with evidence on saving and investing over the life

Table 1. From States to Choices: Simulation Example for One Period

Panel A: State variables

Age	Employment status	Gross Annual income	Bonds	Diversified stocks	Individual stocks	Alternative risky assets
22	Employed	$29,480	$0	$0	$0	$0

Panel B: Raw survey prompt (as written by respondent)

Financial situation	"I'm 23 years old making 30,000 a year, I dont spend a lot besides normal expenses such as food and gas."
Spending advice	"at 30,000 a year what should my spending look like?"
Investment advice	"i'm looking to start investing to get my money up, preferably slow gain. How can i get into this in a smart and somewhat safe way while making 30,000 a year?"

Panel C: Prepared prompt (after variable insertion)

Financial situation	"I'm **22** years old making **$29,480** a year, I dont spend a lot besides normal expenses such as food and gas."
Spending advice	"at **$29,480** a year what should my spending look like?"
Investment advice	"i'm looking to start investing to get my money up, preferably slow gain. How can i get into this in a smart and somewhat safe way while making **$29,480** a year?"

Panel D: LLM advice

"On $29,480/yr (~$2,457/mo gross), aim for a simple, safe setup: 1) Build a base first — Emergency fund: start with $1,000, then build to 3–6 months of essential expenses in a high-yield savings account (HYSA). — Pay off high-interest debt (esp. credit cards) before investing. 2) Spending targets (monthly take-home varies by state/benefits) — Needs (rent, utilities, food, gas, insurance): 55–70% — Wants: 5–15% — Saving/investing: 15–30% (even $50–$200/mo is fine). If rent is high, keep investing smaller but consistent. 3) 'Slow gain' investing (low-cost, diversified) — If you have a 401(k) match, contribute enough to get the full match. — Otherwise (or in addition), open a Roth IRA (Fidelity/Vanguard/Schwab). — Invest in a total-market index fund/ETF or a Target-Date fund (set-and-forget). Keep fees low."

Panel E: Extracted choices

Consumption	Total saving	Bonds contribution	Diversified stocks contribution	Individual stocks contribution	Alternative risky assets contribution
$22,847	$6,633	+$1,000	+$5,633	$0	$0

Panel F: Updated states (beginning of next period)

Age	Employment status	Annual income	Bonds	Diversified stocks	Individual stocks	Alternative risky assets
23	Employed (no transition)	$36,528 (income shocks)	$1,020	$6,247	$0	$0

Notes: This table illustrates the five steps of the simulation for a single individual in one period. Panel A shows the state variables at the start of the period. Panel B shows the raw survey prompt as originally written by a 23-year-old respondent earning approximately $30,000 per year. Panel C shows the same prompt after variable insertion, where boldface values indicate state variables that were substituted into the respondent's text. Panel D shows the full LLM advice. Panel E shows the quantitative choices extracted by the translation LLM call, which must satisfy the budget constraint: consumption + total saving = post-tax income. Panel F shows the state variables at the start of the next period after labor market, income, and asset return shocks are realized.

cycle prior to the widespread adoption of AI financial advice tools, such as the Survey of Consumer Finances (Gomes et al. 2021).

Benchmark II: Life cycle model. The second benchmark is constructed by simulating the life cycle model described above and solving for optimal behavior under the same realizations of exogenous shocks used in the LLM simulation. Because the aggregate stock index is calibrated to be mean-variance efficient, we assume that individuals in this benchmark hold neither non-diversified asset class. Portfolio choice, therefore, reduces to a standard allocation between the risk-free bond and the diversified stock index.

Benchmark III: Academic prompts. The third benchmark replaces the respondent-written survey prompts with a more structured prompt that provides the LLM with explicit information about the individual's current state variables and makes explicit assumptions about the economic environment. The prompt includes individual-specific information on all the state variables, along with baseline assumptions about expected asset returns, taxes, and Social Security rules. It also instructs the LLM to act as a professional financial advisor trained in portfolio theory and life cycle planning. The full prompt and assumptions are presented in Appendix A.6 and Table A3.

1.5 LLM Selection

We choose an OpenAI model as our primary LLM because survey evidence—both from our own sample (Figure E3) and from other recent surveys (Lloyds Banking Group 2025; J.D. Power 2025)—indicates that ChatGPT is the most commonly used LLM for financial advice.[5] Among OpenAI's model family, we use `GPT-5.2` (OpenAI 2025), which was the latest available model at the time we ran our analysis. We set its reasoning effort to "Low" for speed and cost efficiency. For the task in Step 4 of Section 1.3 that translates the LLM's textual advice into quantitative recommendations, we use `GPT-5 Mini`, which is a smaller and faster model in the same family. We choose this smaller model because this task involves deterministic extraction rules rather than open-ended reasoning. As a robustness check, we repeat our main analysis using `Gemini 3 Flash` (Google DeepMind 2025), the second most commonly used model in our survey. We report these results in Appendix E and summarize them in Section 3.4.

[5] Using a closed-source model has clear disadvantages from the perspective of reproducibility. However, we chose to prioritize using a model that households are more likely to use in practice.

2 Describing Prompts and Baseline LLM Advice

This section first describes the qualitative content of the prompts written by our survey respondents and the LLM's corresponding textual responses. We then describe the LLM's baseline advice over the life cycle, and examine its robustness to alternative methodological choices.

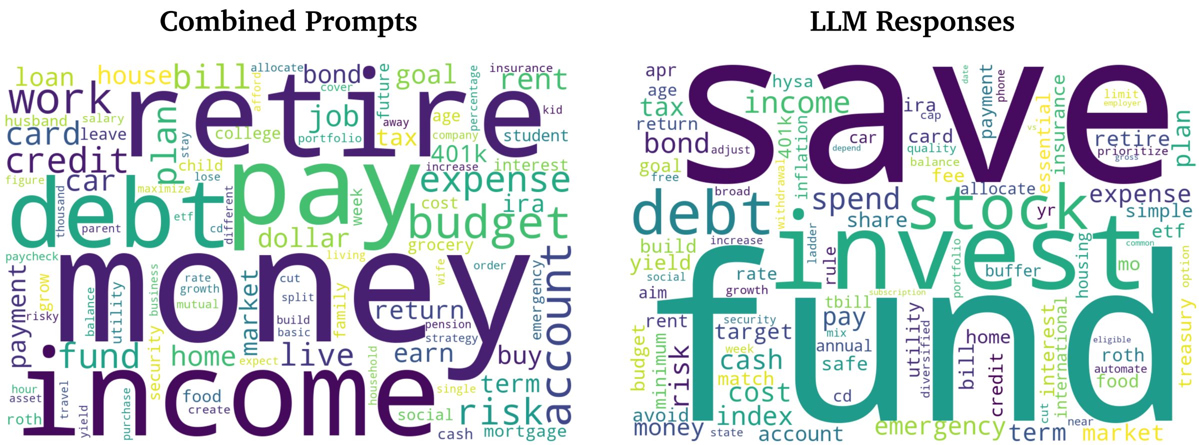
2.1 What Respondents Write and What LLM Advice Recommends

Measuring topics in prompts and advice. We perform a simple textual analysis by matching keywords in each prompt and LLM response to a dictionary of 27 pre-defined topic categories. For each prompt (or LLM response) and category, we record whether any keyword from that category appears and report the share of respondents who mention each category. When comparing prompts with LLM responses, we weight each prompt by the number of times it is drawn and used in the simulation. Full methodological details, including the text preprocessing pipeline (Figure C1), the complete dictionary and topic categories (Table C1), and prompt-level summary statistics (Table E2–Table E4), are in Appendix C.

What respondents write about. Figure 2 shows word clouds for the survey prompts and LLM's responses. The most prominent words in the prompts are "money," "debt," "income," "pay," and "retire." The topics respondents discuss vary across the three prompts they write (Figure E4). When respondents describe their financial situation, the most common categories are Income (35%) and Retirement (30%). When asking for spending advice, Budgeting is the most frequent (45%). When asking for investment advice, Investment Assets (53%) and Risk Preference (40%) are the most common categories.

How LLM advice responds to these prompts. The LLM advice closely reflects the topics survey respondents write about: the five most common topics in the prompts (i.e., Budgeting, Investment Assets, Income, Risk Preference, and Retirement) all appear at substantially higher rates in the LLM's advice (Figure E6). At the same time, the LLM also introduces topics that are less commonly mentioned by respondents. For example, Liquidity appears in 84% of advice responses despite only 6% of respondents mentioning it explicitly, reflecting the model's high propensity to recommend building an emergency fund. Consistent with this pattern, the most prominent words in the LLM advice word cloud are "fund," "invest," "save," "stock," and "debt," along with terms such as "essential," "insurance," and "emergency" (Figure 2). Despite these differences, topic rankings remain broadly aligned between prompts and responses: the rank correlation across all 27 categories is 0.72.

Figure 2. Word Clouds: Respondent Prompts and LLM Advice



Notes: This figure shows word clouds for respondent-written prompts pooled across the three prompt questions (left) and for the corresponding LLM advice (right). Word size is proportional to word frequency after text preprocessing. To make the clouds easier to read, we exclude common stopwords and words that mechanically reflect the wording of the survey questions. Figure E5 shows analogous word clouds separately for each prompt question. Details are provided in Appendix C.

LLM advice provides specific investment recommendations. One important way in which the LLM advice goes beyond respondents' prompts is by naming specific asset classes, financial products, and providers (Figure 3). For example, high-yield savings accounts appear in 59% of advice responses, compared with 3.5% of prompts; bonds appear in 55%, compared with 10%; and Treasury bills, notes, or bonds appear in 40%, compared with 1%. At the product level, fewer than 3% of respondents name any specific ticker, yet LLM advice sometimes recommends branded products: Vanguard investment products appear in 7% of responses; iShares appears in 3.4%; and cryptocurrency tokens such as Bitcoin or Ethereum appear in 1.9%. The fact that this advice often contains specific recommendations about asset classes, account types, commercial products, and providers raises broader questions about how AI financial advice might affect product competition and the regulation of new sources of financial advice.

2.2 Baseline Quantitative Recommendations

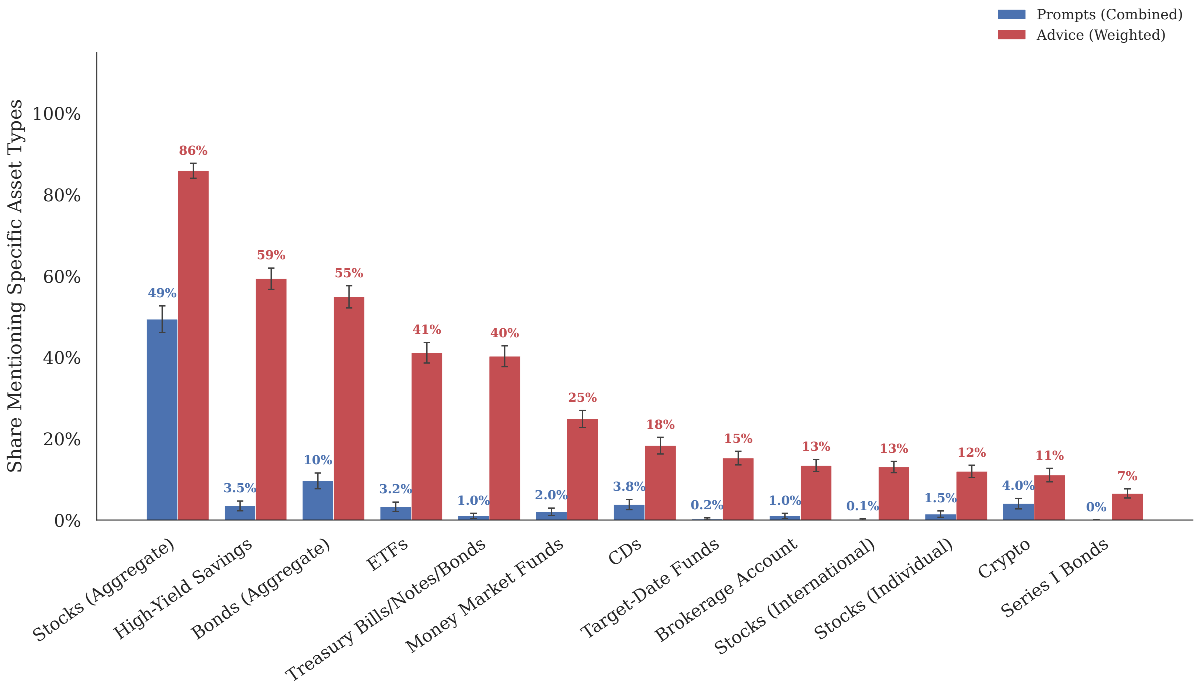
Figure 4 plots the median consumption and diversified equity share by age for individuals following the LLM's advice, along with 25th–75th percentile bands.[6] Table E5 presents additional summary statistics.

Consumption and saving advice. The left panel of Figure 4 shows that, consistent with textbook life cycle theory, the LLM recommends that individuals smooth their income over working life such that consumption is flatter than income. Individuals following the

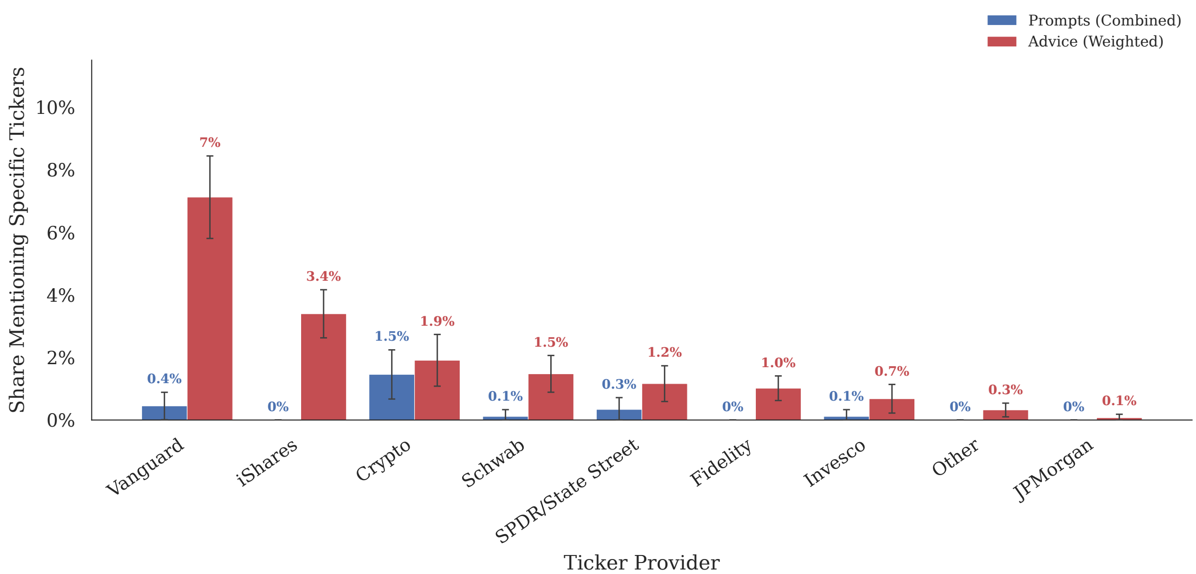
[6]We find quantitatively similar results when using `Gemini 3 Flash`, which are shown in Figure E7.

Figure 3. Asset Classes and Products: Prompts vs. Advice

Panel A: Investment Asset Classes

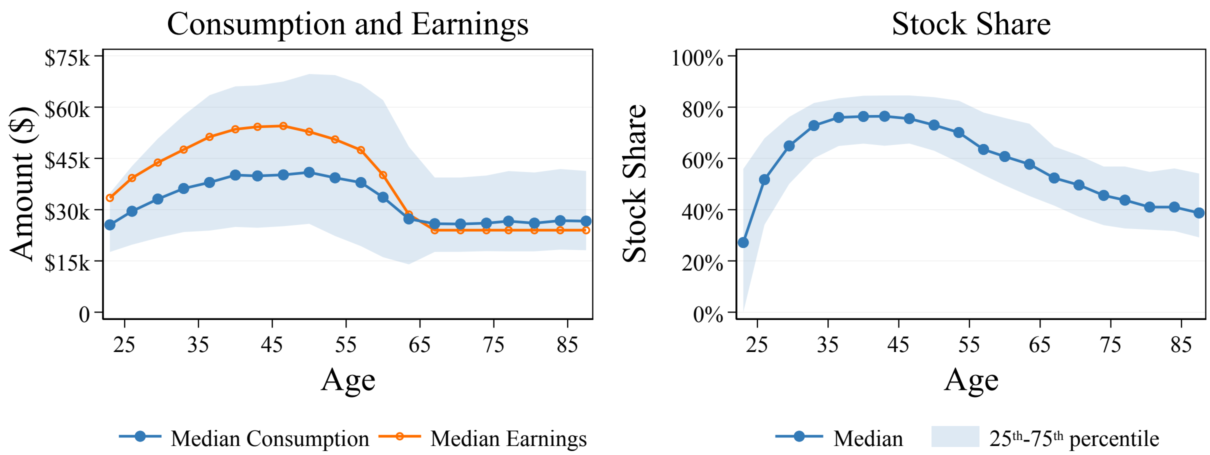


Panel B: Specific Products by Provider



Notes: This figure shows grouped bar charts comparing the prevalence of asset classes and products in prompts and advice. Panel A shows specific investment asset classes with at least 5% prevalence in either prompts or advice, ordered by descending advice prevalence. Panel B shows specific financial products (tickers and fund names) aggregated by issuing provider, ordered by descending advice prevalence. Blue bars show the share of respondents mentioning the category in any of the three prompts. Red bars show the respondent-weighted mean mention rate in the LLM advice. Error bars show 95% confidence intervals.

Figure 4. Life Cycle Profiles of LLM-Recommended Consumption and Stock Shares



Notes: This figure plots the life cycle profiles for individuals following the LLM's recommended choices using the survey prompts, as described in Section 1.4. Ages (22–89) are grouped into 20 equally sized bins. The left panel shows consumption and post-tax earnings, and the right panel shows the stock share. Dots denote median values within each bin, and shaded areas represent the interquartile range (25^{th}–75^{th} percentile). All values are in 2025 dollars.

LLM's advice achieve this by saving more during working life and consuming less than their income, while withdrawing and consuming (slightly) more than their income during retirement. This is reflected in their wealth accumulation, with individuals building up a large stock of over $1 million in assets by the time they retire at age 65 (Table E5).

Investment advice. The right panel of Figure 4 shows that the recommended equity shares are relatively high, around 65% on average, and decline with age after approximately age 45. Qualitatively, these portfolio choices are consistent with textbook theories of portfolio choice with human capital (e.g., Merton 1969; Gomes 2020). While less steep than a typical target-date fund glide path, the LLM's recommended equity allocation in working life is similar to the average equity shares that retirement savers choose when making active portfolio choices (Choukhmane and de Silva 2026). The LLM also recommends small allocations to non-diversified assets (e.g., individual stocks and other risky assets such as crypto, gold, commodities, and collectibles). As shown in Figure E8, participation in these asset classes increases with age, reaching approximately 40% for individual stocks and 75% for other risky assets by retirement. However, conditional portfolio shares remain small, averaging 2–3% of financial wealth, so that the vast majority of the LLM's recommended risky asset allocation is in diversified equities.

3 LLM Advice and Life Cycle Theory

This section evaluates the financial behavior implied by following LLM advice against the prescriptions of life cycle theory. We first show that following the advice would move respondents toward broad theoretical prescriptions: higher participation in diversified equities, equity shares that decline with age, and larger savings buffers. We then document systematic departures from the normative theory's more subtle prescriptions: the advice implies unusually high patience, relies on simple saving and withdrawal heuristics, smooths consumption imperfectly, and allows portfolios to drift passively with realized returns. Finally, we show that a more structured prompt reduces some (but not all) of these departures.

3.1 Fact 1: Following LLM Advice Would Move Individuals Toward Life Cycle Theory

We begin by asking whether following LLM advice would move respondents closer to the broad prescriptions of life cycle theory relative to observed behavior. This comparison is intentionally qualitative: precise quantitative recommendations from a life cycle model depend on the model specification, preference parameters, the calibration of asset and income risk, and other features of the environment. Therefore, we focus on general principles that are robust across calibrations of standard life cycle portfolio choice models: broad participation in diversified equities, equity shares that decline later in life, and the accumulation of precautionary savings buffers (e.g., Gomes 2020).

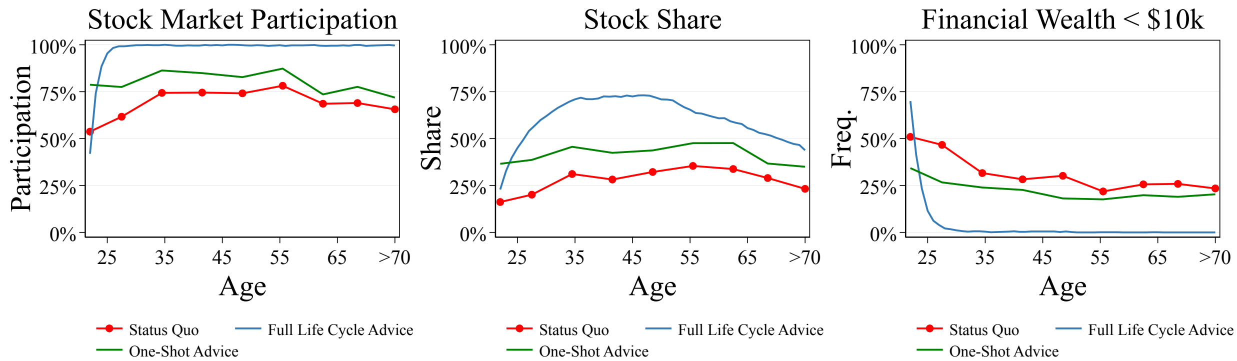
We compare respondents' observed behavior with two versions of LLM-recommended behavior. The first is a one-shot benchmark, which uses the advice generated from each respondent's original survey prompt and current self-reported financial situation, without replacing state variables or simulating subsequent choices.[7] The second is the full life cycle simulation from Section 1.3, in which individuals follow LLM advice each period using prompts drawn and updated to reflect evolving state variables in the life cycle simulations. Figure 5 plots respondents' observed behavior in red, the one-shot advice benchmark in green, and the full life cycle simulation in blue.[8]

Advice would raise participation in diversified equities. As shown in the left panel of Figure 5, one-shot advice already points respondents toward broader stock market

[7] We convert the qualitative advice into numerical choices using each respondent's self-reported age, income, employment status, and asset holdings, together with the same JSON translation procedure described in Step 4 of Section 1.3.

[8] We find quantitatively similar results when using `Gemini 3 Flash`, as shown in Figure E9.

Figure 5. Observed Behavior vs. LLM-Recommended Behavior



Notes: This figure compares respondents' observed financial behavior with the behavior implied by following LLM advice. Red lines show respondents' observed behavior. Green lines show the one-shot advice benchmark, which applies the LLM advice generated from each respondent's own prompt to that respondent's current financial situation. Blue lines show the full life cycle simulation, in which simulated individuals follow LLM advice in each period and prompts are updated to reflect the evolving state variables in the simulation, as described in Section 1.3.

participation. Consistent with the empirical evidence on limited stock market participation, one-third of respondents in our survey report holding no equities, and their average equity share is around 30%. Relative to observed behavior, the green line shows that one-shot advice would raise participation by approximately 5–10 percentage points and raise average equity shares by approximately 10 percentage points. The full life cycle simulation amplifies this movement: following the advice each period leads to near-universal participation and raises equity shares by as much as 40 percentage points relative to observed behavior.

Advice would generate higher equity shares that decline later in life. As shown in the middle panel of Figure 5, recommended portfolio allocations also move closer to the age profile implied by standard life cycle theory. Respondents' observed equity shares are roughly flat by age and, if anything, increase through age 55. By contrast, in the full life cycle simulation, equity shares are higher, rising early in working life as individuals build emergency buffers, then declining after approximately age 45. As we discuss in the next section, however, this decline is quantitatively more modest than what a standard life cycle model predicts.

Advice would build larger savings buffers. The right panel of Figure 5 shows that following LLM advice would lead respondents to build financial buffers more quickly. More than 20% of respondents in every age group report holding less than $10,000 in financial wealth. One-shot recommendations reduce this share in all age groups, but cannot fully offset low accumulated wealth within a single year. In contrast, when we simulate individuals following LLM advice every period, virtually all build a savings buffer above $10,000 by age 30.

Advice responds sensibly to the topics respondents raise. Beyond these average patterns, LLM recommendations also vary systematically with the topics respondents mention in their prompts, often in ways that align with life cycle theory. To document this, we exploit the fact that, within each employment-by-income-by-age prompt bucket, the specific prompt assigned to a simulated individual is randomly drawn. This randomness allows us to compare recommendations generated by different prompts for otherwise similar simulated individuals. Specifically, we regress two outcomes, the net saving rate and the change in risky share, on indicators for each of the dictionary categories defined in Section 2.1, controlling for income, wealth, age fixed effects, and prompt-bucket fixed effects. Figure 6 reports coefficients for categories mentioned in at least 5% of prompts.

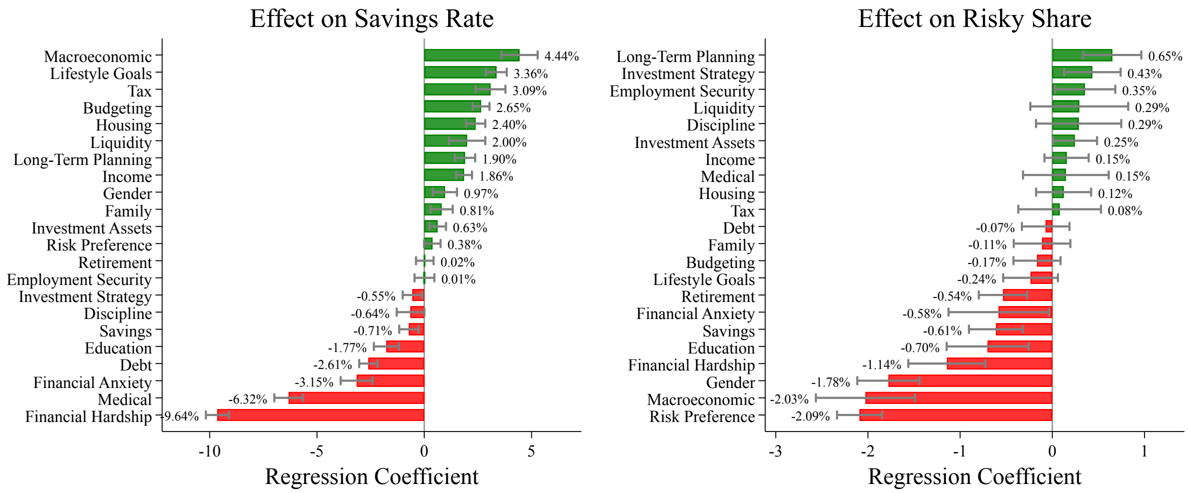
The results suggest that the LLM responds to prompt content in ways that are broadly consistent with life cycle theory. When prompts mention macroeconomic conditions, typically uncertainty about inflation or recessions, the LLM raises the recommended saving rate by around 4pp and lowers the recommended risky share by 2pp, consistent with a greater precautionary motive. Mentions of financial hardship lower the recommended saving rate by around 10pp and the risky share by 1pp, consistent with immediate liquidity needs. Mentions of risk preferences modestly raise the saving rate by 0.4pp but lower the recommended risky share by 2.1pp. This last pattern echoes Rumpf et al. (2026), who find that LLM portfolio recommendations are more sensitive to stated risk preferences than human advisors.

Summary. Taken together, these results show that following LLM advice would move respondents toward broad prescriptions of life cycle theory with higher participation in diversified equities, high equity shares that decline later in life, and larger savings buffers. This is not only an average pattern: the LLM also adjusts its recommendations for saving and risk-taking in response to economically relevant information in respondents' prompts. These findings should be interpreted as the effect of following advice, not the effect of receiving advice. Whether LLM recommendations change actual financial decisions, and how those effects compare to other forms of financial advice, are important questions for future work.

3.2 Fact 2: LLM Advice Departs from Life Cycle Theory in Systematic Ways

We next compare LLM advice to the more quantitative predictions from our calibrated life cycle model. While Fact 1 shows that following the advice would move respondents toward broad principles of life cycle theory, this comparison highlights some systematic departures

Figure 6. Effect of Prompt Categories on Choice Variables



Notes: This figure plots the effect of dictionary category mentions on the net saving rate (left panel) and the change in risky share (right panel). Only dictionary categories mentioned in at least 5% of prompts are included. Each bar reports the coefficient on an indicator for whether the prompt mentions the category, from a regression that also controls for income, wealth, and fixed effects for age and prompt bucket. The dictionary categories are constructed as described in Section 2.1 and listed in Table C1. Error bars show 95% confidence intervals.

from the model's more specific prescriptions. Specifically, the advice implies unusually high patience, relies on simple saving and withdrawal heuristics, smooths consumption imperfectly in response to income shocks, and allows portfolios to drift passively rather than be actively rebalanced.

Advice implies unusually high patience. We summarize the distance between LLM advice and the life cycle model by estimating the preference parameters that would rationalize the LLM's recommendations. Specifically, we use the simulated method of moments (SMM) to estimate the intertemporal discount factor (β) and coefficient of relative risk aversion (γ) that minimize the distance between model-generated and LLM-generated life cycle profiles.[9] We target two sets of moments: the wealth-to-income ratio during working years and the equity share over the full life cycle. Additional details are provided in Appendix D.

The first row of Table 2 reports estimates targeting moments from the baseline advice simulation with survey prompts. The estimated risk aversion, $\hat{\gamma} = 5.3$, is moderate and within the range of standard estimates. By contrast, the estimated discount factor, $\hat{\beta} = 1.034$, is above one. This high value reflects the LLM's recommendation to consume less than income throughout working life, which generates substantial wealth accumulation by

[9]This approach is similar to Cook et al. (2026), who estimate structural preference parameters from LLM behavior in dictator games and job search scenarios. Abdel Haq et al. (2026) pursue an alternative approach, eliciting LLM preferences directly, and show they align closely with those of human subjects.

Table 2. Simulated Method of Moments Estimation Results

Prompt Type	**LLM Model**	$\hat{\beta}$	$\hat{\gamma}$
Survey	`GPT-5.2`	1.034	5.3
Survey	`Gemini 3 Flash`	1.054	5.1
Academic	`GPT-5.2`	0.99	4.7

Notes: This table presents the results of our simulated method of moments estimation. The two columns on the right present the estimated preference parameters $\hat{\beta}$ (intertemporal discount factor) and $\hat{\gamma}$ (coefficient of relative risk aversion) that minimize the sum of squared proportional errors between a set of LLM advice moments and their equivalent moments calculated in the life cycle model. The moments are the 25th, 50th, and 75th percentiles of the wealth-to-income ratio in each working year and the 25th, 50th, and 75th percentiles of the equity share in each year of the full life cycle. For a full description of the estimation procedure, see Appendix D. Survey prompts use the respondent-written prompts from our Prolific survey. Academic prompts use the researcher-designed prompt described in Section 1.4.

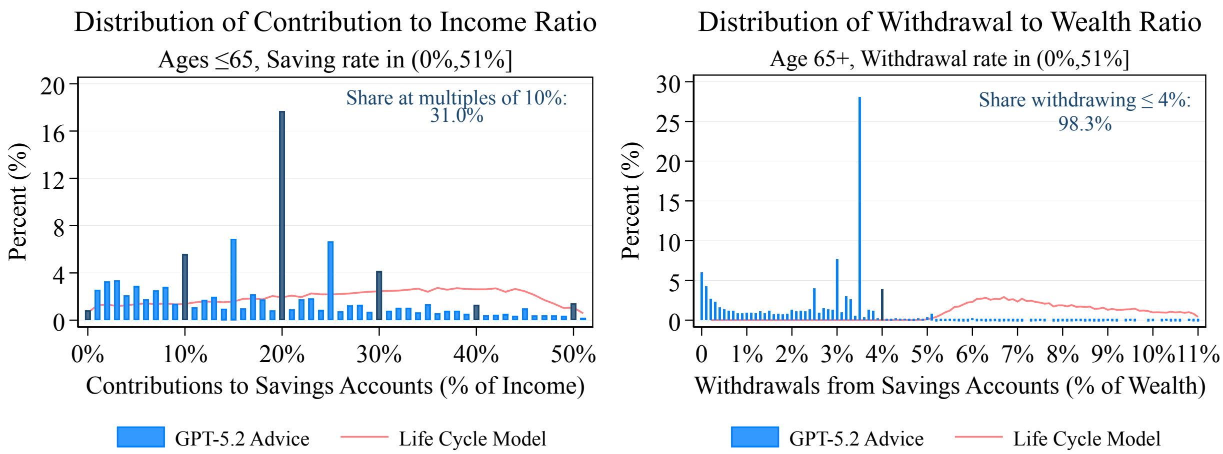
retirement. While bequest motives could, in principle, rationalize this pattern, almost no respondents mention bequests in their prompts. Another possible interpretation is that the LLM advice reflects a form of paternalism: it may recommend a higher saving target than the standard model would prescribe because individuals may systematically undershoot the targets they receive. In that sense, conservative advice can partly offset imperfect implementation. However, if recommendations are followed exactly, a standard life cycle model can rationalize this advice only with an unrealistically high discount factor.

Figure E10 shows the fit of the estimated model on the targeted moments. The model matches the LLM's wealth accumulation during working life and its equity shares after age 45, but cannot match the relatively low and upward-sloping equity shares that the LLM recommends between ages 25 and 45.[10]

Advice relies on simple saving and withdrawal heuristics. We next examine the LLM's individual-level recommendations that underlie the aggregate life cycle profiles. Figure 7 plots the distributions of LLM-recommended saving rates, saving amounts, and withdrawal rates in the baseline advice simulation (blue histograms), along with the corresponding distributions implied by the life cycle model estimated in Table 2 (red lines). The LLM recommendations exhibit bunching at round numbers: 31% of saving rates are multiples of 10%, and 34% of saving amounts are multiples of $5,000. These spikes are absent by construction in the life cycle model, where choices vary continuously with state variables. Withdrawal advice displays an even stronger heuristic pattern. More than 98% of LLM-recommended withdrawals are at or below 4% of assets, echoing the 4% retirement withdrawal rule popularized by Cooley et al. (1998) and commonly used by financial advisors. By contrast, almost no individuals in the life cycle model choose withdrawal rates

[10] This mismatch reflects a standard implication of life cycle portfolio models: absent participation costs (Gomes and Michaelides 2005) or correlated labor income risk (Benzoni et al. 2007; Catherine 2022), optimal equity shares are high early in life.

Figure 7. Saving and Withdrawal Heuristics in LLM Advice: Survey Prompts



Notes: This figure plots the distributions of saving rates, saving amounts, and withdrawal heuristics in both the LLM advice and the life cycle model. The LLM advice is generated using the survey prompts as described in Section 1.4. Blue histograms represent the LLM advice, and dark blue bars highlight heuristic values, with spikes in saving rates at multiples of 10%, saving amounts at multiples of $5,000, and withdrawal-to-wealth ratios at 4%. The red lines represent the corresponding distributions from the life cycle model described in Section 1.2. All values are in 2025 dollars.

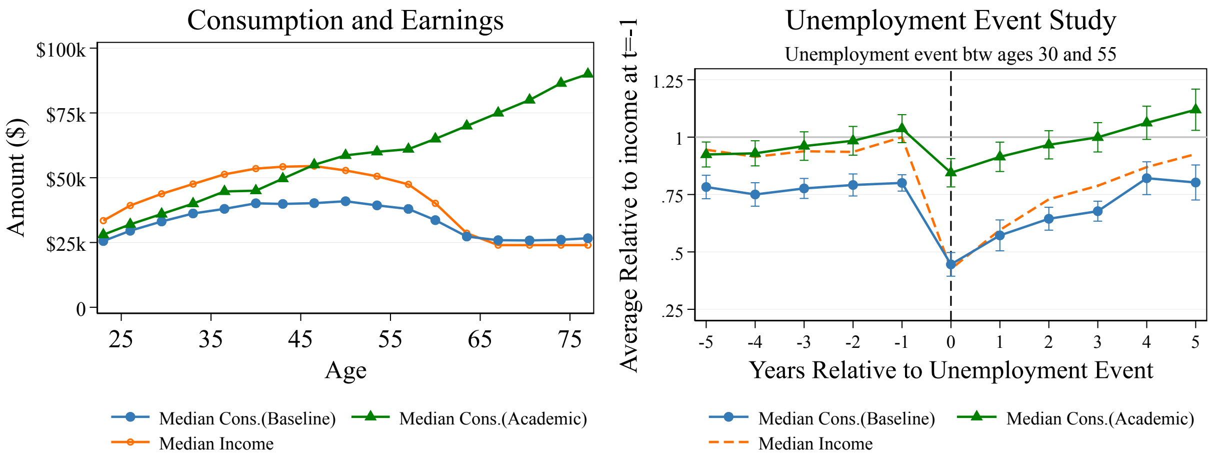
below 4%, consistent with the model prescribing faster decumulation than the LLM.

Consumption is insufficiently smoothed. As shown in the left panel of Figure 8, the LLM's recommended consumption (blue line) closely tracks income (orange line) over the life cycle. The right panel shows that this limited consumption smoothing also appears around simulated transitions into unemployment. After job loss, simulated income falls by roughly 50%, and the LLM's recommended consumption falls by a similar percentage, consistent with empirical evidence on consumption responses to unemployment in Gruber (1997) and Ganong and Noel (2019). In the life cycle model, by contrast, consumption barely falls after job loss because individuals draw down buffer stocks to absorb the shock (Figure E11). The fact that LLM advice smooths consumption insufficiently both over predictable life cycle changes in income and around transitory income shocks is particularly notable, given that simulated individuals hold sizable liquid balances.[11]

Portfolio recommendations exhibit passive drift. Following Calvet et al. (2009), we measure passive portfolio drift as the change in an individual's equity share from period t to $t + 1$ that would occur if the individual made no active portfolio adjustment and simply let returns change the portfolio shares. Figure 9 plots actual changes in equity shares

[11]Results are similar in our baseline specification, where asset holdings are mentioned in the prompt only if the original survey respondent included such information, and in a robustness exercise that appends all relevant state variables to the prompt; see discussion in Section 3.4 and Figure E12.

Figure 8. Consumption Smoothing: Survey vs. Academic Prompts



Notes: This figure compares consumption profiles between the survey and academic prompts. The left panel plots median consumption and median income over the life cycle for each prompt type. The right panel plots the median ratio of consumption relative to its level one year prior to unemployment within a five-year window around the unemployment event, along with median income. All values are in 2025 dollars.

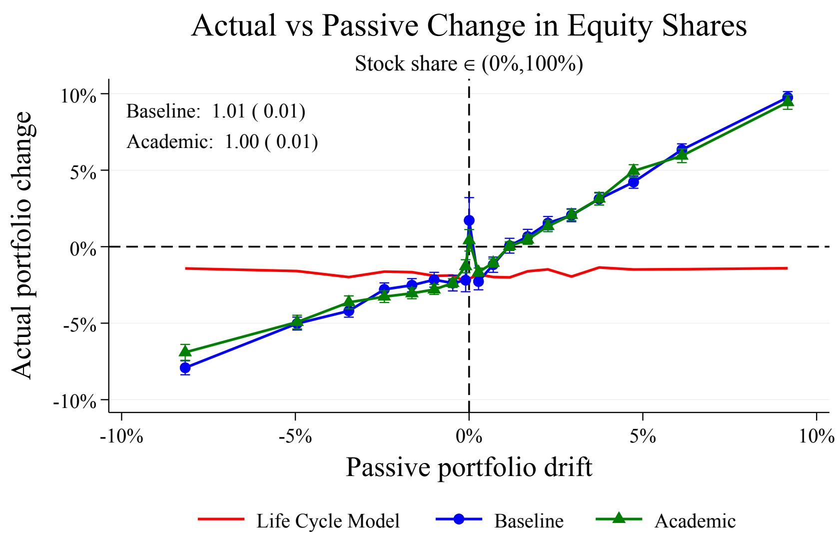
against this passive-drift measure. In the life cycle model with CRRA preferences, these two measures are essentially uncorrelated: target allocations are not sensitive to wealth, and individuals actively choose their portfolio each period. In contrast, LLM-recommended portfolio changes move almost one-for-one with passive drift, exhibiting substantial inertia consistent with evidence from household portfolios (Calvet et al. 2009; Choukhmane and de Silva 2026).[12] This inertia persists even when we skip the translation step and provide the LLM with all relevant state variables, including the existing portfolio allocation (as we discuss later in Section 3.4). This suggests that the inertia in the LLM's advice is not an artifact of respondents omitting information about their current portfolios. Instead, the LLM tends to advise individuals on how to allocate new savings rather than explicitly recommending that they rebalance existing holdings.

3.3 Structured Prompts Reduce Some Departures from Life Cycle Theory

We next examine whether replacing the prompts written by survey respondents with a more structured researcher-designed prompt, which we call the *academic prompt*, moves LLM advice closer to life cycle theory. As described in Section 1.4, the academic prompt provides explicit values for all current state variables, clarifies assumptions about the

[12]Quantitatively, Calvet et al. (2009) estimate a regression coefficient of 0.5 between actual and passive portfolio changes using observational data, suggesting that LLM recommendations exhibit roughly twice as much inertia.

Figure 9. Passive Portfolio Drift in LLM Advice



Notes: This figure compares actual portfolio changes against passive portfolio drift for the LLM advice (using both survey and academic prompts) and the life cycle model. The actual portfolio change is defined as the adjustment in the equity share prior to return realization from one period to the next. Passive portfolio drift is defined as the change in the equity share from before return realization at the beginning of the period to after return realization at the end of the period. Passive portfolio drifts are grouped into 20 equally sized bins. Error bars show 95% confidence intervals. The regression slopes for the survey and academic prompts are reported in the top-left corner. The sample is restricted to observations with stock shares strictly between 0% and 100%. The orange line represents the version where the prompt is combined with the state variables, including the initial portfolio share, and then converted directly into quantitative actions in a JSON format.

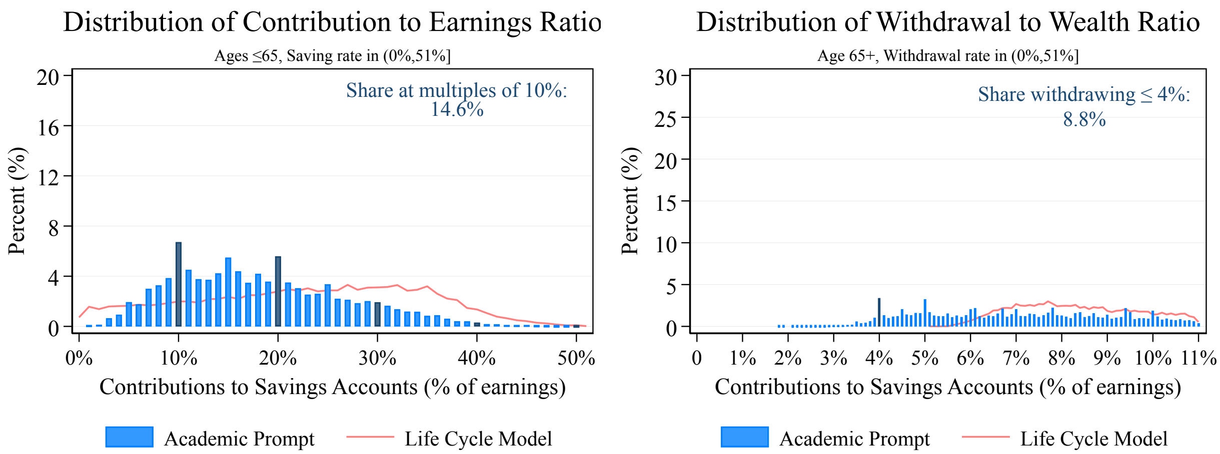
economic environment, and frames the task as providing professional financial advice.[13]

Structured prompts reduce heuristics. The LLM's advice produced in response to the academic prompt exhibits less reliance on simple saving and withdrawal heuristics. Figure 10 shows that the distribution of recommended saving rates exhibits less bunching at round numbers than under respondent-written survey prompts, and withdrawal recommendations are less concentrated at values at or below 4% of assets. These changes translate into a more reasonable estimated discount factor: the academic prompt estimate is $\hat{\beta} = 0.99$, relative to $\hat{\beta} = 1.034$ under survey prompts (third row of Table 2). Portfolio choices, by contrast, remain broadly similar and imply a similar estimate of risk aversion, $\hat{\gamma} = 4.7$.

Structured prompts improve consumption smoothing. The advice produced in response to the academic prompt exhibits greater consumption smoothing over the life cycle and in response to income shocks. The left panel of Figure 8 shows that consumption no

[13]The academic prompt also skips the translation step by instructing the LLM to provide quantitative recommendations directly. As we discuss in Section 3.4, skipping the translation step and providing all state variables while using survey prompts generates advice similar to our baseline. Therefore, the differences in this section can primarily be attributed to the content and framing of the academic prompt, rather than to differences in the translation pipeline or information about state variables. The corresponding life cycle profiles and comparison with observed behavior are shown in Figure E13 and Figure E14.

Figure 10. Saving and Withdrawal Heuristics in LLM Advice: Academic Prompts



Notes: This figure plots the distributions of saving rates, saving amounts, and withdrawal heuristics in both the LLM advice and the life cycle model. The LLM advice is generated using the academic prompt as described in Section 1.4. Blue histograms represent the LLM advice, and dark blue bars highlight heuristic values. The red lines represent the corresponding distributions from the life cycle model described in Section 1.2. All values are in 2025 dollars.

longer tracks income over the life cycle (green line), and individuals decumulate more substantially in retirement. The right panel shows a parallel improvement around transitions into unemployment. With the academic prompt (green), the drop in consumption is smaller and reverses more quickly, consistent with greater use of accumulated savings buffers to absorb income shocks.

Structured prompts do not reduce portfolio inertia. In contrast, structured prompts do not meaningfully reduce portfolio inertia. As shown in Figure 9 (green line), the relationship between actual changes in equity shares and passive portfolio drift remains close to one under both survey and academic prompts. This suggests that the lack of active rebalancing is not primarily due to missing state variables or the informal wording of respondent prompts. Instead, it appears to reflect a more prevalent pattern in the LLM's financial advice.

3.4 Robustness of the Patterns in Baseline LLM Advice

Next, we show that the broad patterns of advice documented above are robust to alternative methodological choices, including using a different model, eliciting quantitative recommendations directly, or repeating identical queries multiple times.

Alternative model. To assess whether our findings are specific to `GPT-5.2`, we rerun the advice-generation step using `Gemini 3 Flash` while retaining `GPT-5 Mini` for the deterministic translation task. The `Gemini 3 Flash` advice produces similar patterns. Relative to respondents' observed behavior, it recommends higher stock market participation and equity shares (Figure E9), and larger savings buffers. It also implies similar structural time and risk preference parameters (Table 2) and a similar model fit (Figure E10, Panel C). Like `GPT-5.2`, `Gemini 3 Flash` relies on simple saving and withdrawal heuristics (Figure E15) and exhibits passive portfolio drift and limited consumption smoothing around job loss (Figure E16). These results suggest that our findings are not specific to a single model.

Eliciting quantitative advice directly. Our baseline simulation first elicits qualitative advice using survey-based prompts and then uses a second LLM call to translate that advice into quantitative choices, with the full set of state variables entering only at the translation stage. As a robustness check of this two-step pipeline, we consider an alternative one-step direct-to-JSON pipeline that elicits quantitative recommendations directly by appending the state variables to the survey prompt and instructing the LLM to return recommendations, without a separate translation step. Figure E12 shows that the one-step direct-to-JSON pipeline reproduces the patterns we document in this section for our baseline advice. Consumption and stock-share life cycle profiles are very similar to the baseline two-step pipeline (Panel A), passive portfolio drift is essentially unchanged, and the consumption response to job loss is nearly identical. Reliance on heuristics persists, though it is somewhat attenuated (Panel B): the share of saving rates at multiples of 10% falls from 31.0% to 18.7%, and the share of withdrawals at or below 4% of assets falls from 98.3% to 92.4%. Thus, our main results are not driven by the separate translation step or by withholding state variables during advice generation.

Repeated queries. LLM advice is inherently stochastic: the same prompt may yield different advice across queries, and the same textual advice may translate into different quantitative recommendations.[14] To quantify the importance of randomness in different steps in our methodology, we repeat the advice-generation and translation steps five times. Figure E17 plots the median range in recommended consumption and stock shares across these repeated queries. When we regenerate the entire pipeline, the median range is around 22% for consumption and 16% for stock shares. When we hold the textual advice fixed and regenerate only the translation step, the corresponding ranges fall to 3% and 4%. Therefore, most of the stochasticity in the LLM's advice comes from differences in advice across repeated identical LLM queries, rather than from translating that advice into quantitative choices.

[14] `GPT-5.2` is a reasoning model and does not allow users to set the temperature to zero. Rumpf et al. (2026) document similar stochasticity in the financial advice provided by LLMs.

4 Heterogeneity in LLM Advice: Supply and Demand

Our analysis thus far has focused on characterizing the average properties of LLM advice. In this section, we examine whether this advice varies systematically across groups of individuals, and how much of this variation arises from demand or supply. Demand-side differences reflect variation in the prompts respondents write: what topics they raise, what information they provide, and how they frame their financial decisions. Supply-side differences reflect how the LLM responds to the same or similar prompts when observable characteristics change. We first show that simulated individuals accumulate more wealth when advice is generated from prompts written by respondents with higher financial literacy or prior AI experience, or by men. We then use randomized variation in prompt labels to decompose differences in advice by gender, age, and income into demand and supply components.

4.1 Fact 3: LLM Advice Varies Systematically Across Groups of Individuals

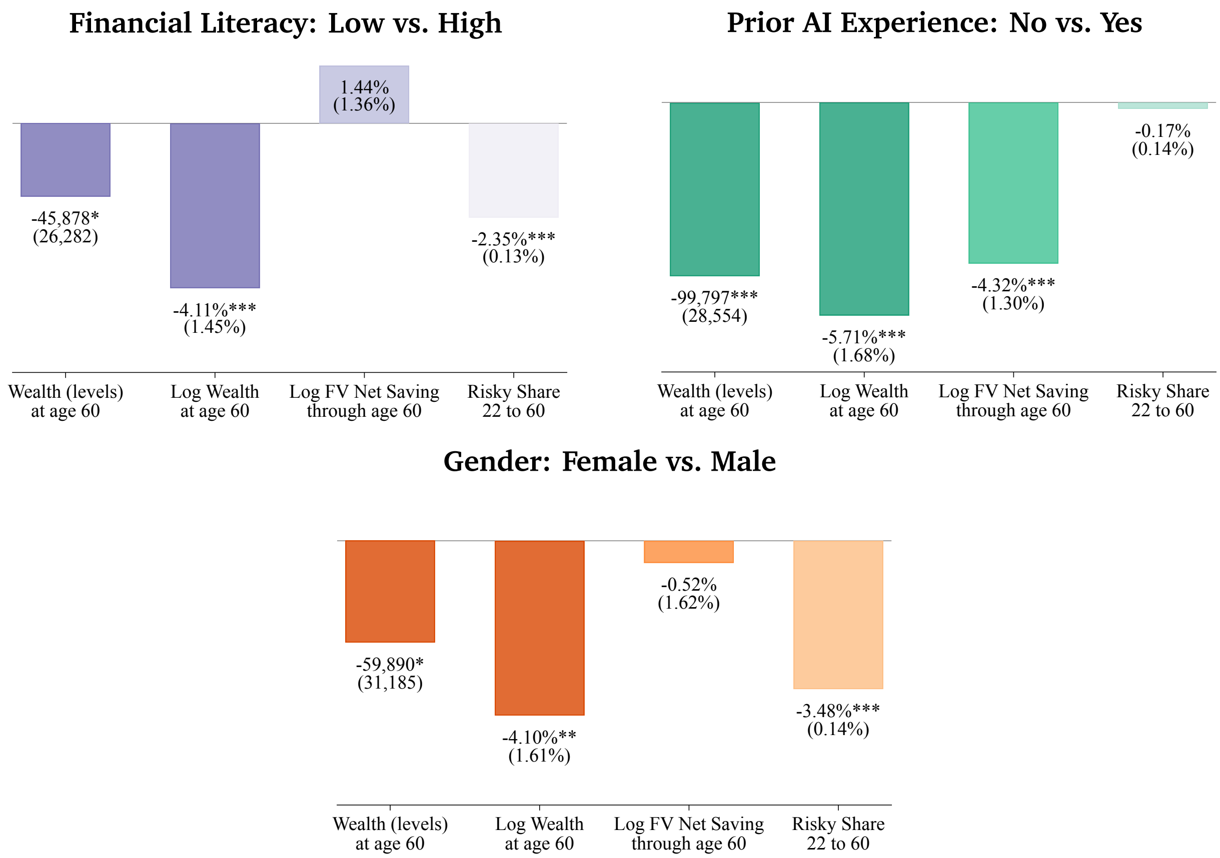
To quantify heterogeneity in LLM advice, we repeat the life cycle simulation separately for groups of survey respondents based on their financial literacy, prior AI experience, and gender. For each characteristic, we split respondents into two groups and simulate life cycle paths using only prompts written by respondents in one group at a time. Importantly, all simulations use the same realizations of exogenous shocks (i.e., income, employment, mortality, and asset returns), so any differences in simulated outcomes are driven entirely by differences in the prompts written by each group and the LLM's responses.

Figure 11 summarizes the resulting differences in life cycle outcomes. In each panel, the first two bars report the implied difference in average wealth at age 60 across the two groups, in levels and in logs. The next two bars decompose these wealth differences into differences in cumulative savings, measured as the future value of net saving compounded at the risk-free rate, and differences in portfolio allocation, measured by the average risky share between ages 22 and 60.[15]

Low-financial-literacy prompts accumulate less wealth. The top-left panel of Figure 11 splits respondents into high- and low-literacy groups based on their performance on the Big Five financial literacy test (Lusardi and Mitchell 2023). Prompts written by respondents

[15]We compound net saving at the risk-free rate rather than at ex-post portfolio returns, so that this measure is not directly affected by portfolio choices. Additionally, we stop at age 60 to ensure that these differences entirely reflect working-life advice.

Figure 11. Heterogeneity in LLM Financial Advice



Notes: This figure compares life cycle outcomes when simulating LLM advice using prompts written by different groups of individuals. All exogenous shocks are held fixed across groups. The top-left panel compares prompts written by individuals with high versus low financial literacy (as measured by a Big Five financial literacy test), the top-right panel compares prompts written by individuals who have versus have not previously used AI for financial advice, and the bottom panel compares prompts written by men versus women. Within each panel, the four bars show (from left to right) the difference in wealth at age 60 (in levels), the difference in log wealth at age 60, the difference in the log future value of net saving through age 60 (computed at the risk-free rate), and the difference in the average risky asset share between ages 22 and 60. All values are in 2025 dollars. Standard errors in parentheses.

who answer at least one of these questions incorrectly generate advice that results in wealth at age 60 that is \$46,000, or 4.1%, lower on average. This difference is driven primarily by portfolio choice: prompts from the less financially literate group lead to a 2.35pp lower average allocation to risky assets between ages 22 and 60. Differences in cumulative savings go in the opposite direction and would, if anything, partially offset the wealth difference, since the less financially literate group is advised to save more through age 60. The word clouds in Figure E18 suggest that this difference in investment advice reflects both what the two groups ask about in their prompts and the textual advice they receive; Figure E19 provides illustrative examples.

Prompts without AI experience accumulate less wealth. In our survey, 48% of respondents report having used an AI tool for financial advice or information in the past three months (Figure E20), a share consistent with other recent industry surveys (Lloyds Banking

Group 2025; J.D. Power 2025). The top right panel of Figure 11 shows that prompts written by respondents who have not previously used AI for financial advice generate recommendations that result in wealth at age 60 that is \$100,000, or 5.7%, lower on average. Unlike the financial literacy result, this difference is driven almost entirely by saving behavior: the future value of cumulative net saving is 4.32% lower in the non-AI-user simulation, while the difference in average risky asset shares is small and not statistically significant.[16] Figure E21 provides illustrative prompts from each group, and word clouds in Figure E22 show the corresponding differences in the LLM's textual advice.

Prompts from women accumulate less wealth. The bottom panel of Figure 11 compares advice given by the LLM to men and women. Prompts written by women generate recommendations that result in wealth at age 60 that is \$60,000, or 4.1%, lower on average than prompts written by men. This difference is mainly driven by portfolio choice: the average risky asset share recommended to women is 3.48pp lower than the share recommended to men, while the future value of cumulative net saving is nearly identical across the two groups. Figure E23 shows that this difference in risky-share recommendations appears across all age and income groups, and that men are also advised to rebalance their portfolios more actively. This gender difference in risky-share advice is consistent with Foltyn and Olsson (2026), who find an average gender gap in equity allocations in response to researcher-designed scenarios of 1.8pp across 33 LLMs. It is also consistent with patterns observed in observational data (Agnew et al. 2003) and in professional financial advice (Bucher-Koenen et al. 2025).

4.2 Gender Differences Reflect Both Demand and Supply

The heterogeneity results above raise a natural question: do differences in LLM advice reflect differences in what respondents ask (demand), or differences in how the LLM responds to otherwise similar prompts (supply)? We answer this question by focusing on gender specifically because it has been a focus of study in the literature on human financial advice (Bucher-Koenen et al. 2025), and because our prompt design allows us to experimentally separate demand- from supply-side drivers of gender differences in a way that is difficult in human-advisor settings (Reuter and Schoar 2024).

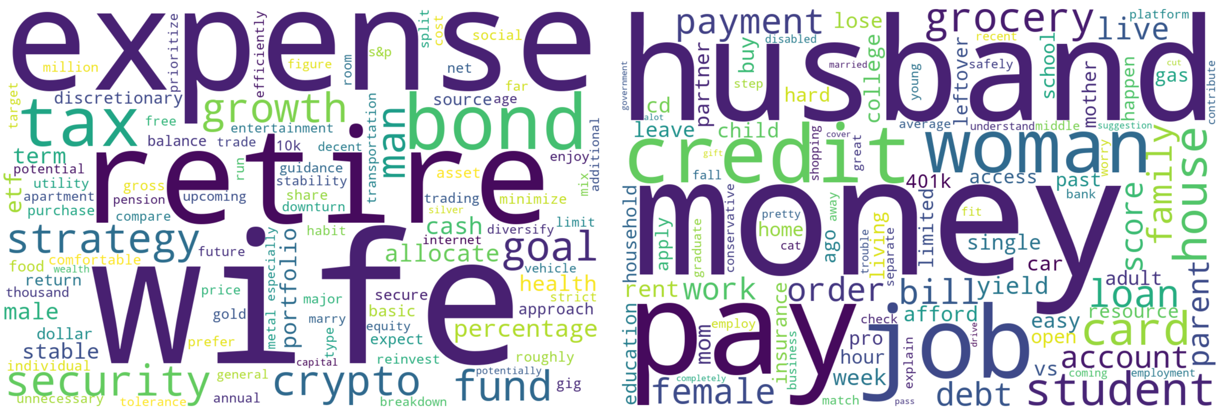
Men and women write prompts and receive advice with different language. Figure 12 shows word clouds of terms that are overrepresented in prompts (Panel A) written by men

[16] Although AI use and financial literacy are correlated, the distinct mechanisms behind the wealth gaps (i.e., risky allocation versus cumulative saving) suggest that these two comparisons capture different sources of heterogeneity in the advice.

Figure 12. Gender Differences in Prompts and Advice

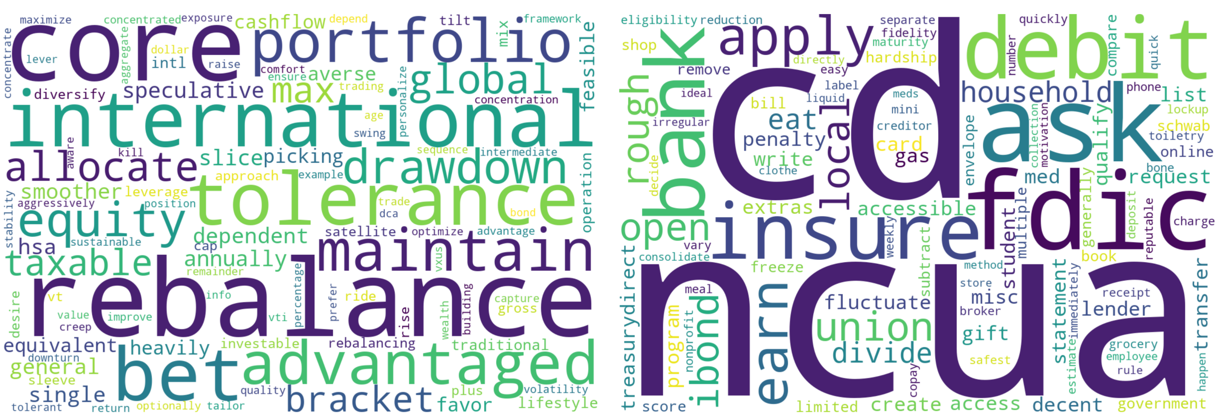
Panel A: Words Overrepresented in Prompts by Gender

Men **Women**



Panel B: Words Overrepresented in Advice by Gender

Men **Women**



Notes: This figure shows word clouds of terms that are differentially used by men versus women. Panel A shows words overrepresented in the prompts written by men (left) versus women (right). Panel B shows words overrepresented in the LLM's advice to men (left) versus women (right). Word size is proportional to the difference in frequency between groups.

versus women and in the corresponding advice (Panel B). Men are much more likely to mention stocks and discuss investment strategies, while women are more likely to explicitly mention their gender, discuss debt, and mention safe, liquid investment options such as certificates of deposit (CDs), deposits, and credit unions. The LLM's responses also differ systematically, with more aggressive investment language in advice to men. One of the most common words overrepresented among men is "rebalance," consistent with the gender differences in recommended portfolio rebalancing shown in Figure E23.

Randomized labels to separate demand from supply. To separate demand from supply, we use the 83% of prompts that do not explicitly reveal the respondentâĂŹs gender, whether directly or through gendered self-descriptions (e.g., prompts written by a woman referencing being a âĂIJmotherâĂİ or a âĂIJwifeâĂİ), and randomly insert either "I am a man" or "I am a woman" at the beginning of each prompt. Because the inserted label is randomized, its effect identifies the supply channel: how the LLM responds to gender holding the content of the prompt fixed. The effect of the prompt author's gender, holding the inserted label fixed, identifies the demand channel: how prompts written by men and women differ in ways that affect advice.

Using these labeled prompts, we simulate life cycles as in the baseline and estimate

$$Y_{i,t} = \beta_D \,\text{FemaleAuthor}_i + \beta_S \,\text{FemaleLabel}_{i,t} + \delta^{\text{bucket}}_{i,t} + \varepsilon_{i,t} \tag{1}$$

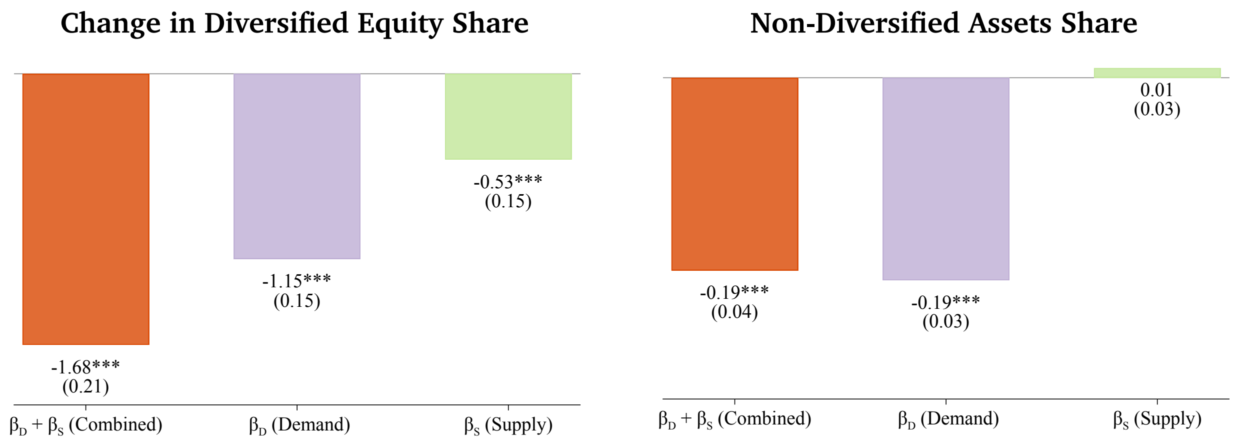
where $Y_{i,t}$ is the outcome of interest for individual i and age t, FemaleAuthor_i indicates that the prompt used to generate advice was written by a woman, and $\text{FemaleLabel}_{i,t}$ indicates that the prompt used to generate advice was randomly assigned a female label. The fixed effects $\delta^{\text{bucket}}_{i,t}$ correspond to the age × income × employment cells within which prompts are randomly drawn, ensuring that comparisons are made among prompts assigned to similar simulated individuals. The coefficient β_D captures demand (the effect of prompts written by women rather than men, holding labeled gender fixed), while β_S captures supply (the effect of a gender label, holding prompt content fixed).

Gender gap in diversified holdings is mostly demand-driven. Figure 13 shows the results from estimating (1) with the change in diversified equity share as the dependent variable. Moving from a prompt written by a man and assigned a male label to a prompt written by a woman and assigned a female label reduces the recommended risky share by 1.68pp. Two-thirds of this difference is demand-driven: drawing a prompt written by a woman reduces the recommended risky share by $\hat{\beta}_D = -1.15$pp relative to drawing a prompt written by a man, holding the randomized gender label fixed.[17] The remaining one-third is supply-driven: holding prompt content fixed, a female label reduces the recommended risky share by $\hat{\beta}_S = -0.53$pp. Therefore, both what respondents write and how the LLM responds to stated gender contribute to the gender gap in portfolio advice, with demand accounting for more of the difference.

Gender gap in non-diversified holdings is entirely demand-driven. We next estimate (1) with non-diversified risky assets as the dependent variable, which include individual stocks, crypto, gold, commodities, and collectibles, and account for approximately 1% of baseline

[17]We use the change in risky share rather than its level because of the substantial inertia documented in Figure 9. Results are quantitatively similar when using the level of risky shares.

Figure 13. Decomposing the Gender Difference: Demand vs. Supply



Notes: This figure decomposes gender differences in LLM-recommended portfolio choices into demand and supply components using the specification in (1). The sample is restricted to the 83% of survey prompts that do not explicitly mention gender. For each prompt, we randomly insert either "I am a man" or "I am a woman" at the beginning. The left panel uses the change in diversified equity share as the dependent variable; the right panel uses the change in the share allocated to non-diversified risky assets, defined as individual stocks, crypto, gold, commodities, and collectibles. In each panel, the left bar reports the combined effect of a prompt written by a woman and assigned a female label rather than a prompt written by a man and assigned a male label ($\hat{\beta}_D + \hat{\beta}_S$). The middle bar reports the demand effect: the effect of a prompt written by a woman rather than a man, holding the randomized label fixed ($\hat{\beta}_D$). The right bar reports the supply effect: the effect of assigning a female rather than male label, holding prompt content fixed ($\hat{\beta}_S$). Standard errors are in parentheses.

allocations. In contrast to the gender gap in diversified holdings, the right panel of Figure 13 shows that the gender difference in non-diversified holdings is entirely demand-driven ($\hat{\beta}_D = -0.19$pp, $\hat{\beta}_S \approx 0$pp). We also estimate the same specification for the net saving rate, where Figure 11 shows essentially no gender gap. Figure E24 confirms that this near-zero gap does not mask offsetting forces: both the demand coefficient ($\hat{\beta}_D = -0.88$pp) and the supply coefficient ($\hat{\beta}_S = -1.76$pp) are negative, though neither is statistically distinguishable from zero.

Interpretation. Distinguishing supply- vs. demand-driven differences in LLM advice matters because the two channels have distinct interpretations. Supply-driven differences, in which the LLM gives different advice to identical prompts labeled as coming from a man versus a woman, may reflect implicit inference: the model may use gender as a signal for characteristics correlated with gender, such as risk preferences or labor-market risk (Croson and Gneezy 2009). But even when such correlations exist, gender is a coarse signal, and this type of inference would ideally be made explicit to users (Filippin and Crosetto 2016). Alternatively, supply-driven differences may reflect algorithmic bias, with the model reproducing gender stereotypes present in its training data. Demand-driven differences have a different interpretation: they reflect real differences in what men and women choose to write about, but model design can still determine how much to amplify or dampen those differences when providing advice.

4.3 Age and Income Differences Are Largely Supply-Driven

Next, we apply a similar supply-demand decomposition to understand variation in LLM advice by age and income, two canonical exogenous state variables in life cycle models. Unlike the gender decomposition, this exercise uses the variable-insertion step in our baseline simulation rather than randomized labels. Among prompts that explicitly mention age (23%) or income (38%), our baseline simulation replaces the respondent's stated values with the simulated individual's current age or income within each employment-income-age bucket. This creates two sources of variation: the author's actual age or income, which captures demand-side differences in the prompts people write, and the inserted age or income, which captures the LLM's supply-side response to the value shown in the prompt.

Formally, for age and income, we estimate the following specification among working-life observations

$$Y_{i,t} = \beta_D \, \text{AuthorVar}_i + \beta_S \, \text{InsertedVar}_{i,t} + \delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t} \tag{2}$$

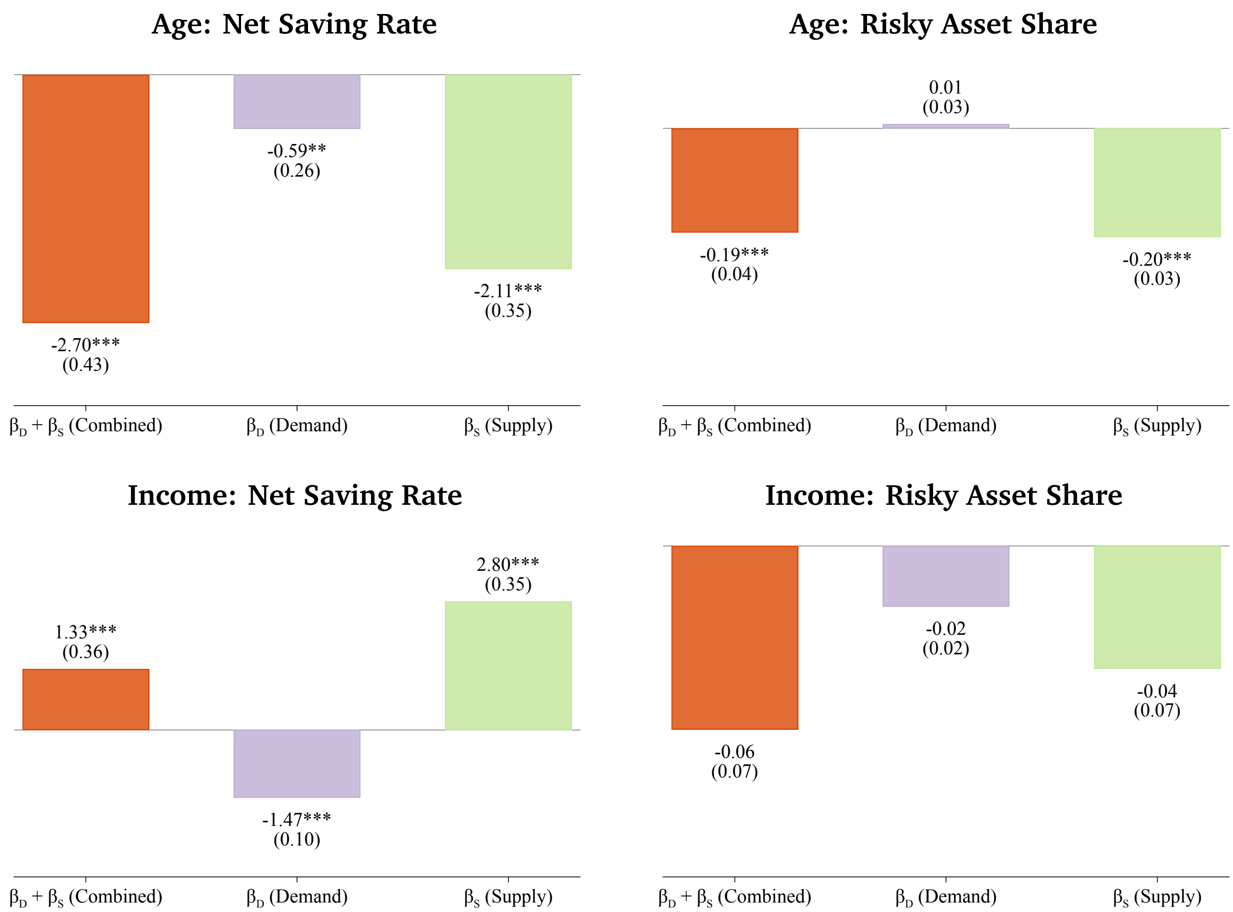
where $Y_{i,t}$ is the outcome of interest for individual i and age t, AuthorVar_i is the actual age or income of the prompt that is used to generate advice, $\text{InsertedVar}_{i,t}$ is the value of age or income inserted into the prompt used to generate advice, and $\delta_{i,t}^{\text{bucket}}$ are prompt-bucket fixed effects like in (1). As before, β_D captures demand, while β_S captures supply.[18]

The top row of Figure 14 shows that age differences are mostly supply-driven. Each additional year of inserted age lowers the recommended net saving rate by 2.11pp and the recommended risky share by 0.20pp. These patterns are consistent with the basic life cycle prediction that saving and equity shares decline with age. By contrast, the demand-side effects of author age are much smaller: −0.59pp for net saving and approximately zero for risky shares. Therefore, most of the age profile of advice reflects the LLM adjusting the advice to the age shown in the prompt, rather than differences in what younger versus older respondents write about.

The bottom row of Figure 14 shows that income affects saving advice, but has little effect on investment advice. For net saving, the demand and supply effects have opposite signs. A $10,000 increase in the prompt author's actual income lowers the recommended saving rate by 1.47pp, while a $10,000 increase in the income inserted into the prompt raises it by 2.80pp. The supply channel dominates, producing a positive combined effect of 1.33pp

[18] Figure E25 repeats the decomposition using the one-step direct-to-JSON pipeline that elicits quantitative recommendations directly, as described in Section 3.4. Because this specification appends state variables, including age and income, to all prompts, it allows us to use the full sample rather than only prompts that explicitly mention age or income. The qualitative patterns are similar: age effects remain predominantly supply-driven, and income continues to affect saving but not risky shares.

Figure 14. Decomposing Age and Income Differences: Demand vs. Supply



Notes: This figure decomposes age and income differences in LLM-recommended saving and portfolio choices into demand and supply components using the specification in (2). The sample is restricted to working-life observations (age < 65) and to prompts that explicitly mention the relevant variable: 23% of prompts for age and 38% for income. The top row uses age, measured in years, as the regressor; the bottom row uses income, measured in $10,000 units. The left column uses the net saving rate as the dependent variable; the right column uses the change in the risky share. In the age panels, the combined effect is the effect of a prompt written by someone one year older and shown to the LLM with an inserted age that is also one year older ($\hat{\beta}_D + \hat{\beta}_S$); the demand effect is the effect of a prompt written by someone one year older, holding the inserted age fixed ($\hat{\beta}_D$); and the supply effect is the effect of showing the LLM an inserted age that is one year older, holding prompt content fixed ($\hat{\beta}_S$). For income, the same definitions apply to a $10,000 increase in prompt-writer income and/or inserted income.

per \$10,000. By contrast, both demand and supply effects on risky shares are close to zero. Thus, the LLM responds to higher stated income by recommending higher saving, but income-related differences in the prompts respondents write partly offset this effect.

5 Conclusion

Recent surveys suggest that the share of people using large language models (LLMs) for personal financial advice now rivals or exceeds the share who consult a human financial advisor (Lloyds Banking Group 2025; J.D. Power 2025; Gallup 2025). In this paper, we develop and implement a method to describe quantitatively the personal financial advice provided by LLMs. Applying this method to `GPT-5.2` and `Gemini 3 Flash`, we document three facts. First, following LLM advice would move most respondents closer to broad principles of life cycle theory: larger savings buffers, broader participation in diversified equity funds, and age-declining equity shares. Second, the advice departs from more demanding prescriptions of life cycle theory: it implies unusually high patience, relies on simple saving and withdrawal heuristics, smooths consumption imperfectly after income shocks, and allows portfolios to drift passively with realized returns. Third, the advice varies systematically across groups, including by gender, financial literacy, and prior AI experience. These differences reflect both demand (what respondents write) and supply (how the model responds to otherwise similar prompts).

These findings suggest that generative AI has the potential to offer an affordable, widely accessible source of financial guidance that could help overcome the significant costs, biases, and conflicts of interest associated with traditional human financial advisors (Reuter and Schoar 2024). The fact that LLM advice generally aligns with the broad prescriptions of life cycle theory is promising, especially because these general-purpose models are not explicitly optimized to improve household financial decisions. At the same time, our results show that the content of AI financial advice depends not only on model capabilities, but also on what users ask. These demand-side constraints are likely to constraint the quality of the advice users receive even as models improve. These demand-side constraints are likely to constraints the quality of the financial advice they can provide, even as model improve over time. The departures from life cycle theory that we document therefore provide concrete diagnostics to guide model development. More broadly, our methodology, which combines survey responses with simulations, provides a systematic way to track how financial advice changes as models evolve, as users learn to write different prompts, and as firms and regulators decide whether AI advice should mirror, amplify, or mitigate heterogeneity in what individuals ask.

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Internet Appendix to "AI Financial Advice: Supply, Demand, and Life Cycle Implications"

For Online Publication Only

Contents

Appendix A. Prompt Construction and Design

This appendix provides additional details on how the respondent-written prompts described in Section 1.1 were processed and integrated into the life cycle simulation framework. The survey questions are shown in Figure E1 and sample demographics are in Table E1.

A.1 Survey Prompt Variable Insertion

This section describes the rules we use to encode life cycle state variables in respondent-written survey prompts. Whenever a prompt mentions a state variable used in the life cycle model, we replace that mention with a placeholder that can be updated to the simulated individual's state. These replacements cover income, age, wealth, tenure, and stock allocations. Table A2 lists representative replacement patterns for each state variable.

Household rules. When a survey prompt reports household income or household wealth, we divide the reported amount by two unless the prompt specifies additional working adults. If income or wealth is split across accounts in which a partner contributes to some categories but not others, we weight the replacement amounts by the simulated agent's implied contribution. For example, suppose a respondent reports X in an individual brokerage account and a combined Y in a household 401(k). We then insert {total_wealth*$X/(X+Y/2)$} for the brokerage account and {total_wealth*$Y/(X+Y/2)$} for the 401(k), respectively. If the prompt explicitly mentions a partner's income, we leave that income unchanged.

Income calculations. If the prompt mentions an hourly income rate and a number of hours worked per week, denoted by N, we insert hourly income as {income/(52*N):,.2f}. If the number of hours is given as a range, we use the midpoint. If the prompt does not specify hours, we set $N = 40$ unless the job is described as part-time, in which case we set $N = 20$. When income comes from multiple sources, we preserve the ratio of income from each source. When the prompt mixes multiple time frames, such as monthly and hourly pay, we annualize each amount, compute the implied ratios, and then rescale to the relevant frequency. If income is reported as a range, we use the midpoint.

If replacing a range with a single value would create an ungrammatical sentence, we replace both bounds of the range with the same value instead.

Wealth stock calculations. Many survey prompts describe financial wealth as dollar amounts held in several accounts. If the prompt mentions a single account using an ambiguous term such as "savings," we interpret that amount as total wealth. If the prompt mentions a single account that clearly corresponds to non-stock assets, diversified stock assets, individual stock holdings, or other risky assets, we interpret the amount as wealth in that asset class. For prompts with multiple accounts, we use the following allocation rule. Let $\{D_n\}_{n\in N_D}$ denote dollar amounts in diversified stock accounts, $\{I_n\}_{n\in N_I}$ dollar amounts in individual stock accounts, $\{S_n\}_{n\in N_S}$ dollar amounts in non-stock accounts, and $\{O_n\}_{n\in N_O}$ dollar amounts in other risky asset accounts. Let $\{M_n\}_{n\in N_M}$

denote mixed accounts, such as retirement or brokerage accounts, whose portfolio shares are ambiguous. We define total wealth as $W = \sum_{n \in N_D} D_n + \sum_{n \in N_I} I_n + \sum_{n \in N_S} S_n + \sum_{n \in N_O} O_n + \sum_{n \in N_M} M_n$, let θ_x be the share of wealth held in asset class $x \in \{\text{divstock, indstock, nonstock, other}\}$, and define $\theta = \theta_D + \theta_I$ as the stock-market share. For each object, the corresponding life cycle value is denoted with a tilde (e.g., $\tilde{W}$). We distinguish two cases.

- **Scenario 1: No mixed accounts.**
  - We assign each asset class according to the account type described in the prompt. This preserves ratios within asset classes, but not ratios across asset classes.
  - $\tilde{D}_n = \frac{D_n}{W}\tilde{W}$
  - $\tilde{I}_n = \frac{I_n}{W}\tilde{W}$
  - $\tilde{S}_n = \frac{S_n}{W}\tilde{W}$
  - $\tilde{O}_n = \frac{O_n}{W}\tilde{W}$
- **Scenario 2: Mixed accounts.**
  - We first allocate non-mixed accounts up to the relevant simulated asset-class shares. We then allocate the remaining wealth to mixed accounts in proportion to their original account balances.
  - $\tilde{D}_n = \min(\frac{D_n}{W}, \tilde{\theta}_D)\tilde{W}$
  - $\tilde{I}_n = \min(\frac{I_n}{W}, \tilde{\theta}_I)\tilde{W}$
  - $\tilde{S}_n = \min(\frac{S_n}{W}, \tilde{\theta}_S)\tilde{W}$
  - $\tilde{O}_n = \min(\frac{O_n}{W}, \tilde{\theta}_O)\tilde{W}$
  - $\tilde{M}_n = (\tilde{W} - \sum_{n \in N_D} \tilde{D}_n - \sum_{n \in N_I} \tilde{I}_n - \sum_{n \in N_S} \tilde{S}_n - \sum_{n \in N_O} \tilde{O}_n)\frac{M_n}{\sum_{n \in N_M} M_n}$

When the prompt refers to stock-market investments without identifying whether they are diversified or individual stock holdings, we combine them into one investment account. We classify holdings as diversified when the prompt mentions exchange-traded funds, mutual funds, index funds, or tickers for tradable assets that fall into these categories. We classify holdings as individual stocks when the prompt refers to specific stocks, a small bundle of stocks, or stocks chosen by the writer. In this case, the same allocation rule applies with I_n interpreted as each investment account and without a separate D_n term.

One-time lump sum payments. When a survey prompt refers to a one-time lump sum payment, such as an inheritance, settlement award, or house sale, we replace the value only if the prompt describes the payment as having already occurred. If the prompt describes the payment as expected but not yet received, we leave the value unchanged. When the value is replaced, we first compare it with the midpoint of the respondent's financial-wealth bucket, using $50,000 by default when the respondent selected "I don't know/prefer not to answer." Let m be this midpoint, W the wealth state variable from the model, and z the value of the original lump sum. We calculate the inserted value, x, as follows:

$$x \equiv f(z, m) = \begin{cases} \{\text{total\_wealth*z/m:,.0f}\}, & m \geq z, \\ \{\text{total\_wealth*z/(z+m):,.0f}\}, & m < z, \end{cases} \tag{3}$$

Excluded variables. We leave several potential state variables unchanged in the survey prompts. This group mainly includes references to rent, groceries, bills, and debt. In the life cycle model, consumption represents cash outflows, so descriptions of payments toward necessities and debt enter the spending description and allow the LLM advice to reflect the prompt author's stated circumstances.

A.2 Prompt Selection

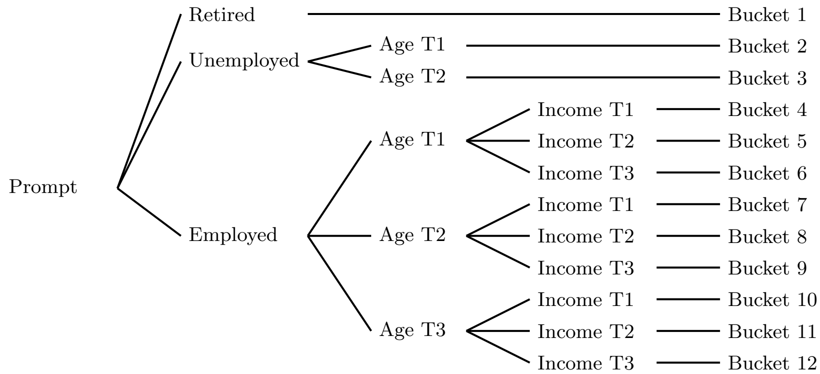
As described in Section 1.1, out of an initial 1,000 complete Qualtrics survey responses, 952 respondents passed our authenticity and specificity checks. Specifically, we removed two types of responses: 33 responses that failed Prolific's platform-level authenticity checks and 15 responses that a manual spot check deemed unusable. To pass the manual spot check, each of the three prompts had to allude to its respective topic in the following ways: the first prompt was required to say something about the respondent's situation, the second to refer to general spending or saving, and the third to investing.

A.3 Prompt Bucketing

In order for the survey prompts to better reflect the state variables in the choice problem, they are divided into 12 different buckets. These buckets are partitioned by a combination of employment status, age, and income. Prompts are first sorted by the employment status of the writer. Within each employment type except for retirement, prompts are further sorted by the age of the writer into halves for the unemployed and terciles for the employed. For just the employed prompts, each age tercile is further divided into income terciles. See Figure A1 for more details on this bucketing system, and for the distribution of prompts across buckets see Table A1.

In the survey that collects the prompts, the income variables are discrete multiple choice responses. The coarseness of these options results in many ties that make tercile divisions unbalanced. In general, ties are defaulted into the lower tier. In the rare scenario where many respondents choose the highest possible option, ties are defaulted into the higher tier.

Figure A1. Prompt Bucketing Tiers



Notes: This figure outlines the tiered system used to determine prompt buckets. Survey prompts are divided based on the employment status of the writer. Within each employment status except for retirement, prompts are bucketed by age tercile. For just those prompts whose writer was employed, the age buckets are further divided by income terciles. In total, there are 12 buckets. To assign a life cycle choice problem to a prompt, the state variables are used to sort the choice problem into one of the 12 buckets. The age, income, and wealth cutoffs are heuristic thresholds based on the survey sample. A prompt is then randomly selected from the appropriate bucket.

Table A1. Prompt Bucket Distribution by Heterogeneity Cut

Prompt Distribution by Bucket and Cut											
Employment Status	Age Bucket	Income Bucket	Bucket Number	Baseline	Financial Literacy		AI Usage		Gender		No Explicit Gender
					Low	High	No	Yes	Female	Male	
Retired			1	132	68	64	97	35	57	75	97
Unemployed	1		2	71	55	16	42	32	30	42	63
	2		3	68	54	14	35	30	27	40	57
Employed	1	1	4	117	90	57	60	61	58	63	105
	1	2	5	60	31	19	29	29	38	25	51
	1	3	6	56	24	14	20	33	32	25	46
	2	1	7	84	59	51	53	69	38	44	69
	2	2	8	94	37	14	25	23	45	45	72
	2	3	9	49	48	25	36	25	25	23	39
	3	1	10	101	65	31	53	40	40	59	88
	3	2	11	77	25	29	16	50	48	16	67
	3	3	12	43	42	20	31	28	28	29	35
			Total	952	598	354	497	455	466	486	789

Notes: This table displays the distribution of prompts by heterogeneity cut. For each column, the entire set of prompts is split into two sets based on financial literacy, AI usage, or gender. Within each set, prompts are then split into the 12 buckets displayed in Figure A1.

Table A2. Mapping between State Variables and Respondent-Written Prompts

Original text	Replacement text
	Age
“X years old”	“{age} years old”
“in my Xs”	“in my {math.floor(age/10)*10}s”
“I retire in X years”	“I retire in {65 - age} years”
	Income
“$X a year before taxes”	“${pretaxincome:,.0f} a year before taxes”
“$X a month before taxes”	“${pretaxincome/12:,.0f} a month before taxes”
“$X biweekly before taxes”	“${pretaxincome/26:,.0f} biweekly before taxes”
“$X a week before taxes”	“${pretaxincome/52:,.0f} a week before taxes”
“$X a day before taxes”	“${pretaxincome/(52*5):,.0f} a day before taxes”
“$X an hour before taxes”	“${pretaxincome/(52*40):,.2f} an hour before taxes”
“$X a year after taxes”	“${income:,.0f} a year after taxes”
“$X a month after taxes”	“${income/12:,.0f} a month after taxes”
“$X biweekly after taxes”	“${income/26:,.0f} biweekly after taxes”
“$X a week after taxes”	“${income/52:,.0f} a week after taxes”
“$X a day after taxes”	“${income/(52*5):,.0f} a day after taxes”
“$X an hour after taxes”	“${income/(52*40):,.2f} an hour after taxes”
	Stock Allocation
“X% of wealth in savings”	“{100 - current_stock_allocation}% of wealth in savings”
“X% of wealth in stock”	“{current_stock_allocation}% of wealth in stock”
	Miscellaneous
“X years into a job”	“{tenure} years into a job”
“past annual income of $X”	“past annual income of ${avgpastincome:,.0f}”

Notes: This table lists representative replacement patterns used to update respondent-written prompts with simulated state variables. The left column gives examples of original prompt text; the right column gives the corresponding placeholder text inserted before generating LLM advice.

A.4 LLM Advice Pipeline

The process of converting LLM responses to consumption and portfolio decisions uses a two-step pipeline. As mentioned in the previous section, each agent-age pair is exogenously assigned a prompt bucket based on a combination of their age, employment status, and income. In the first step, a prompt from the bucket is randomly drawn. The resulting prompt is then sent to the LLM with the following line concatenated at the end: ‘‘`Respond in <=200 words.`’’. The LLM advice output text is saved. In the second step, the LLM advice is passed back into the LLM with an additional deterministic instruction block that describes how to convert the advice to a JSON output that can be fed back into the life cycle model framework.

A.5 LLM Prompts

This subsection presents the full text of the two prompts used in our simulation pipeline. The academic prompt (Section 1.4) is used to elicit financial advice from `GPT-5.2`, while the JSON translation prompt is used to convert the LLM's textual advice into quantitative choices via `GPT-5 Mini`, as described in Step 4 of Section 1.3.

A.6 Academic Prompt

The following is the full text of the academic prompt used to elicit financial advice from `GPT-5.2`, as described in Section 1.4. Individual-specific state variables (age, employment status, income, account balances) are inserted into the User section of the prompt each period.

PROMPT EXAMPLE

SYSTEM

You are a U.S. financial advisor acting in the best interest of your clients. You are academically trained in household finance, modern portfolio theory, and life cycle planning. Your objective is to produce high-quality personalized financial advice tailored to my circumstances under these baseline assumptions:

- Normal life expectancy, living expenditures, retirement age, employment risk, and income risk
- Current U.S. tax law and Social Security rules stay unchanged
- Risk-free savings earn 2.0% real return annually
- Real stock returns match the 60-year U.S. total stock market historical average
- I am single with no dependents and have no bequest motive. I do not care about wealth after death

Given my **age, employment status, annual unemployment benefits, average annual post-tax income (since age 22), taxable account balances, and total net wealth**, give the best financial advice for allocating my resources across annual spending and the following four taxable accounts:

- D=diversified stock (index funds/ETFs/mutual funds; "S&P 500/total market")
- I=individual stocks (single-company stocks or a small bundle (<5))
- N=nonstock (cash/savings/HYSA/money market/CDs/bonds/treasuries/buffer/emergency fund)
- O=other (anything not in D, I, N; crypto/gold/commodities/collectibles)

Based on my information and these assumptions, think step by step about your best advice for addressing the following questions (but do not output your reasoning):

1. How much should I consume this year in dollar amounts taking into account all my living expenses over the year?
2. Should I contribute to, withdraw from, or keep unchanged each of my 4 taxable accounts?
3. Should I transfer money between each of my four accounts?

INTERNAL COMPUTATION STRUCTURE (do not output):

INPUTS: sN,sD,sI,sO,total_wealth,income_post_tax,avgpastincome,tenure.

VARIABLES (annual):
Contribution (cN,cD,cI,cO); withdrawals (wN,wD,wI,wO); transfers tXY for X,Y in (N,D,I,O), X≠Y.
C = cN+cD+cI+cO
W = wN+wD+wI+wO

FUNDING
Contributions (cN,cD,cI,cO) use income_post_tax
Withdrawals (wN, wD, wI, wO) fund spending; reduce holdings
Transfers (tXY) move existing holdings between buckets

SOURCE LIMITS
wN+tND+tNI+tNO<=sN
wD+tDN+tDI+tDO<=sD
wI+tIN+tID+tIO<=sI
wO+tON+tOD+tOI<=sO

BUDGET IDENTITY
consume_amt = income_post_tax + W - C
Constraints: consume_amt>=0; C<=income_post_tax+W

OUTPUT INSTRUCTIONS
Provide this advice ONLY as JSON with double quotes with the format described below without other explanations. All values must be numeric (no dollar signs, no commas, no text). If no action is taken in a category, return 0.

Before producing the JSON, verify internally that:

- The budget identity and all source constraints are satisfied
- All contribution, withdrawal, and transfer values must be >= 0.

If any constraint is violated, correct the values before output.

Prompt Example

System (continued)

OUTPUT JSON KEYS (exactly these, in this order): {
"consume_amt": <dollar amount to spend this year; must satisfy the budget identity and be >=0>,
"nonstock_contrib": <dollar amount contributed to NONSTOCK using income>,
"nonstock_withdraw": <dollar amount withdrawn from NONSTOCK to spend>,
"divstock_contrib": <dollar amount contributed to DIVERSIFIED STOCK using income>,
"divstock_withdraw": <dollar amount withdrawn from DIVERSIFIED STOCK to spend>,
"indstock_contrib": <dollar amount contributed to INDIVIDUAL STOCK using income>,
"indstock_withdraw": <dollar amount withdrawn from INDIVIDUAL STOCK to spend>,
"other_contrib": <dollar amount contributed to OTHER using income>,
"other_withdraw": <dollar amount withdrawn from OTHER to spend>,
"transfer_nonstock_to_divstock": <dollar amount transferred from NONSTOCK to DIVERSIFIED STOCK>,
"transfer_divstock_to_nonstock": <dollar amount transferred from DIVERSIFIED STOCK to NONSTOCK>,
"transfer_nonstock_to_indstock": <dollar amount transferred from NONSTOCK to INDIVIDUAL STOCK>,
"transfer_indstock_to_nonstock": <dollar amount transferred from INDIVIDUAL STOCK to NONSTOCK>,
"transfer_divstock_to_indstock": <dollar amount transferred from DIVERSIFIED STOCK to INDIVIDUAL STOCK>,
"transfer_indstock_to_divstock": <dollar amount transferred from INDIVIDUAL STOCK to DIVERSIFIED STOCK>,
"transfer_nonstock_to_other": <dollar amount transferred from NONSTOCK to OTHER>,
"transfer_other_to_nonstock": <dollar amount transferred from OTHER to NONSTOCK>,
"transfer_divstock_to_other": <dollar amount transferred from DIVERSIFIED STOCK to OTHER>,
"transfer_other_to_divstock": <dollar amount transferred from OTHER to DIVERSIFIED STOCK>,
"transfer_indstock_to_other": <dollar amount transferred from INDIVIDUAL STOCK to OTHER>,
"transfer_other_to_indstock": <dollar amount transferred from OTHER to INDIVIDUAL STOCK>
}

User

HERE IS MY INFORMATION:

- age: 39 years old
- I have been working at the same firm for 2 years
- My annual take-home pay income (after taxes) is $43,188 this year
- My average annual income since age 22 is $16,329. Going forward, my average income will determine my social security benefits in retirement.
- Amount held in taxable diversified stock assets (mutual funds, stock ETFs, stock index funds, diversified brokerage holdings): $13,536
- Amount held in taxable individual stocks (singular stocks or a small bundle (<5)): $0
- Amount held in taxable non-stock assets (cash, checking, savings, HYSA, money market, CDs, bonds, Treasuries, emergency fund): $15,208
- Amount held in other assets (e.g., crypto, gold, commodities, collectibles): $0

Respond with JSON only. No prose.

Table A3. Academic Prompt: Assumptions and Individual-Specific Statements

Panel A: Baseline Assumptions	
Academic Prompt Assumption	**Life Cycle Model Implementation**
"Normal life expectancy, living expenditures, retirement age, employment risk, and income risk"	Mortality risk calibrated to SSA life tables; exogenous retirement at age 65; persistent income risk and employment transitions calibrated to SIPP data
"Current U.S. tax law and Social Security rules stay unchanged"	2025 Federal income tax schedule (single filer) and Social Security benefit formula
"Risk-free savings earn 2.0% real return annually"	2% annual risk-free return
"Real stock returns match the 60-year U.S. total stock market historical average"	Log-normal returns: 6.4% equity premium, 20% standard deviation
"I am single with no dependents and have no bequest motive. I do not care about wealth after death."	One individual per household and zero utility from bequests

Panel B: Individual-Specific Statements	
Category	**LLM Prompt Statement**
Case 1: Employed	
Employment	"I have started a new job this year" / "I have been working at the same firm for {tenure} years"
Current Income	"My annual take-home pay income (after taxes) is `$X` this year"
Past Income	"My average annual income since age 22 is `$Y`. Going forward, my average income will determine my Social Security benefits in retirement"
Case 2: Unemployed	
Employment	"I am currently unemployed"
Current Income	"My annual unemployment benefit income (after taxes) is `$X` this year"
Case 3: Retired (Age ≥ 65)	
Employment	"I am retired"
Current Income	"My annual Social Security benefit income (after taxes) is `$X` this year"

Notes: Panel A shows the correspondence between the baseline assumptions in the prompt and their calibration in the life cycle model. Panel B shows the individual-specific statements that are inserted into the prompt each period based on the simulated individual's state variables (see Section 1.3).

A.7 JSON Translation Prompt

The following is the full text of the JSON translation prompt used to convert textual LLM advice into quantitative choices via `GPT-5 Mini`, as described in Step 4 of Section 1.3.

PROMPT EXAMPLE

USER

MAP (4 assets):
- D=diversified stock (mutual funds/stock ETFs/stock index funds/diversified brokerage holdings)
- I=individual stock (singular stocks or a small bundle (<5))
- N=nonstock (cash/checking/savings/HYSA/money market/CDs/bonds/Treasuries/emergency fund)
- O=other (anything not in the other 3 categories; e.g., crypto, gold, commodities, collectibles)

Use income unless stated. Transfers only if advice explicitly says rebalance/move/sell/buy existing holdings.
Inputs: sN, sD, sI, sO, total_wealth, income_post_tax, ADVICE_TEXT.
stock_share=(sD+sI)/total_wealth.

Stock-split default: if advice says `"stocks"` but doesn't distinguish D vs I and doesn't change split:
D_share=sD/(sD+sI) if sD+sI>0 else 1; I_share=1-D_share.
Any stock_total -> D=round(D_share*stock_total), I=stock_total-D. No D<->I transfers unless explicitly instructed.

Extraction priority:
1) Annual $ amounts. If only `"stocks $"`: set stock_total and split via D_share/I_share unless the split changed.
2) Monthly $ → annual: annual = 12 * m. If "until N hits target then switch": infer targetN, m_pre, m_post (or compute annual using months=min(12,ceil(max(0,targetN-sN)/mNpre))).
3) Ranges → midpoint.
4) % of income_post_tax → total: total round(pct * income_post_tax); apply stated split. If only stocks vs. nonstocks: stock_total =round(s * total), N=total-stock_total, then split stock_total via D_share/I_share.
5) Else: all zeros.

Variables: contributions cN, cD, cI, cO; withdrawals wN, wD, wI, wO; transfers tXY for X,Y in {{N,D,I,O}}, X not equal to Y.
Let C=cN+cD+cI+cO, W=wN+wD+wI+wO. Source limits: wN+(tND+tNI+tNO)<=sN; wD+(tDN+tDI+tDO)<=sD; wI+(tIN+tID+tIO)<=sI; wO+(tON+tOD+tOI)<=sO.
Budget: consume = income_post_tax + W - C >=0 and C <= income_post_tax + W. All amounts >=0.

Deterministic repair (order):
A) If any source limit violated: reduce that asset's outgoing transfers first (in output-key order), then its withdrawal.
B) If C > income_post_tax + W: scale contributions proportionally so C=income_post_tax+W (round N then D then I; O absorbs remainder).
C) Set consume by budget; if consume<0, reduce contributions proportionally (same rounding order) until consume=0.

OUTPUT JSON KEYS (exactly these, in this order): {{
"consume_amt": <dollar amount to spend this year; must satisfy the budget identity and be >=0>,
"nonstock_contrib": <dollar amount contributed to NONSTOCK using income>,
"nonstock_withdraw": <dollar amount withdrawn from NONSTOCK to spend>,
"divstock_contrib": <dollar amount contributed to DIVERSIFIED STOCK using income>,
"divstock_withdraw": <dollar amount withdrawn from DIVERSIFIED STOCK to spend>,
"indstock_contrib": <dollar amount contributed to INDIVIDUAL STOCK using income>,
"indstock_withdraw": <dollar amount withdrawn from INDIVIDUAL STOCK to spend>,
"other_contrib": <dollar amount contributed to OTHER using income>,
"other_withdraw": <dollar amount withdrawn from OTHER to spend>,
"transfer_nonstock_to_divstock": <dollar amount transferred from NONSTOCK to DIVERSIFIED STOCK>,
"transfer_divstock_to_nonstock": <dollar amount transferred from DIVERSIFIED STOCK to NONSTOCK>,
"transfer_nonstock_to_indstock": <dollar amount transferred from NONSTOCK to INDIVIDUAL STOCK>,
"transfer_indstock_to_nonstock": <dollar amount transferred from INDIVIDUAL STOCK to NONSTOCK>,
"transfer_divstock_to_indstock": <dollar amount transferred from DIVERSIFIED STOCK to INDIVIDUAL STOCK>,
"transfer_indstock_to_divstock": <dollar amount transferred from INDIVIDUAL STOCK to DIVERSIFIED STOCK>,
"transfer_nonstock_to_other": <dollar amount transferred from NONSTOCK to OTHER>,
"transfer_other_to_nonstock": <dollar amount transferred from OTHER to NONSTOCK>,
"transfer_divstock_to_other": <dollar amount transferred from DIVERSIFIED STOCK to OTHER>,
"transfer_other_to_divstock": <dollar amount transferred from OTHER to DIVERSIFIED STOCK>,
"transfer_indstock_to_other": <dollar amount transferred from INDIVIDUAL STOCK to OTHER>,
"transfer_other_to_indstock": <dollar amount transferred from OTHER to INDIVIDUAL STOCK>}}

Appendix B. Life Cycle Model Details

This appendix provides the full specification and calibration of the life cycle model described in Section 1.2.

B.1 Demographics and Preferences

Each period corresponds to one year, and working life starts at $t = 0$ and lasts for T_w periods. Retirement starts at $t = T_w$, and agents can live at most T periods. Before their certain death in period $t = T$, investors face age-dependent mortality risk with survival probability in period $t + 1$ conditional on survival in period t denoted by m_t. We denote an investor's age as $a_t = t + a_0$, where a_0 is the age investors enter working life. Investors have time-separable CRRA utility over consumption streams. We denote investors' annualized time discount factor as β and relative risk aversion as γ.

B.2 Labor Market

At any point in time, investors can be in one of four employment states, denoted emp_t: E = employed by the same employer as in the previous period, JJ = employed by a different employer than in the previous period, U = unemployed in the current period, and Ret = retired. The fact that investors face uncertainty about their future employment status, in addition to earnings risk, is an important feature of our model because it introduces deviations in income shocks from normality, which Braxton et al. (2021) highlight as important empirically.

Employment: $emp_t = E$. While working, investors earn an exogenous income w_t. The log income process consists of a deterministic component that is cubic in age, a stochastic component that follows an AR(1) process with normally distributed innovations, and a transitory shock that is normally distributed:

$$\begin{aligned} \ln w_t =& \delta_0 + \delta_1 a_t + \delta_2 a_t^2 + \delta_3 a_t^3 + \eta_t + \varepsilon_t, \quad \eta_t = \rho \eta_{t-1} + \xi_t^E, \\ & \xi_0^E \sim N(0, \sigma_{\xi_0}^2), \quad \xi_t^E \sim N(0, \sigma_\xi^2), \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2) \qquad \forall t > 0. \end{aligned} \tag{4}$$

Investors' tenure status evolves according to $ten_t = ten_{t-1} + 1$ if they remain employed by the same employer. We assume that the initial distribution of ξ_0^E is different from that of ξ_t^E for $t > 0$ to account for heterogeneity in the initial period incomes.

Job transition: $emp_t = JJ$. While in the employed state (E), an investor may make a job-to-job transition with probability $\pi^{JJ}(t, ten_t)$ that depends on both their age and tenure at the current job. After a job-to-job transition, income evolves according to:

$$\begin{aligned} \ln w_t = \delta_0 + \delta_1 a_t + \delta_2 a_t^2 + \delta_3 a_t^3 + \eta_t + \varepsilon_t, \quad \eta_t = \rho \eta_{t-1} + \xi_t^{JJ}, \\ \xi_t^{JJ} \sim N(\mu^{JJ}, \sigma_\xi^2), \qquad \varepsilon_t \sim N(0, \sigma_\varepsilon^2) \end{aligned} \tag{5}$$

This earnings process captures a wage premium associated with switching jobs. Investors' tenure is reset to $ten_t = 0$ following a job-to-job transition.

Unemployment: $emp_t = U$**.** While in the employed state (E), an investor may become unemployed with probability $\pi^{EU}(t, ten_t)$ that depends on both their age and tenure at their current job. When investors are unemployed, they receive unemployment benefits equal to $ui_t = ui(\eta_t)$, where $ui(\eta_t)$ is described below. If investors become employed at $t+1$ after being unemployed in period t, income at $t+1$ evolves according to

$$\begin{gathered}\ln w_{t+1} = \delta_0 + \delta_1 a_{t+1} + \delta_2 a_{t+1}^2 + \delta_3 a_{t+1}^3 + \eta_{t+1} + \varepsilon_{t+1}, \\ \eta_{t+1} = \rho\eta_t + \xi_{t+1}^U, \quad \xi_{t+1}^U \sim N(-\mu^{EU}, \sigma_\xi^2), \quad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2).\end{gathered} \tag{6}$$

This earnings process captures the persistent wage reduction associated with experiencing unemployment.

Retirement: $emp_t = Ret$**.** In period $t = T_w$, all investors retire deterministically. During retirement in periods $t \in [T_w, T-1]$, investors earn public pension benefits denoted by ss_t, which are described below.

B.3 Savings Account

Investors start with zero assets at $t = 0$ and cannot borrow. They can accumulate assets inside a savings account that can be invested in one of two financial assets for the life cycle model and one of four financial assets for the prompts to the LLMs. First, there is a risk-free bond that has a constant gross return of $R_t^B = R_f$ per year. Second, there is a risky asset that corresponds to a diversified stock market index and pays a stochastic i.i.d. gross return of R_t^D per year, where

$$\ln R_t^D = \ln R_f + \mu_D + u_t^D, \quad u_t^D \sim \mathcal{N}(0, \sigma_D^2). \tag{7}$$

Additionally, there are two non-diversified assets, each with its own stochastic gross return, R_t^I and R_t^O, corresponding to the individual stock bundle and other risky assets (e.g., crypto, gold, commodities, and collectibles), respectively. Their joint return process is defined relative to the diversified market return:

$$\ln R_t^x = \alpha_x + b_x\left(\ln R_t^D - \ln R_f\right) + \sigma_{\varepsilon,x} u_{x,t}, \quad u_{x,t} \sim \mathcal{N}(0,1), \quad \forall x \in \{I, O\}, \tag{8}$$

where $u_{I,t}$ and $u_{O,t}$ are drawn independently of each other and of the market shock u_t^D in (7). The balance of the savings account, denoted by L_t, then evolves according to:

$$L_{t+1} = \left(L_t + s_t^l\right)\left[1 + (R_{t+1}^\theta - 1)(1 - \tau_c)\right], \quad L_0 = 0, \tag{9}$$

where s_t^l is the net saving that the investor places in this account, τ_c is the rate of capital taxation, and R_t^θ is the investors' portfolio return that depends on their portfolio share in the risky assets.

$$R_{t+1}^\theta = R_f + \theta_t^D \left(R_{t+1}^D - R_f\right) + \theta_t^I \left(R_{t+1}^I - R_f\right) + \theta_t^O \left(R_{t+1}^O - R_f\right), \quad \theta_t^D + \theta_t^I + \theta_t^O \in [0,1]. \tag{10}$$

When solving the life cycle model, we impose $\theta_t^I = \theta_t^O = 0$ because, by construction, neither non-diversified asset can improve the Sharpe ratio of a portfolio that combines the bond and diversified stock index (see Appendix B.6). While this is an approximation because the CAPM would not arise in equilibrium in our model, we make this assumption for parsimony when solving the model. In the LLM simulation, however, all four assets are available, and the LLM is free to allocate wealth to the non-diversified assets.

B.4 Government

Unemployment benefits. Investors receive an unemployment benefit of $ui(\eta_t)$ when their employment ends. This benefit depends on the labor productivity, η_t, from the last period in which the agent was employed.

Retirement benefits. After retirement, investors receive Social Security benefits, denoted by $ss_t = ss(ae_{T_w})$. ae_{T_w} is the investor's average lifetime earnings at the time of retirement. Let ae_t denote the average of wages up to and including period t. Initialize $ae_1 = w_1$. Then

$$ae_{t+1} = \begin{cases} \frac{w_{t+1} + t \cdot ae_t}{t+1}, & \text{if } t < T_w, \\ ae_{T_w} & \text{else.} \end{cases} \tag{11}$$

Taxation. Investors face a nonlinear income tax schedule $tax_i(\cdot)$, which depends on their taxable income. For employed individuals or those in job transitions, their taxable income is their wages. For unemployed or retired individuals, their taxable income is the benefits they receive from the government.

B.5 Summary of Investors' Problem

Investors face a dynamic optimization problem with seven state variables: a_t = age, η_t = labor productivity, ε_t = transitory income shock, emp_t = employment status, ten_t = tenure, ae_t = average lifetime income, and L_t = savings. Denote the vector of these state variables as $\mathbf{x}_t$. Investors have three controls: c_t = consumption, θ_t^D = portfolio share, and s_t^l = savings. In choosing these controls, we restrict investors from borrowing and engaging in any margin trading (i.e., short-selling or taking leveraged positions):

$$L_t \geq 0, \quad \theta_t^D \in [0,1]. \tag{12}$$

When simulating LLM advice, the only difference is that there are five controls, including the two non-diversified asset shares θ_t^I and θ_t^O.

B.6 Calibration

We calibrate our model parameters following Choukhmane and de Silva (2026) where possible. We provide a brief overview here and refer the reader to that paper for a more detailed discussion.

Demographics. We set the length of one period in the model to one year and set $a_0 = 22$, $T_w = 43$, and $T = 68$, such that workers enter working life at age 22, retire at 65, and live their final year of life at 89. For each age, we calibrate mortality risk to match the 2015 U.S. Social Security Actuarial Life Tables. We use the equivalence scale estimated in Lusardi et al. (2017) to capture changes in household composition over the life cycle.

Labor income process. We use data from the Survey of Income and Program Participation (SIPP) to estimate parameters of the labor income process and transition probabilities at the annual frequency. This income process has several components. First, we estimate an earnings process for workers staying in the same job, corresponding to (4), which contains a deterministic and stochastic component.[1] Second, we use data on employment transitions from SIPP to estimate the median salary increase following a job-to-job transition, μ^{JJ}, and the median salary decrease when workers transition back to employment after an unemployment spell, $-\mu^{EU}$. Third, we use SIPP microdata to estimate the three transition probabilities between the three working-age labor market states. Finally, we set the initial unemployment rate equal to 22%, which is the share of age-22 individuals in SIPP who are unemployed.

Tax and benefit system. Investors' tax liability, $tax_i(\cdot)$, is calculated according to the 2025 U.S. federal income tax schedule for single filers who claim only the standard deduction. We calculate Social Security benefits according to the 2025 formula with a Supplemental Security Income program floor. Unemployment benefits are computed with a replacement rate of 40%. We set the capital return tax rate, τ_c, to 21%.

Bond and diversified stock returns. We set the net risk-free rate to be constant at 2% to match the average one-year Treasury rate between 2015 and 2025. We set the equity premium to be 6.4%, which is equal to the average inflation-adjusted return on the CRSP Value-Weighted Index between 1925 and 2006 minus our 2% risk-free rate.[2] We set the volatility of log stock returns to 20%, which matches that of the CRSP Value-Weighted Index. We assume that asset returns are uncorrelated with shocks to labor income and employment transition probabilities.

[1] Relative to Choukhmane and de Silva (2026), we also estimate the variance of the transitory shock as part of this estimation.

[2] We adjust for inflation using the CPI.

Non-diversified asset returns. Both non-diversified assets are calibrated to have the same underlying return process, but independent idiosyncratic shock realizations, so they represent distinct risky positions. We calibrate this process such that neither non-diversified asset can improve the Sharpe ratio of a portfolio that combines a bond and a diversified stock market index, implying that a CAPM investor would optimally hold zero portfolio shares in both assets. Formally, we start by taking the average arithmetic mean return and standard deviation, μ_x and σ_x, for individual stocks from Bessembinder (2018), which gives $\mu_x = 0.1474$ and $\sigma_x = 0.819$. Assuming the CAPM holds and denoting μ_m as the arithmetic mean diversified market return, we can compute the non-diversified asset beta using:

$$\beta_x^* = \frac{(1+\mu_x) - R_f}{(1+\mu_m) - R_f}. \tag{13}$$

Next, note that log return parameters for the non-diversified asset are given by:

$$\sigma_{x,\log} = \sqrt{\log\left(1 + \left(\frac{\sigma_x}{1+\mu_x}\right)^2\right)}, \tag{14}$$

$$\mu_{x,\log} = \log(1+\mu_x) - \tfrac{1}{2}\,\sigma_{x,\log}^2. \tag{15}$$

We assume $\beta_x = b_x$, and then compute the idiosyncratic log volatility using

$$\sigma_{\varepsilon,x} = \sqrt{\sigma_{x,\log}^2 - b_x^2 \sigma_D^2} \tag{16}$$

and the log intercept as

$$\alpha_x = \mu_{x,\log} - b_x \mu_D, \tag{17}$$

where μ_D is the mean log excess return of the diversified asset in (7).[3] This then implies that, given a simulated market gross return R_t^D and an independent draw $u_{x,t} \sim \mathcal{N}(0,1)$, we can construct each non-diversified asset's return as

$$R_t^x = \exp\Big(\alpha_x + b_x\big(\log R_t^D - \log R_f\big) + \sigma_{\varepsilon,x}\, u_{x,t}\Big). \tag{18}$$

[3]We assume $b_x = \beta_x$ for simplicity. In reality, the arithmetic beta implied by a log-loading b_x is $\frac{\text{cov}(R_t^x, R_t^D)}{\text{var}(R_t^D)} = \frac{1+\mu_x}{1+\mu_m}\frac{\exp(b_x\sigma_D^2)-1}{\exp(\sigma_D^2)-1}$, which implies $b_x = \frac{1}{\sigma_D^2}\log\left(1 + \beta_x^* \frac{1+\mu_m}{1+\mu_x}\left(\exp(\sigma_D^2) - 1\right)\right)$. However, given our calibration, these two are quite similar.

Appendix C. Textual Analysis Details

This appendix provides methodological details for the dictionary-based textual analysis described in Section 2.1. Figure C1 provides a schematic overview of the text processing pipeline.

Text preprocessing. Each free-text response is processed through a three-step pipeline. First, finance-specific terms are normalized using pattern-based rewriting (e.g., "401(k)" becomes "401k," "t-bills" becomes "tbill"). Second, the text is tokenized and lemmatized using spaCy's `en_core_web_sm` model, which splits text into individual words and reduces them to root forms (e.g., "worried" becomes "worry," "savings" becomes "saving"). We use the small spaCy model because testing across all three model sizes (`sm`, `md`, `lg`) showed effectively identical results: fewer than 2% of tokens differed, and dictionary prevalence shifted by at most 0.3 percentage points. Third, a domain-specific root-merging step collapses financial variants that spaCy misses (e.g., "retirement" to "retire," "savings" to "save").

Finance-relevant alphanumeric strings—including account types (401k, IRA, Roth), fund tickers (VOO, VTI, SCHD, QQQ), and acronyms (ETF, HYSA, REIT)—are detected via a curated list of special tokens and preserved as-is without lemmatization. Plural forms are normalized (e.g., "401ks" to "401k," "ETFs" to "ETF").

The output is one sequence of root-form words per respondent per prompt. Stopwords are *not* removed at this stage, preserving consecutive multi-word phrases containing common words (e.g., "credit card," "long term," "financial goal") for dictionary matching.

Dictionary construction. We define 27 topic dictionaries covering the following categories: Debt, Savings, Retirement, Housing, Investment Strategy, Income, Employment Security, Budgeting, Education, Family, Gender, Financial Hardship, Financial Anxiety, Risk Preference, Medical, Long-Term Planning, Tax, Lifestyle Goals, Macroeconomic, Investment Assets, Providers, Products, Liquidity, Discipline, Diversification, Insurance, and Inheritance. Each category contains a curated list of keywords, including both single words and multi-word phrases. The complete dictionary is listed in Table C1.

Keywords are processed through the same root-form pipeline as respondent text, ensuring consistent matching. Multi-word keywords (e.g., "credit card," "high yield savings account") are matched by checking for consecutive words in the respondent's processed text. Categories are not mutually exclusive: keywords can appear in multiple dictionaries (e.g., "mortgage" appears in both Debt and Housing), reflecting the multidimensional nature of financial topics.

Prevalence measurement. For each respondent, prompt, and dictionary category, we record a binary indicator equal to one if any keyword from the category appears in the respondent's processed text. Category prevalence is the share of respondents with a positive indicator. Confidence intervals use the normal approximation for binomial proportions ($\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$, clamped to $[0, 1]$).

Figure E26 reports the combined respondent-level prevalence (any mention across all three prompts) for each category.

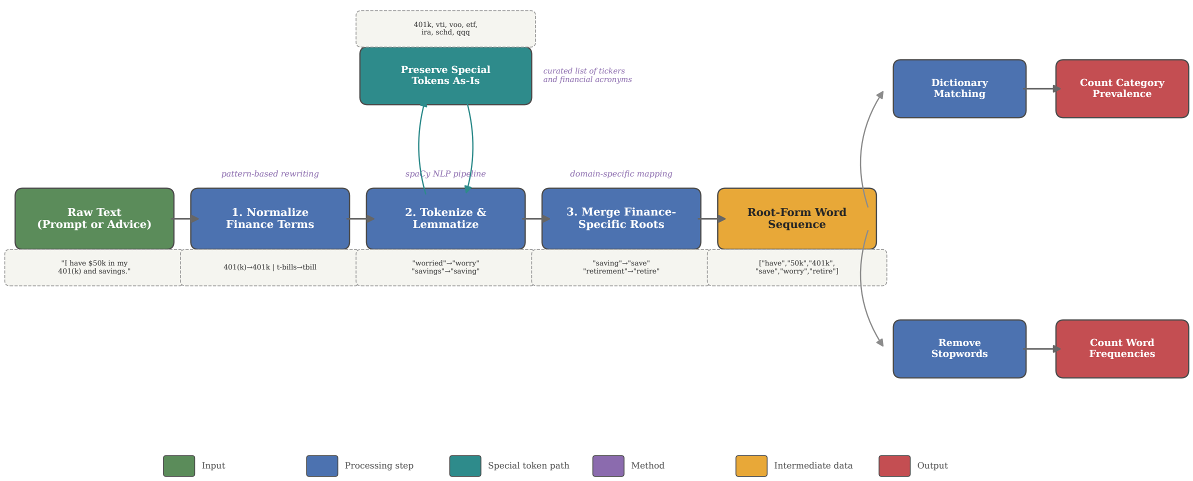
Prompts-versus-advice comparison. To compare respondent prompts with AI-generated advice, we match respondent IDs across the two corpora, yielding $n = 944$ matched pairs (8 of 952 respondents lack advice responses). The estimands differ by source: prompt prevalence is a respondent-level binary indicator (any mention across the three prompts), while advice prevalence is the respondent-weighted mean mention rate across sampled advice responses. Advice confidence intervals use respondent-level standard errors ($\mathrm{sd}(\mathrm{rate}_i)/\sqrt{N}$). We report Spearman rank correlations between prompt and advice prevalence across categories to summarize the degree of topic-ranking agreement.

Asset, provider, and product breakdowns. Within the Investment Assets, Providers, and Products dictionaries, we define grouped keyword display maps that aggregate related keywords under interpretable labels (e.g., all Vanguard fund tickers under "Vanguard," all S&P 500 tracking funds under "Stocks (Aggregate)"). These groupings allow finer-grained comparison of specific asset classes, financial service providers, and individual products between prompts and advice. Respondents are deduplicated across synonyms within each display group. Figure E27 reports the full set of investment asset groups, including categories below the 5% prevalence threshold used in the main text.

Word clouds. Word clouds provide a visual summary of word frequency, with word size proportional to raw count. Stopwords—a union of scikit-learn's default English stopwords, custom generic terms, and survey-instrument words (e.g., "financial," "advice," "situation")—are removed only for word-cloud construction, not for dictionary matching. Prompt-specific tautological terms (e.g., "spend" and "invest" in the investment advice prompt) are additionally removed from the prompt word clouds: the combined prompt word cloud (Figure 2) excludes the union of all prompt-specific stopwords, while the individual prompt word clouds (Figure E5) exclude only the stopwords specific to each prompt.

Summary statistics. Table E2, Table E3, and Table E4 report descriptive statistics for each prompt by demographic subgroup: sample size, mean word count, mean character count, and the share of responses that mention specific numbers or dollar amounts. Number- and dollar-amount detection runs against the raw text (before preprocessing): *has_number* equals one if the text contains any digit, and *has_dollar* equals one if the text contains "$" or the word "dollar." The financial situation prompt is the longest (mean 43 words), with 65% of respondents mentioning specific numbers and 34% mentioning dollar amounts. The spending and investment advice prompts are shorter (mean 27 words each), with lower rates of quantitative detail.

Figure C1. Text Processing Pipeline



Notes: Schematic of the text preprocessing pipeline. Raw text (prompt or advice) passes through three processing steps: (1) pattern-based normalization of finance terms, (2) spaCy tokenization and lemmatization, and (3) domain-specific root merging. Finance-relevant alphanumeric tokens (e.g., 401k, VOO, ETF) are preserved as-is via a special token path. The resulting root-form word sequence feeds into two outputs: dictionary matching (which counts category prevalence without stopword removal) and word frequency counting (which removes stopwords for word-cloud display).

Table C1. Dictionary Categories and Keywords

Category	Keywords
Debt	bankruptcy, car loan, consolidate, credit, credit card, credit card payment, credit score, debt, debt payment, loan, loan payment, loan principal, minimum payment, mortgage, mortgage payment, owe, pay debt, payoff, personal loan, refinance, student debt, student loan
Savings	checking account, emergency fund, emergency savings, high yield savings, high yield savings account, nest egg, rainy day, save money, savings account, set aside
Retirement	401k, annuity, ira, pension, required minimum distribution, retire, retirement, retirement account, retirement savings, rmd, roth, roth ira, social security, ssdi, ssi
Housing	apartment, buy house, condo, down payment for house, down payment on house, home down payment, home equity, home loan, house, house down payment, house fund, housing, mortgage, mortgage down payment, own home, property, rent, rent payment
Investment Strategy	allocate, capital appreciation, compound interest, compounding, day trade, day trading, dividend, growth, portfolio, profit, return
Income	annual income, bonus, disability, disposable income, earning, fixed income, household income, income, limited income, low income, monthly income, passive income, paycheck, salary, stipend, take home, wage
Employment Security	change job, employment, freelance, gig, job, job stability, jobless, layoff, main provider, provider, self-employed, sole provider, stable employment, stable income, stable job, unemployed, unemployment
Budgeting	afford, bill, budget, cost, cut back, discretionary, entertainment, essential, expense, food, groceries, living expenses, monthly budget, necessities, spend less, spending plan, subscription, utilities
Education	college, degree, education, school, student, tuition

Category	Keywords
Family	boyfriend, brother, childcare, children, dad, daughter, dependent, divorced, family, father, fiance, fiancee, girlfriend, grandchildren, grandfather, grandmother, household, husband, kids, married, mom, mother, parent, partner, sister, son, spouse, widow, widower, wife
Gender	boyfriend, dad, father, female, gentleman, girlfriend, grandfather, grandmother, husband, lady, male, man, mom, mother, pregnant, single dad, single father, single mom, single mother, stay at home dad, stay at home mom, widow, widower, wife, woman
Financial Hardship	am broke, bankruptcy, barely, be broke, concerned about bills, concerned about money, difficult, ends meet, financial strain, go broke, hardship, paycheck to paycheck, poverty, shortfall, struggle, tight, worried about bills, worried about money
Financial Anxiety	afraid, anxiety, anxious, concern, fear, hesitant, nervous, overwhelm, panic, panicking, stress, stressed, stressful, uncertain, unsure, worry
Risk Preference	aggressive, conservative, gamble, guaranteed, high risk, low risk, moderate risk, risk adverse, risk averse, risk free, risk profile, risk tolerance, risk-averse, risky, safer, secure, speculative, volatile
Medical	copay, dental, doctor, health, health insurance, healthcare, medical, medication, medicine, premium, prescription, therapy
Long-Term Planning	financial goal, financial security, future, goal, horizon, long term, retire early, retirement goal, short term, timeline
Tax	capital gains, property tax, tax, tax bracket, tax-advantaged, tax-free, taxable
Lifestyle Goals	car, car insurance, car lease, car payment, commute, new car, transportation, travel, trip, vacation, vehicle
Macroeconomic	cost of living, crash, downturn, economy, fed funds rate, federal reserve, high prices, inflation, interest rate, market crash, price increase, purchasing power, recession, rising prices

Category	Keywords
Investment Assets	ada, altcoin, annuity, avax, balanced fund, bitcoin, blue chip, blue chip stocks, bnb, bond, bond etf, bond fund, bond index, bond mutual fund, brokerage, brokerage account, btc, cardano, cash equivalent, cd, cds, chainlink, commodities, commodity, corporate bond, crypto, dividend etf, dividend fund, dividend stock, dogecoin, etf, eth, ethereum, gold, government bond, growth fund, growth stock, high yield account, high yield savings, high yield savings account, high-yield savings, high-yield savings account, hysa, ibond, income fund, index etf, index fund, individual stock, international stock, junk bond, large cap, litecoin, ltc, market fund, matic, mid cap, money market, money market account, money market fund, muni bond, municipal bond, mutual fund, penny stock, precious metal, preferred stock, real estate, reit, reit etf, rental property, ripple, s&p, savings bond, sector fund, silver, single stock, small cap, sol, solana, stablecoin, stock, stock etf, stock fund, stock market, stock mutual fund, t-bill, target date, target date fund, target-date, target-date fund, tbill, three fund portfolio, three-fund portfolio, total bond market, total international, total market, total stock market, treasury, treasury bill, treasury bills, treasury bond, treasury note, usdc, usdt, value fund, value stock, xrp
Providers	acorns, ally, ally bank, ameriprise, bank of america, betterment, bilt, bofa, capital one, cash app, cashapp, charles schwab, chase account, chase bank, chase card, chase checking, chase credit, chase freedom, chase sapphire, chase savings, coinbase, discover bank, discover financial, discover online savings, e*trade, edward jones, etrade, fidelity, goldman sachs, jpmorgan chase, m1 finance, marcus, merrill, merrill lynch, morgan stanley, navy federal, paypal, robin hood, robinhood, schwab, sofi, synchrony, td ameritrade, td bank, usaa, vanguard, wealthfront, webull, wells fargo, zelle
Products	agg, avax, avuv, bil, binance coin, bitcoin, bnb, bnd, btc, cardano, chainlink, dgro, dogecoin, eth, ethereum, fskax, ftihx, fxaix, fxnax, fzilx, fzrox, gld, iau, ief, itot, ivv, ixus, jepi, jepq, litecoin, matic, polygon, qqq, qqqm, ripple, schb, schd, schh, schp, schz, sgov, slv, sol, solana, soxx, spaxx, splg, spy, swisx, swppx, swtsx, swvxx, tlt, usdc, usdt, vbtlx, vfiax, vgit, vgsh, vgt, vig, vmfxx, vnq, voo, vt, vti, vtiax, vtip, vtsax, vtwax, vug, vwo, vxus, vym, xlk, xrp

Category	Keywords
Liquidity	accessible, available funds, buffer, cash buffer, cash on hand, cash reserve, cushion, easy access to funds, emergency cash, emergency fund, emergency savings, hold cash, liquid, liquid cash, liquidity, rainy day, withdraw, withdrawal
Discipline	cut back, discipline, frivolous, guilty, habit, impulsive, indulge, overspend, restrictive, self discipline, shopping, splurge, unnecessary, wasteful
Diversification	asset allocation, asset mix, balance between different investment options, balance between investment options, balance between investments, balanced portfolio, between stocks and bonds, diversification, diversified, diversify, mix, portfolio allocation, rebalance, split between, stocks and bonds
Insurance	auto insurance, car insurance, coverage, deductible, homeowner insurance, life insurance, renter insurance
Inheritance	beneficiary, estate planning, inherit, inheritance, trust fund

Appendix D. Additional Details on Simulated Method of Moments Estimation

To estimate the preference parameters β and γ for each version of the LLM output, we use simulated method of moments on a discrete grid. Our estimation uses a set of 324 moments. We use the 25th, 50th, and 75th percentiles of the wealth-to-income ratio from ages 23 to 64, and the 25th, 50th, and 75th percentiles of equity share from ages 23 to 88. Let $\mathcal{W}_{i,t}$ and $\theta_{i,t}$ be the wealth-to-income ratio and equity share, respectively, for agent i at time t. The former is defined as $\mathcal{W}_{i,t} \equiv L_{i,t}/\text{tax}_i(w_{i,t})$.

For the wealth-to-income ratio moments, we omit the first year as no wealth has been accumulated, and we omit the retirement period as our results show that the LLM does not decumulate wealth sufficiently fast. For the equity share moments, we omit the first and last periods of life.

We estimate the preferences by solving the life cycle model on a 31-by-31 grid of β and γ values ranging from 0.9 to 1.2 and 1.1 to 17, respectively. In the β dimension, our grid is equispaced with steps of 0.01. In the γ dimension, we space the grid with intervals of 0.3 from 1.1 to 5, then increase the intervals to 0.5 until 10, and then use steps of 1. After solving for the life cycle moments on this grid, we then calculate an error term between the targeted moments and every point on the grid. The error calculation is summarized by the following equation:

$$Z(\beta,\gamma) = X_{\text{LC}}(\beta,\gamma) - X_{\text{LLM}}, \qquad \text{SMM}_{\text{error}}(\beta,\gamma) = Z(\beta,\gamma)' \cdot W \cdot Z(\beta,\gamma) \tag{19}$$

In the above equation, X_{LLM} is the vector of LLM moments and $X_{\text{LC}}(\beta,\gamma)$ is the corresponding vector of life cycle moments when solving the life cycle model using β and γ as the inputted preference parameters. W is the weighting matrix. For our estimation, we use the identity matrix as the weighting matrix as it has better small sample properties than the optimal weighting matrix.

Appendix E. Additional Tables and Figures

Figure E1. Survey Questions for Eliciting LLM Prompts

Prompt 1: Describe your situation

Write a description of your situation, including whatever details about yourself you think would help the AI tool give you useful financial advice. **Assume the AI tool does not have any information about you unless you provide it here.**

Prompt 2: Ask for spending advice

In your own words, write a prompt **asking an AI tool for advice on how much to spend** over the coming year.

Assume the AI tool already has the information you provided in your last response.

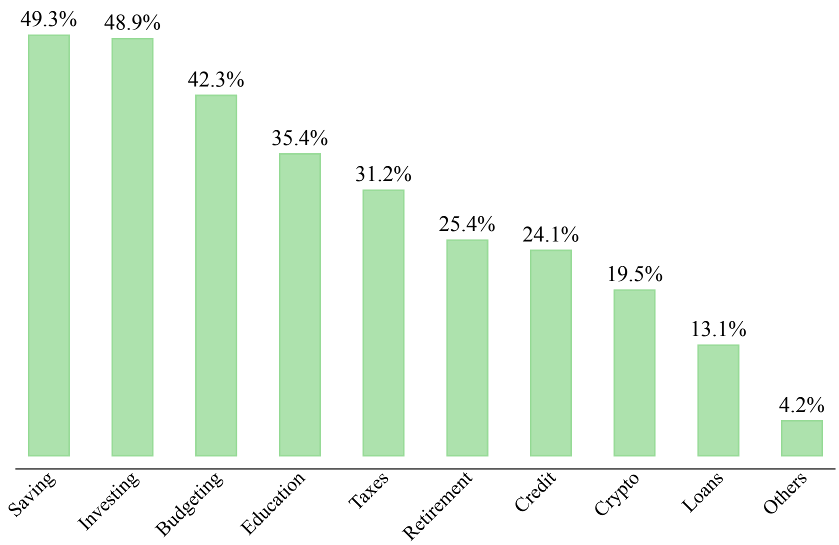
Prompt 3: Ask for investment advice

In your own words, write a prompt **asking an AI tool for advice on how to invest** your savings between stocks and safer options.

Assume the AI tool already has the information you provided in your last response.

Notes: This figure displays the three survey questions used to collect respondent-written LLM prompts. The final prompt incorporated into the life cycle model framework is a concatenation of the three responses with state variables inserted.

Figure E2. Personal Finance Topics Discussed with AI Tools



Notes: This figure shows the personal finance topics that survey respondents report having used AI tools to help with, among the $N = 455$ respondents who have previously used AI for financial advice or information. Respondents could select multiple topics.

Table E1. Demographic Comparison: CPS vs. Prolific Survey

	CPS Data	Prolific Survey
Male	.49	.50
20–29 years old	.17	.17
30–39 years old	.18	.20
40–49 years old	.17	.16
50–59 years old	.16	.21
60–69 years old	.16	.18
70–79 years old	.11	.08
80+ years old	.05	.00
$0–25,000	.35	.26
$25,001–50,000	.25	.27
$50,001–75,000	.16	.20
$75,001–100,000	.10	.12
$100,001–150,000	.08	.10
$150,000+	.06	.05
White	.76	.64
Black/African-American	.13	.11
Asian/Asian-American	.07	.06
Mixed	.02	.11
Other	.02	.08
Employed	.62	.73
Unemployed	.03	.12
Retired	.36	.14

Notes: This table compares the demographics of the Prolific survey to the Current Population Survey (CPS) conducted by the U.S. Census Bureau. The CPS data used in this analysis come from the March 2025 CPS. The race categories represent all of the race options collected by Prolific.

Table E2. Prompt Summary Statistics: Financial Situation

Subgroup	Level	N	Mean Word Count	Mean Char Count	Share Numbers	Share Dollars
All	All	952	43.0	218.3	0.6523	0.3435
Income	Low	490	43.1	218.3	0.6122	0.3367
Income	Mid	296	42.6	217.3	0.7095	0.3446
Income	High	130	40.6	205.9	0.6846	0.3923
Financial Literacy	Low	253	40.3	201.3	0.5968	0.3320
Financial Literacy	Mid	345	45.3	230.9	0.6725	0.3478
Financial Literacy	High	354	42.6	218.1	0.6723	0.3475
AI Use	No	497	40.9	206.7	0.6117	0.3159
AI Use	Yes	455	45.2	231.0	0.6967	0.3736
Gender	Female	486	44.8	225.9	0.6255	0.3333
Gender	Male	466	41.0	210.3	0.6803	0.3541
Employment	Unemployed	139	48.5	251.2	0.5324	0.2950
Employment	Retired	132	39.8	203.0	0.6591	0.3333
Employment	Employed	681	42.5	214.5	0.6755	0.3554

Notes: This table provides descriptive statistics on the first prompt that individuals write describing their financial situation. "Share Numbers" is the share of responses containing any digit. "Share Dollars" is the share containing "$" or the word "dollar." Different rows correspond to different subgroups of individuals, defined as follows. Income groups: Low (<$50k), Mid ($50k–$100k), High (>$100k). Income is missing for 36 respondents who declined to report it. Financial Literacy groups: Low (0–3 correct), Mid (4 correct), High (5 correct out of 5 quiz questions). AI Use: whether the respondent has previously used AI for financial advice.

Table E3. Prompt Summary Statistics: Spending Advice

Subgroup	Level	N	Mean Word Count	Mean Char Count	Share Numbers	Share Dollars
All	All	952	26.7	137.0	0.3109	0.1702
Income	Low	490	27.2	137.8	0.3122	0.1673
Income	Mid	296	26.1	134.9	0.3007	0.1588
Income	High	130	25.5	133.7	0.3308	0.2000
Financial Literacy	Low	253	24.6	123.5	0.2767	0.1779
Financial Literacy	Mid	345	27.5	141.0	0.3159	0.1594
Financial Literacy	High	354	27.4	142.7	0.3305	0.1751
AI Use	No	497	24.7	125.5	0.3038	0.1730
AI Use	Yes	455	28.9	149.5	0.3187	0.1670
Gender	Female	486	26.7	135.8	0.3189	0.1975
Gender	Male	466	26.6	138.2	0.3026	0.1416
Employment	Unemployed	139	29.5	149.7	0.3597	0.1871
Employment	Retired	132	24.1	123.5	0.3106	0.1818
Employment	Employed	681	26.6	137.0	0.3010	0.1645

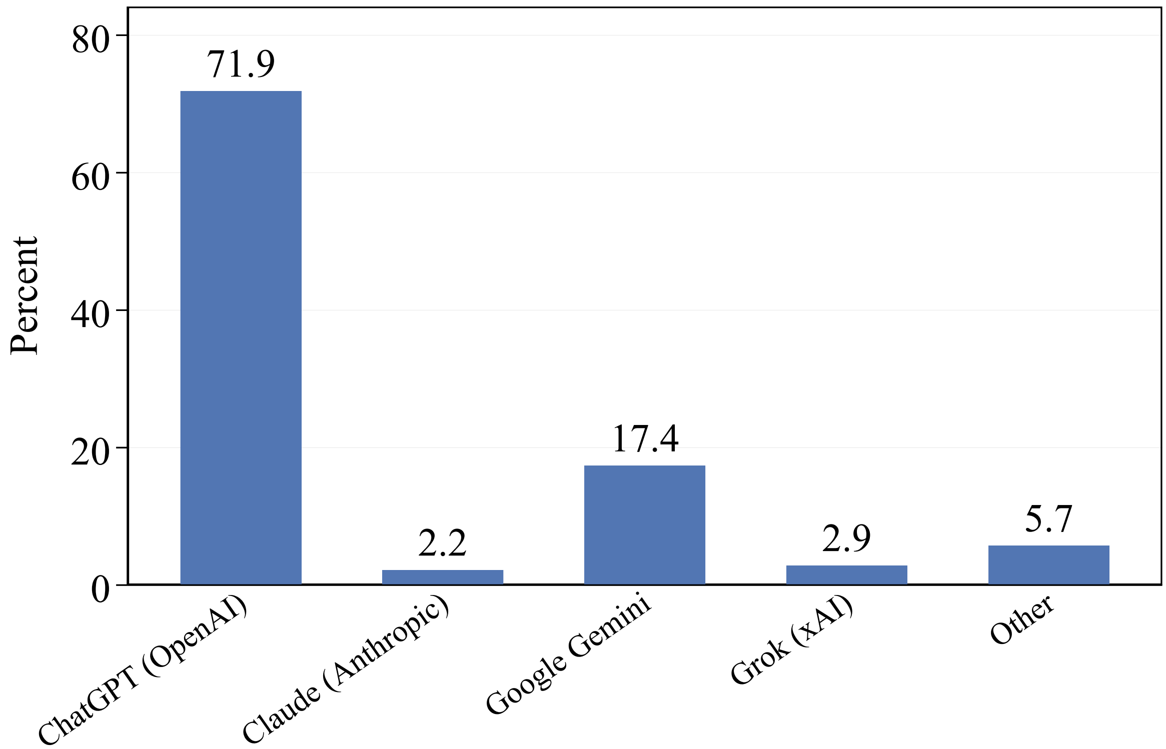
Notes: This table reproduces Table E2 for the second prompt that individuals write, in which they ask for spending advice.

Table E4. Prompt Summary Statistics: Investment Advice

Subgroup	Level	N	Mean Word Count	Mean Char Count	Share Numbers	Share Dollars
All	All	952	27.0	142.0	0.2773	0.1187
Income	Low	490	26.7	139.1	0.2388	0.1082
Income	Mid	296	27.3	144.2	0.2872	0.1182
Income	High	130	27.2	144.9	0.4000	0.1769
Financial Literacy	Low	253	23.8	123.2	0.1897	0.0949
Financial Literacy	Mid	345	28.0	145.7	0.3188	0.1304
Financial Literacy	High	354	28.5	151.8	0.2994	0.1243
AI Use	No	497	24.8	129.2	0.2254	0.0966
AI Use	Yes	455	29.6	156.0	0.3341	0.1429
Gender	Female	486	26.6	137.5	0.2654	0.1235
Gender	Male	466	27.5	146.6	0.2897	0.1137
Employment	Unemployed	139	27.1	141.4	0.2086	0.1007
Employment	Retired	132	23.3	122.7	0.2197	0.0833
Employment	Employed	681	27.8	145.9	0.3025	0.1292

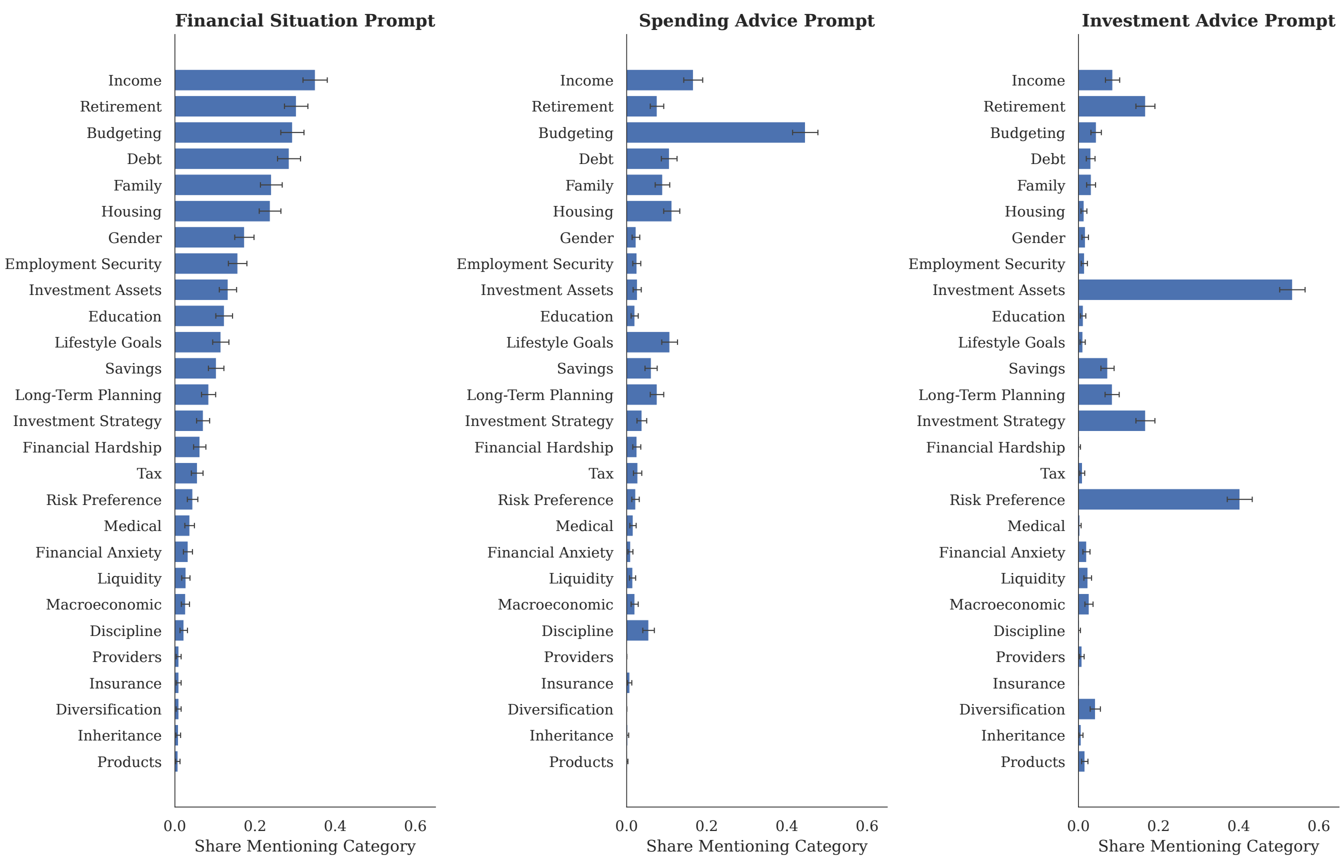
Notes: This table reproduces Table E2 for the third prompt that individuals write, in which they ask for investment advice.

Figure E3. AI Model Usage Among Survey Respondents



Notes: This figure shows the fraction of survey respondents who have used AI for financial advice, broken down by model (e.g., ChatGPT, Gemini, Claude). The sample consists of respondents who reported prior AI use in our Prolific survey.

Figure E4. All Dictionary Categories by Prompt

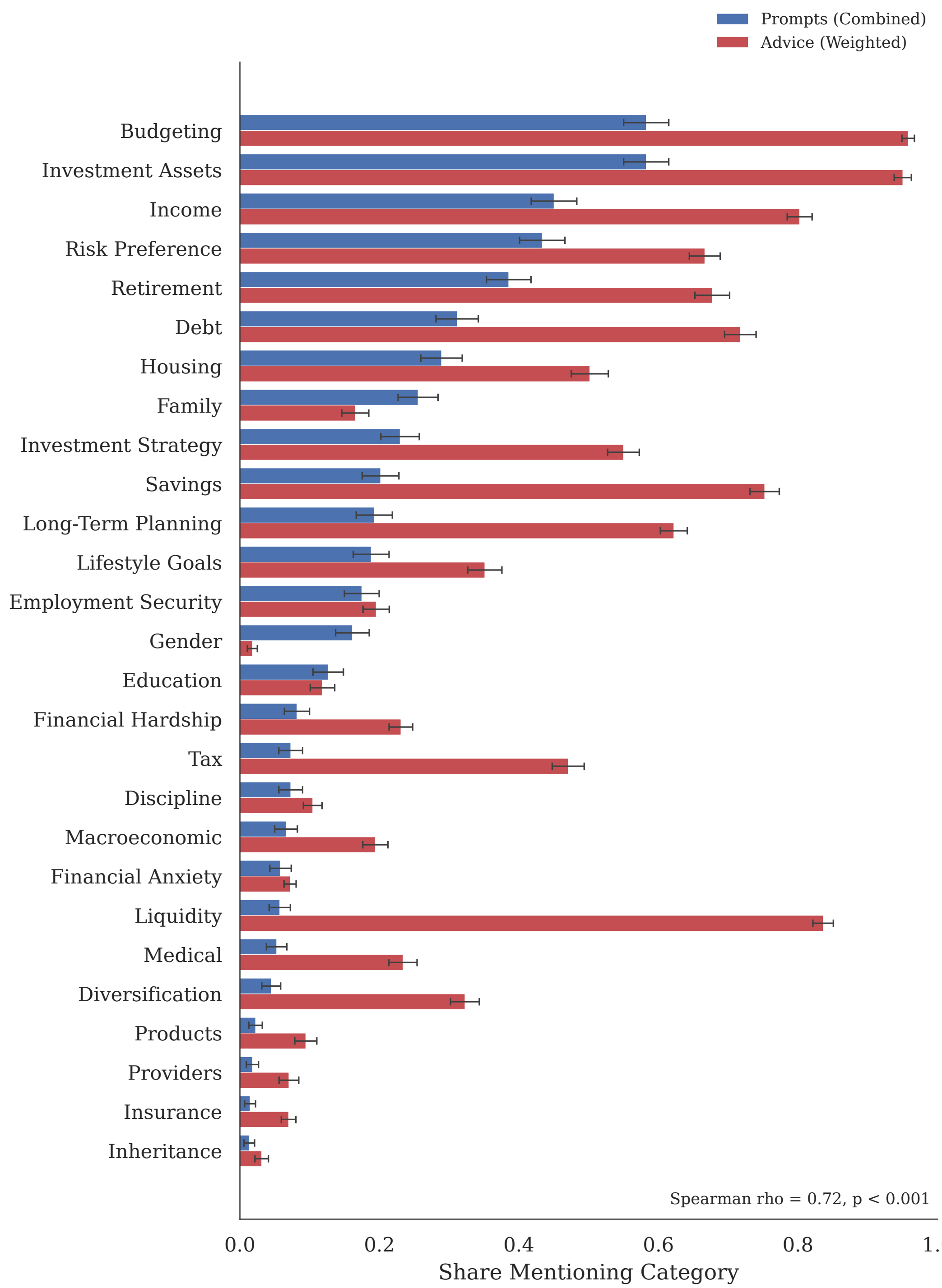


Notes: Each panel of this figure shows the prevalence of the different dictionary categories for one of the three prompts. The bars represent the share of respondents whose prompt text contains at least one keyword from the category, along with 95% confidence intervals. Categories are ordered by decreasing prevalence in the financial situation prompt.

Figure E5. Word Clouds by Individual Prompt

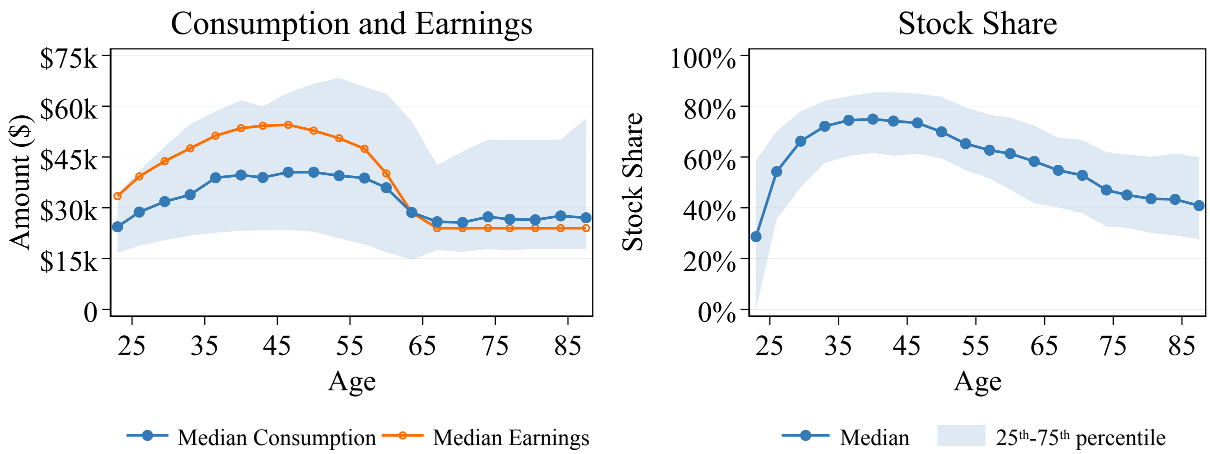
Notes: This figure reproduces the analysis in Figure 2, separating out the three prompts that individuals write.

Figure E6. All Dictionary Categories: Prompts vs. Advice



Notes: This figure shows the prevalence of the different dictionary categories in the combined prompts that individuals write (blue) versus the respondent-weighted mean mention rate in the LLM's advice (red). Categories are ordered by descending combined-prompt prevalence. Error bars show 95% confidence intervals. The Spearman rank correlation between the two sets of bars is shown in the bottom right of the plot.

Figure E7. Life Cycle Profiles of LLM-Recommended Consumption and Stock Shares: `Gemini 3 Flash`



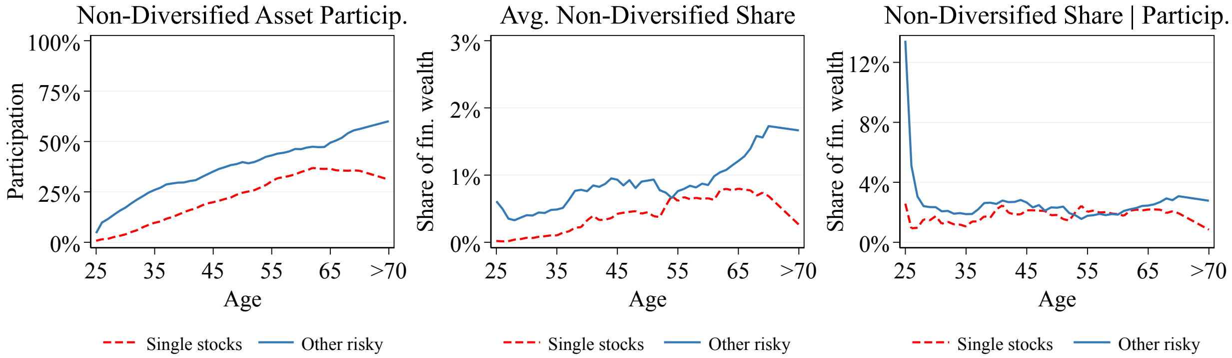
Notes: This figure plots the life cycle profiles for individuals following the LLM's recommended choices using `Gemini 3 Flash` with the survey prompts, as described in Section 1.4. Ages (22–89) are grouped into 20 equally sized bins. The left panel shows consumption and post-tax earnings, and the right panel shows the stock share. Dots denote median values within each bin, and shaded areas represent the interquartile range (25^{th}–75^{th} percentile). The estimated preference parameters are $\hat{\beta} = 1.054$ and $\hat{\gamma} = 5.1$. All values are in 2025 dollars.

Table E5. Summary Statistics from LLM and Life Cycle Model Simulations

Static	Employment Status	Age					Total
		22–29	30–39	40–49	50–59	60–64	
Panel A: Simulated Inputs							
Post-tax Earnings ($)	Employed	44,145	60,884	68,857	66,335	60,191	60,782
	Unemployed	16,449	22,017	22,838	21,782	19,249	20,456
Employment rate (%)	-	90%	91%	91%	85%	61%	86%
Avg. Tenure (years)	-	1.9	3.1	3.9	4.0	3.3	3.3
Panel B: LLM Endogenous Choices							
Consumption ($)	Employed	34,637	47,804	53,918	54,914	53,437	48,842
	Unemployed	13,363	19,294	21,539	24,952	27,202	22,816
Liquid Wealth ($)	Employed	46,964	215,949	618,130	1,350,137	2,147,938	701,998
	Unemployed	29,315	163,963	487,529	1,233,073	1,931,464	1,032,009
Stock Participation (%)	Employed	87%	100%	100%	100%	100%	97%
	Unemployed	84%	99%	100%	100%	100%	98%
Equity Share (%)	Employed	46%	70%	73%	67%	60%	65%
	Unemployed	43%	66%	72%	67%	60%	62%
Net Saving Rate	Employed	21%	19%	18%	14%	7%	17%
(% of earnings)	Unemployed	17%	9%	-2%	-32%	-62%	-25%
Panel C: Life Cycle Model (LCM) Endogenous Choices							
Consumption ($)	Employed	25,450	43,469	64,936	85,830	99,249	59,578
	Unemployed	19,687	33,185	48,613	63,660	77,342	55,546
Liquid Wealth ($)	Employed	70,262	334,886	736,493	1,124,386	1,315,676	644,512
	Unemployed	44,536	241,214	583,203	881,494	1,041,613	684,088
Stock Participation (%)	Employed	100%	100%	100%	100%	100%	100%
	Unemployed	79%	100%	100%	100%	100%	97%
Equity Share (%)	Employed	100%	97%	83%	68%	58%	84%
	Unemployed	79%	99%	82%	66%	60%	73%
Net Saving Rate	Employed	40%	23%	-4%	-48%	-98%	-7%
(% of earnings)	Unemployed	-25%	-62%	-131%	-251%	-390%	-220%

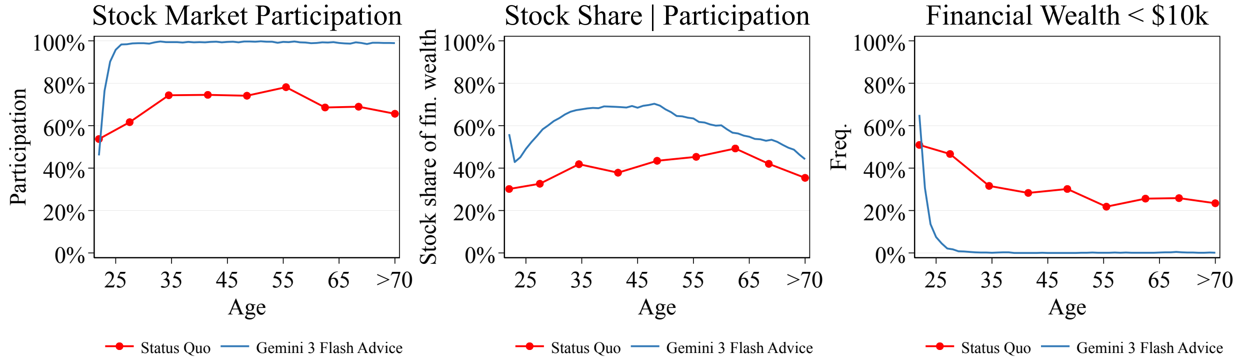
Notes: This table reports mean values for key variables from both the LLM and life cycle model simulations, broken down by age group (22–29, 30–39, 40–49, 50–59, 60–64). The Total column shows averages over the entire working life (ages 22–64). Panel A presents exogenous inputs that are identical across both simulations. Post-tax earnings correspond to wage and unemployment insurance earnings, employment rate is the share of individuals working, and tenure is the average number of years at the current employer. Panels B and C show endogenous outcomes from the LLM and life cycle model, respectively, broken down by employment status. Net saving rate is defined as (post-tax earnings minus consumption) divided by post-tax earnings. All monetary values are in 2025 dollars.

Figure E8. Non-Diversified Asset Participation and Portfolio Shares



Notes: This figure plots the life cycle profiles of non-diversified asset holdings—individual stocks and other risky assets (e.g., crypto, gold, commodities, collectibles)—for individuals following the LLM's advice using the survey prompts, as described in Section 1.4. The left panel shows unconditional participation rates by age; the middle panel shows the average portfolio share of financial wealth allocated to each asset class; and the right panel shows the average portfolio share conditional on participation. Ages (22–89) are grouped into 20 equally sized bins.

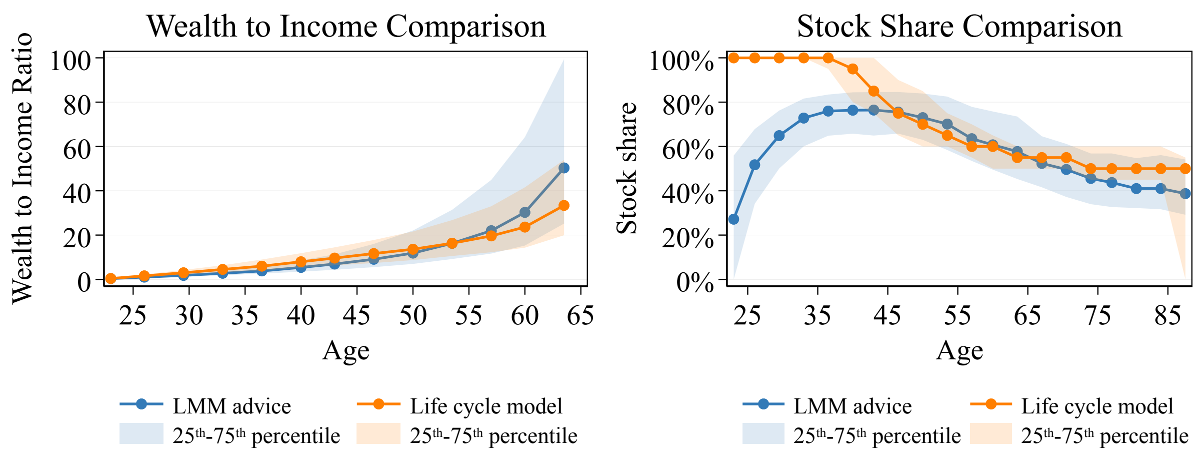
Figure E9. Observed Behavior vs. LLM-Recommended Behavior: `Gemini 3 Flash`



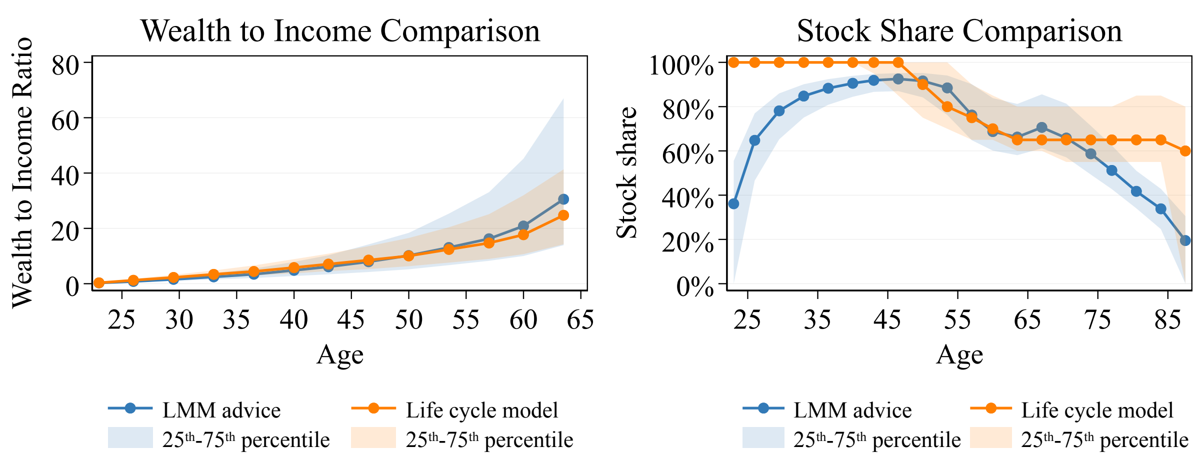
Notes: This figure compares survey respondents' observed financial behavior with the LLM's recommendations using `Gemini 3 Flash`. Observed behavior, as reported by individuals in our Prolific survey, is shown in red. The blue line represents the full life cycle simulation, with state variables replaced and updated as described in Section 1.3. Ages (22–89) are grouped into 20 equally sized bins. All values are in 2025 dollars.

Figure E10. Life Cycle Model Fit to LLM Advice Moments

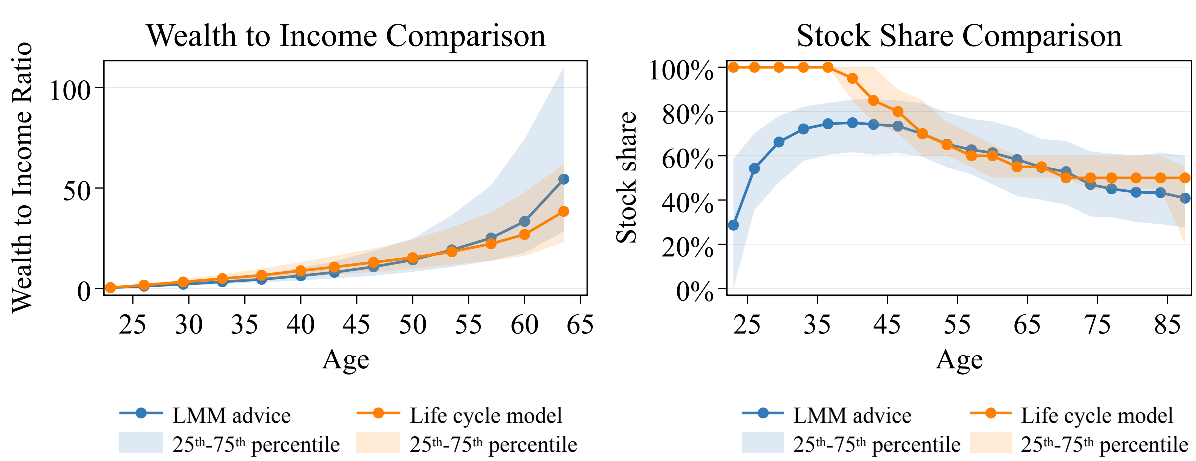
Panel A: Survey Prompts (`GPT-5.2`)



Panel B: Academic Prompts (`GPT-5.2`)

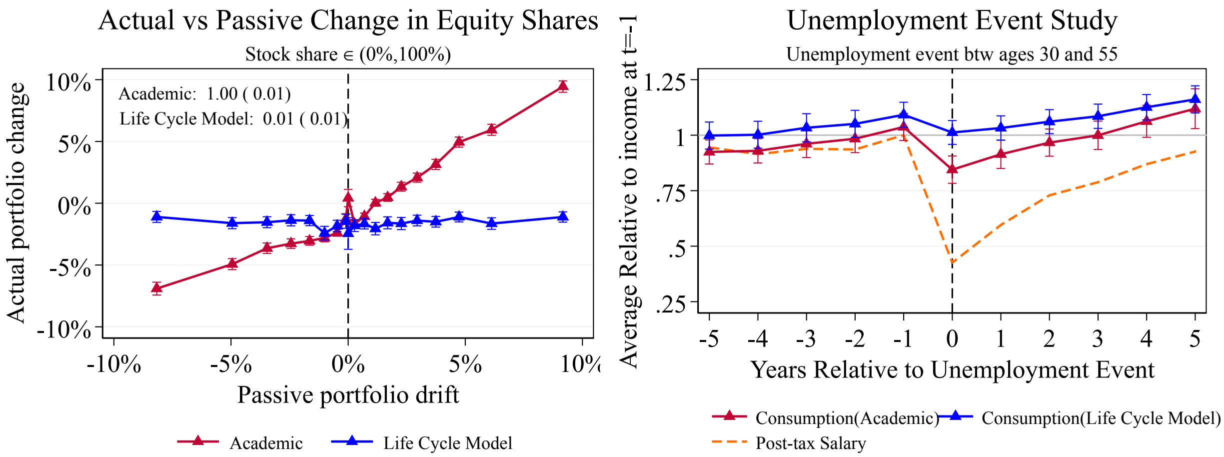


Panel C: Survey Prompts (`Gemini 3 Flash`)



Notes: This figure shows the fit of the estimated life cycle model (with the SMM-estimated $\hat{\beta}$ and $\hat{\gamma}$ from Table 2) to the LLM advice moments. Panel A corresponds to survey prompts with `GPT-5.2` ($\hat{\beta}$ = 1.034, $\hat{\gamma}$ = 5.3), Panel B to academic prompts with `GPT-5.2` ($\hat{\beta}$ = 0.99, $\hat{\gamma}$ = 4.7), and Panel C to survey prompts with `Gemini 3 Flash` ($\hat{\beta}$ = 1.054, $\hat{\gamma}$ = 5.1). Each panel plots the targeted moments (wealth-to-income ratio in working life and equity share in the full life cycle by age) for both the LLM advice and the estimated model.

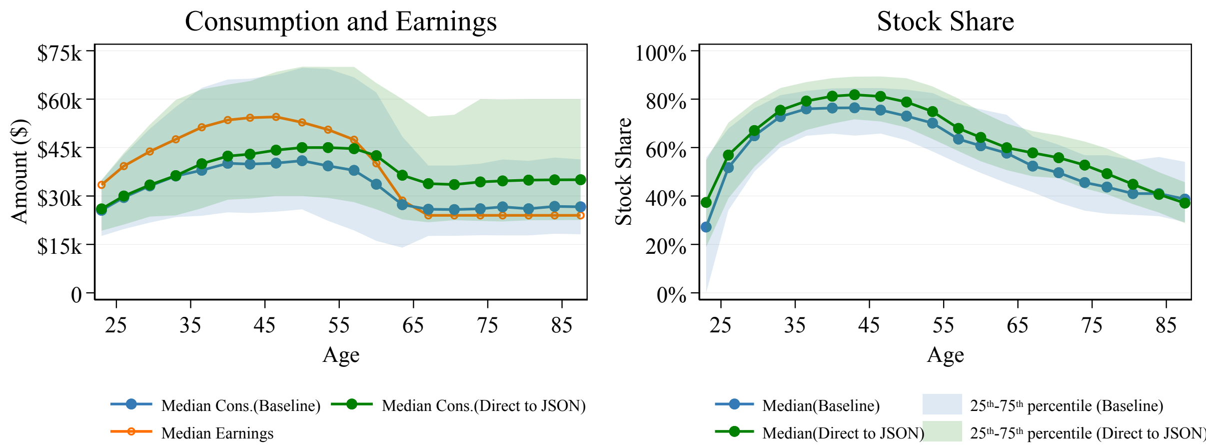
Figure E11. Responsiveness of LLM Advice to Shocks: Academic Prompt



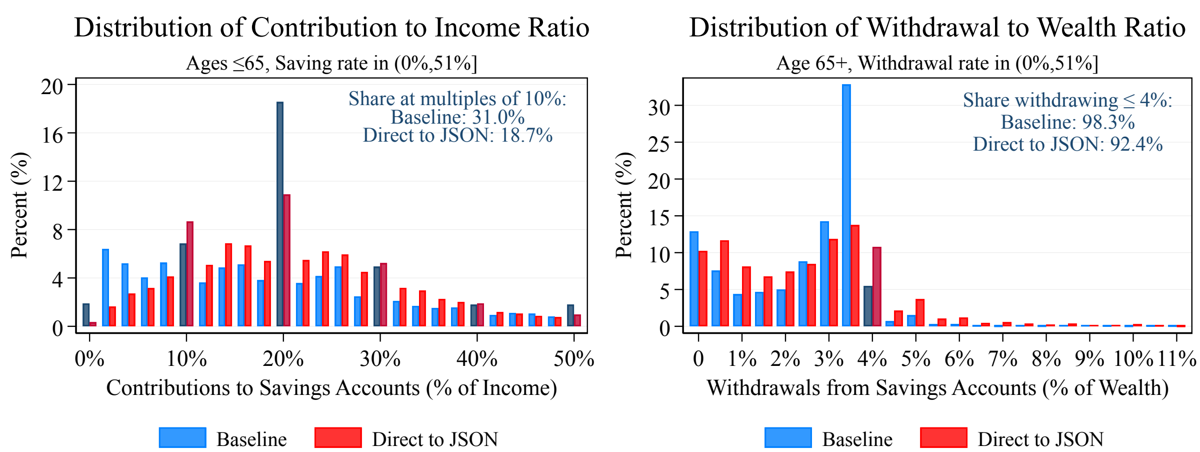
Notes: This figure compares the responsiveness to shocks between the LLM advice and the life cycle model using the academic prompt described in Section 1.4. Blue dots denote the LLM advice, red triangles denote the life cycle model results, and the orange dashed line indicates the post-tax earnings path implied by the life cycle model described in Section 1.2. The left panel presents a binscatter plot of average actual portfolio change against passive portfolio drift. Passive portfolio drifts are grouped into 20 equally sized bins. The right panel plots the unemployment event study, constructed in the same way as the right panel of Figure 8.

Figure E12. Baseline Two-Step Pipeline vs. One-Step Direct-to-JSON Pipeline

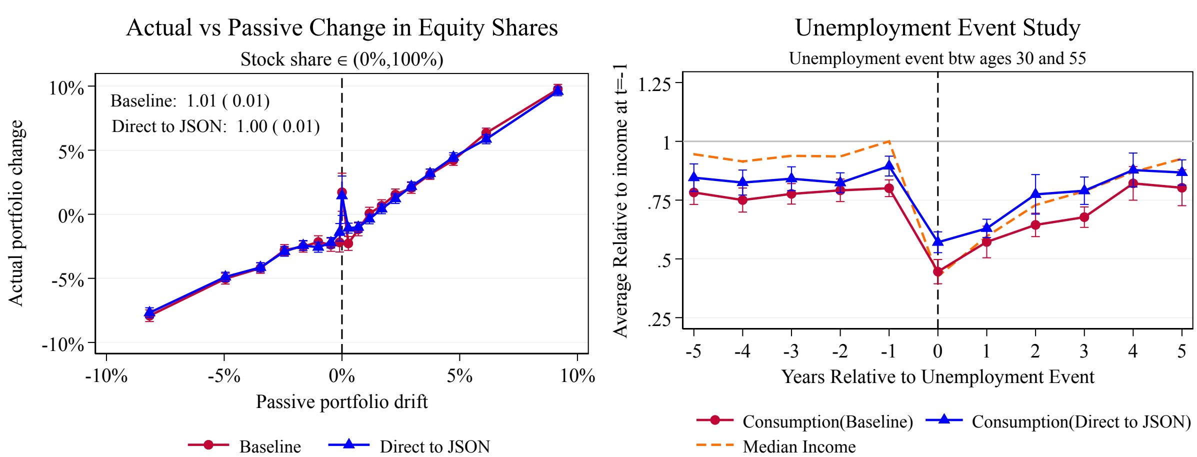
Panel A: Consumption and Stock Share Life Cycle Profiles



Panel B: Saving and Withdrawing Heuristics

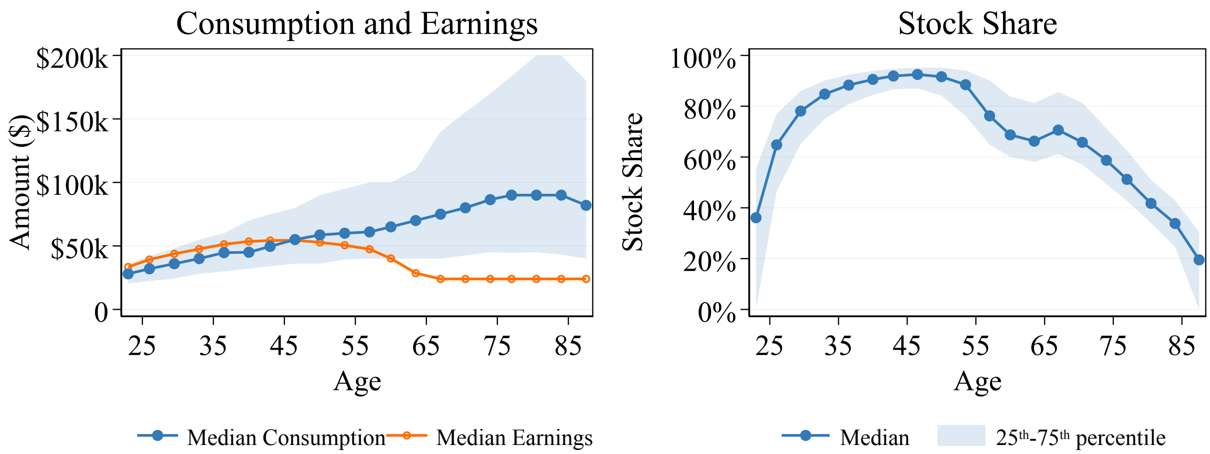


Panel C: Shock Response



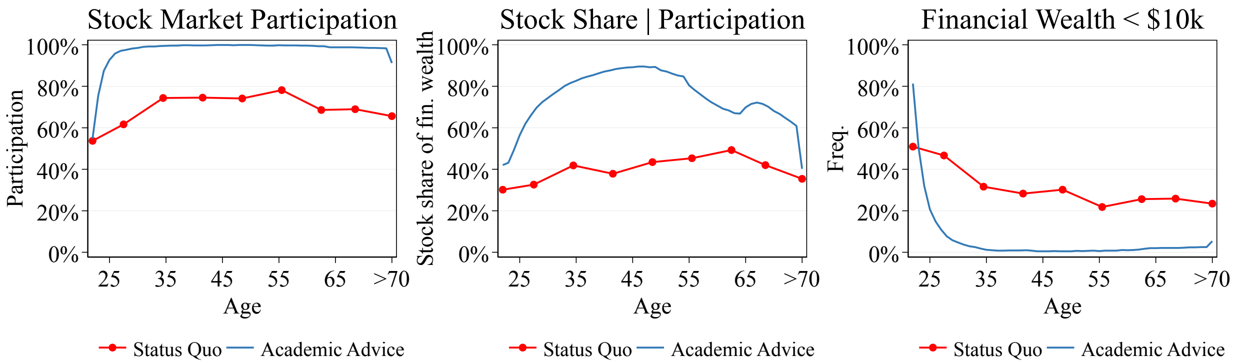
Notes: This figure compares the two-step pipeline with the one-step direct-to-JSON pipeline. Both use `GPT-5.2` and survey prompts. In the two-step pipeline, survey prompts first return qualitative advice; a second LLM call extracts quantitative actions from that advice. In the one-step direct-to-JSON pipeline, survey prompts are combined with the JSON translation instructions to elicit quantitative advice directly. In the two-step pipeline, state variables enter only in the translation step. In the one-step direct-to-JSON pipeline, state variables are passed together with the prompt and translation instructions.

Figure E13. Life Cycle Profiles of LLM-Recommended Consumption and Stock Shares: Academic Prompt



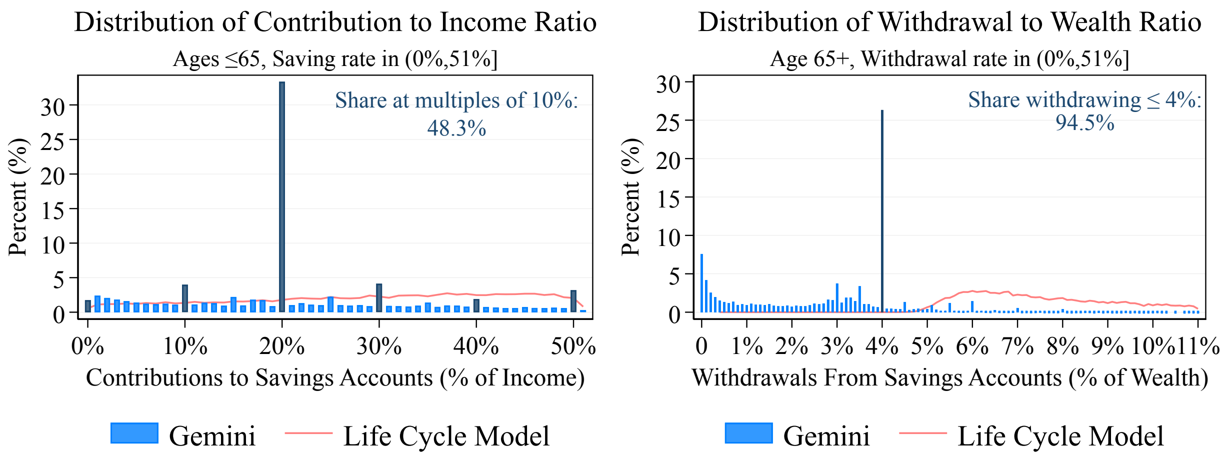
Notes: This figure plots the life cycle profiles for individuals following the LLM's recommended choices using the academic prompt, as described in Section 1.4. Ages (22–89) are grouped into 20 equally sized bins. The left panel shows consumption and post-tax earnings, and the right panel shows the stock share. Dots denote median values within each bin, and shaded areas represent the interquartile range (25^{th}–75^{th} percentile). The estimated preference parameters are $\hat{\beta} = 0.99$ and $\hat{\gamma} = 4.7$. All values are in 2025 dollars.

Figure E14. Observed Behavior vs. LLM-Recommended Behavior: Academic Prompt



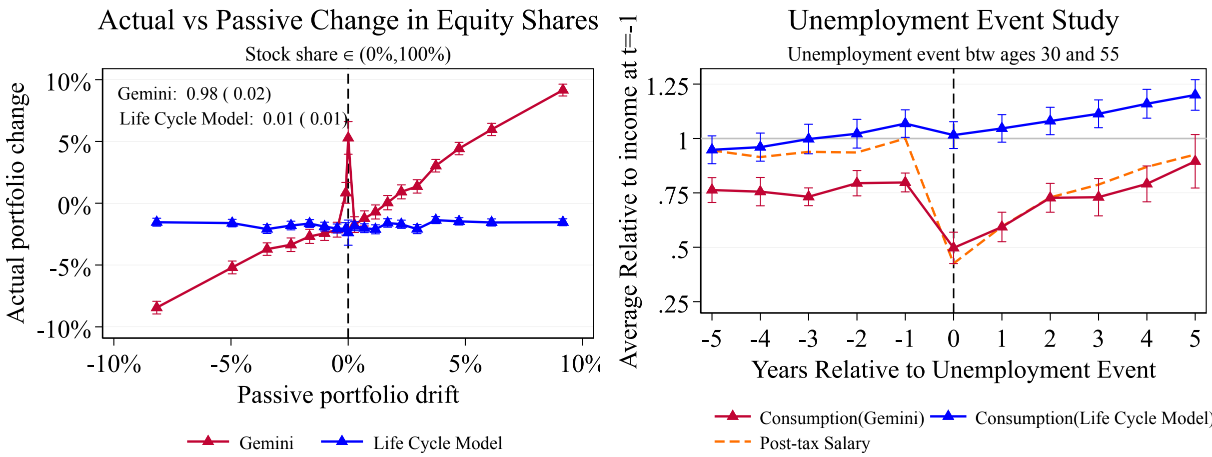
Notes: This figure compares survey respondents' observed financial behavior with the LLM's recommendations using the academic prompt described in Section 1.4. Observed behavior, as reported by individuals in our Prolific survey, is shown in red. Ages (22–89) are grouped into 20 equally sized bins. All values are in 2025 dollars.

Figure E15. Saving and Withdrawal Heuristics in LLM Advice: `Gemini 3 Flash`



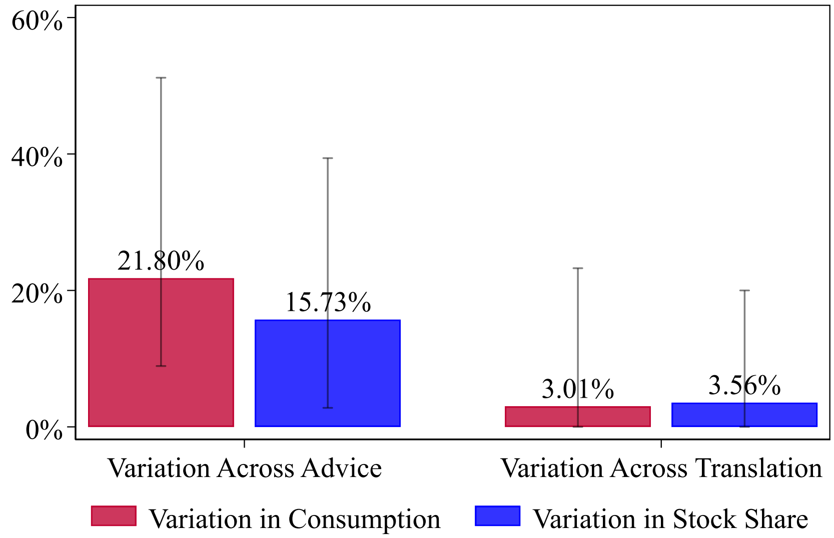
Notes: This figure plots the distributions of saving rates, saving amounts, and withdrawal heuristics in both the LLM advice and the life cycle model. The LLM advice is generated using `Gemini 3 Flash` with the survey prompts as described in Section 1.4. Blue histograms represent the LLM advice, and dark blue bars highlight heuristic values. The red lines represent the corresponding distributions from the life cycle model described in Section 1.2. All values are in 2025 dollars.

Figure E16. Responsiveness of LLM Advice to Shocks: `Gemini 3 Flash`



Notes: This figure compares the responsiveness to shocks between the LLM advice and the life cycle model using `Gemini 3 Flash`. Blue dots denote the LLM advice, red triangles denote the life cycle model results, and the orange dashed line indicates the post-tax earnings path implied by the life cycle model described in Section 1.2. The left panel presents a binscatter plot of average actual portfolio change against passive portfolio drift. Passive portfolio drifts are grouped into 20 equally sized bins. The right panel plots the unemployment event study, constructed in the same way as the right panel of Figure 8.

Figure E17. Variation in LLM Response across Advice and Translation



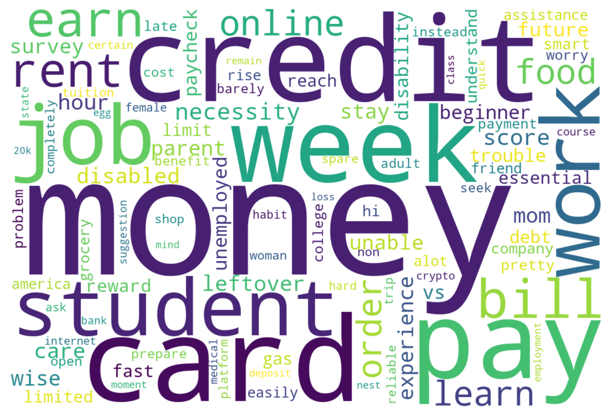
Notes: This figure shows the variation in quantitative choices for the same prompt across advice generation and JSON translation. For each bar, the same prompt or advice is passed through the respective parts of the pipeline five times. The red bars plot the median range in consumption scaled by the mean consumption advice. The blue bars plot the median range in equity share. The left pair shows the variation over different generations of advice with the same prompt, thus capturing the stochasticity of the full LLM pipeline. The right pair shows the variation over different translations of the same advice and captures the stochasticity of just the JSON translation step.

Figure E18. Differences in Prompts and Advice by Financial Literacy

Panel A: Words Overrepresented in Prompts by Financial Literacy

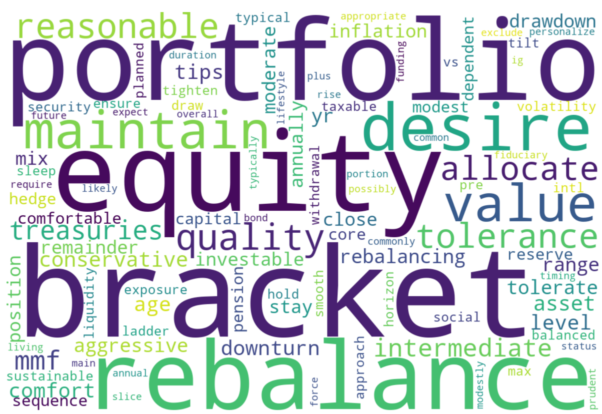
High Financial Literacy

Low Financial Literacy

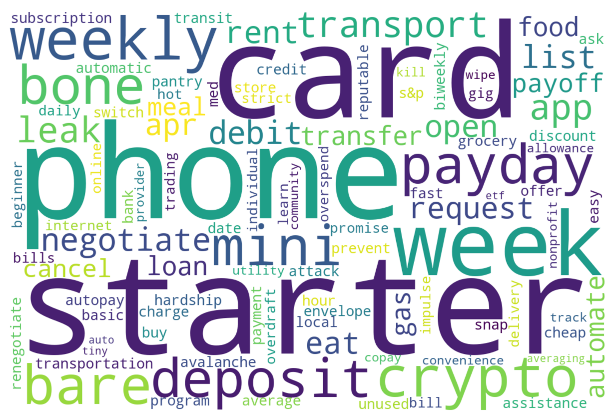


Panel B: Words Overrepresented in Advice by Financial Literacy

High Financial Literacy



Low Financial Literacy



Notes: This figure shows word clouds of terms that are differentially used by individuals with high versus low financial literacy, as measured by a Big Five financial literacy test. Panel A shows words overrepresented in the prompts written by individuals with high financial literacy (left) versus low financial literacy (right). Panel B shows words overrepresented in the LLM's advice to individuals with high (left) versus low (right) financial literacy. Word size is proportional to the difference in frequency between groups.

Figure E19. Example Prompts by Financial Literacy

High Financial Literacy User:

I am a short term futures trader usually executing **3 to 4 trades per daily session**. I do not hold any positions after the market closes. I **trade the futures micro contracts of indexes, crude oil, gold, currencies and bonds**. I also place option trades on the SPX. I am looking to imptove my strategy which is based on Cumalative Delta and Orderflow. My account size is 10,000 USD

I'd like to know how much to spend over the coming year.

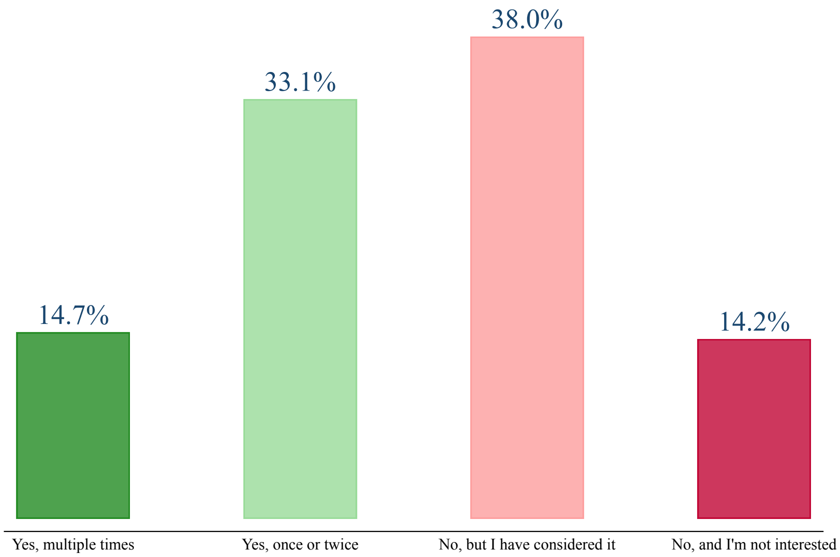
I'd like to know how to invest to improve my strategy and reduce my risk exposure.

Low Financial Literacy User:

Hello....Im in need of some financial advice. I need to invest some money into something that will make me money in the future. [...] I have current stocks invested, however, **I know very little about stocks** and how and when to buy/sell, and **what stocks I need to buy and/or avoid**? Some advice would be very helpful! Thank you! :)

Notes: This figure shows example prompts written by individuals with high and low financial literacy, as measured by a Big Five financial literacy test administered in our Prolific survey.

Figure E20. Prior AI Use for Financial Advice Among Survey Respondents



Notes: This figure shows the distribution of responses to the question "In the past 3 months, have you used an AI tool to get financial advice or information?" in our Prolific survey (N = 952).

Figure E21. Example Prompts by Prior AI Experience

AI User:

Assume you are a CFA with 20+ years of experience. I am needing advice to help create a sustainable investment portfolio using E*trade or similar self-select investment platforms. I have $5K to invest currently. I have another $10k in savings and am already investing in an employer sponsored retirement account contributing the max allowing amount.

Using your CPA knowledge, and our investment strategy determined previously, how much would I bel able to continually spend to add to my investment portfolio?

What additional investment vehicles should I be looking add for my further spending? Should reinvest in high return vehicles or expand and diversify?

AI Non-User:

I would like the AI tool to create a budget for my new pay rate as I am starting a new job. I will be making $20 an hour and working 40-45 hours a week. I pay $1000 in rent & around $500 in utilities/ household need bills.

How much could I spend freely with this salary while maintaining funding for rent, household needs, & $200 savings a month?

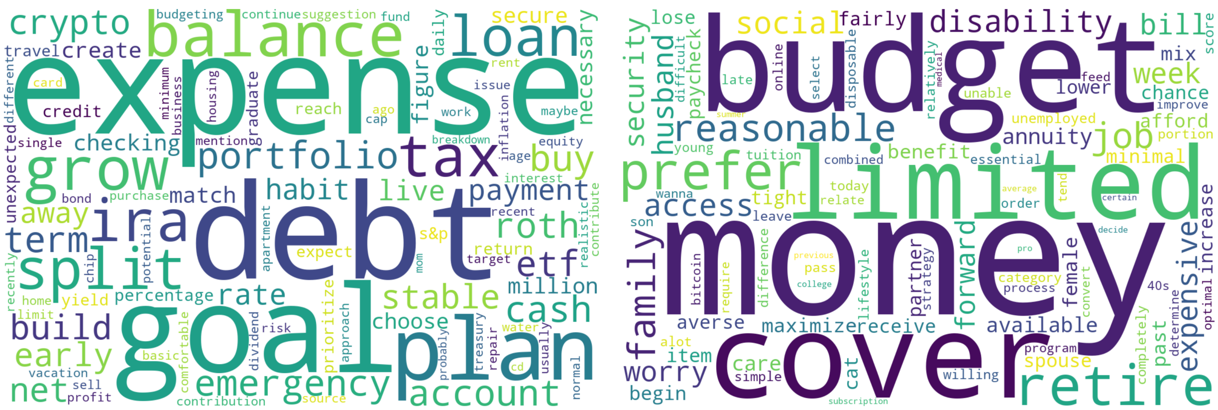
Where should I invest starting with $50 & consistently adding $25 a month after?

Notes: This figure shows example prompts written by individuals who have and have not previously used AI for financial advice, as reported in our Prolific survey.

Figure E22. Differences in Prompts and Advice by Prior AI Experience

Panel A: Words Overrepresented in Prompts by Prior AI Experience

Prior AI Use | **No Prior AI Use**

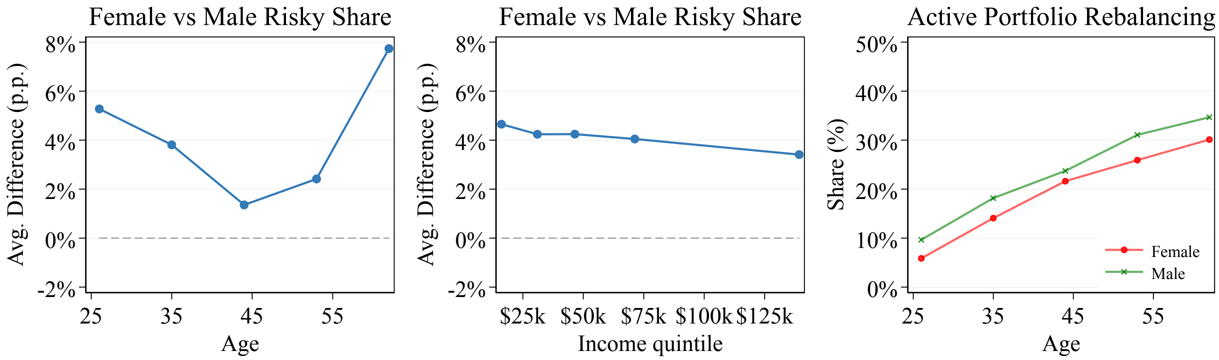


Panel B: Words Overrepresented in Advice by Prior AI Experience

Prior AI Use | **No Prior AI Use**

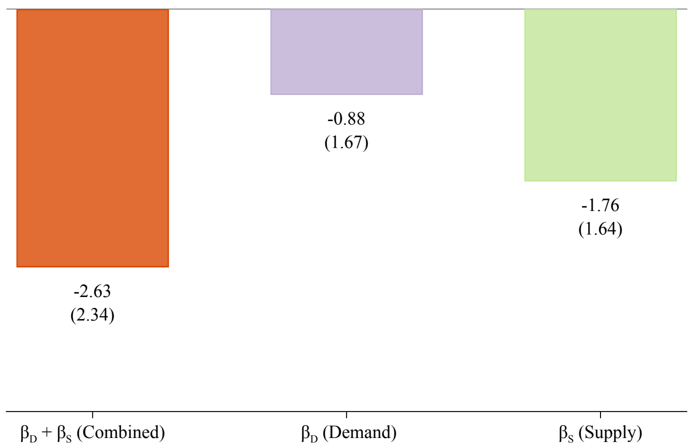
Notes: This figure shows word clouds of terms that are differentially used by individuals who have versus have not previously used AI for financial advice. Panel A shows words overrepresented in the prompts written by individuals with prior AI experience (left) versus without (right). Panel B shows words overrepresented in the LLM's advice to individuals with prior AI experience (left) versus without (right). Word size is proportional to the difference in frequency between groups.

Figure E23. Gender Gap in Recommended Equity Shares and Portfolio Rebalancing



Notes: This figure examines the gender gap in LLM-recommended equity shares by age and income. The left panel plots the average difference in recommended risky asset shares (men minus women) by age. The middle panel plots the same difference by income quintile. The right panel plots the share of individuals whose LLM-recommended portfolio change constitutes active rebalancing (rather than passive drift), separately for men and women by age.

Figure E24. Decomposing the Gender Difference in Net Saving Rate: Demand vs. Supply

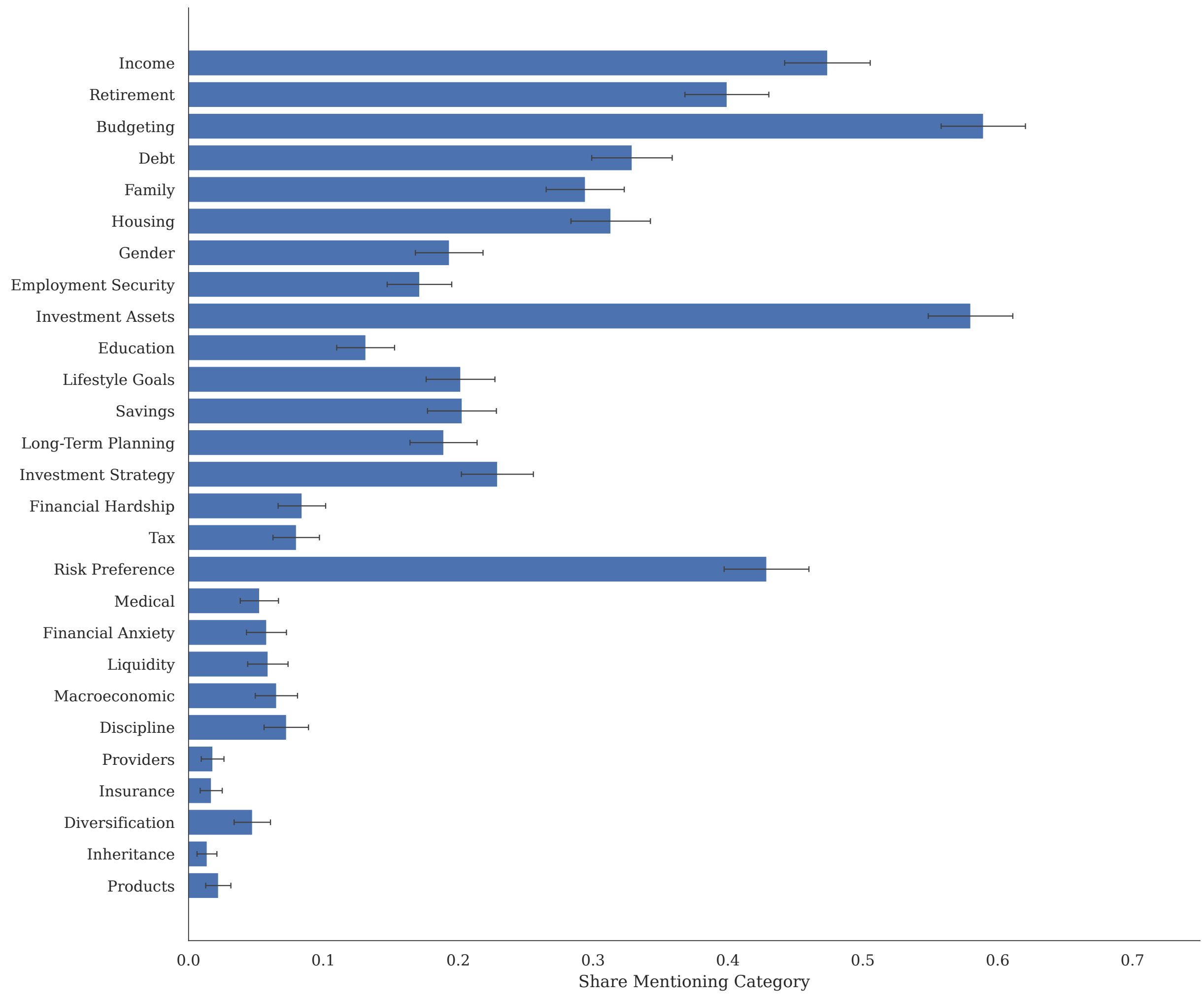


Notes: This figure decomposes the gender difference in LLM-recommended net saving rates into demand and supply components. We restrict to the 83% of survey prompts that do not explicitly mention gender and randomly insert "I am a man" or "I am a woman" at the beginning of each prompt. We then estimate $Y_{i,t} = \beta_D \, \text{FemaleAuthor}_i + \beta_S \, \text{FemaleLabel}_{i,t} + \delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t}$, where FemaleAuthor_i indicates a prompt written by a woman, $\text{FemaleLabel}_{i,t}$ indicates a randomly assigned female label, and $\delta_{i,t}^{\text{bucket}}$ are age × income × employment cell fixed effects. Net saving is calculated as the difference between total contributions to savings and total withdrawals, divided by income. The left bar shows the combined effect $\hat{\beta}_D + \hat{\beta}_S$, the middle bar shows the demand effect $\hat{\beta}_D$, and the right bar shows the supply effect $\hat{\beta}_S$. Standard errors in parentheses.

Figure E25. Decomposing Age and Income Differences in the Direct-to-JSON Pipeline: Demand vs. Supply

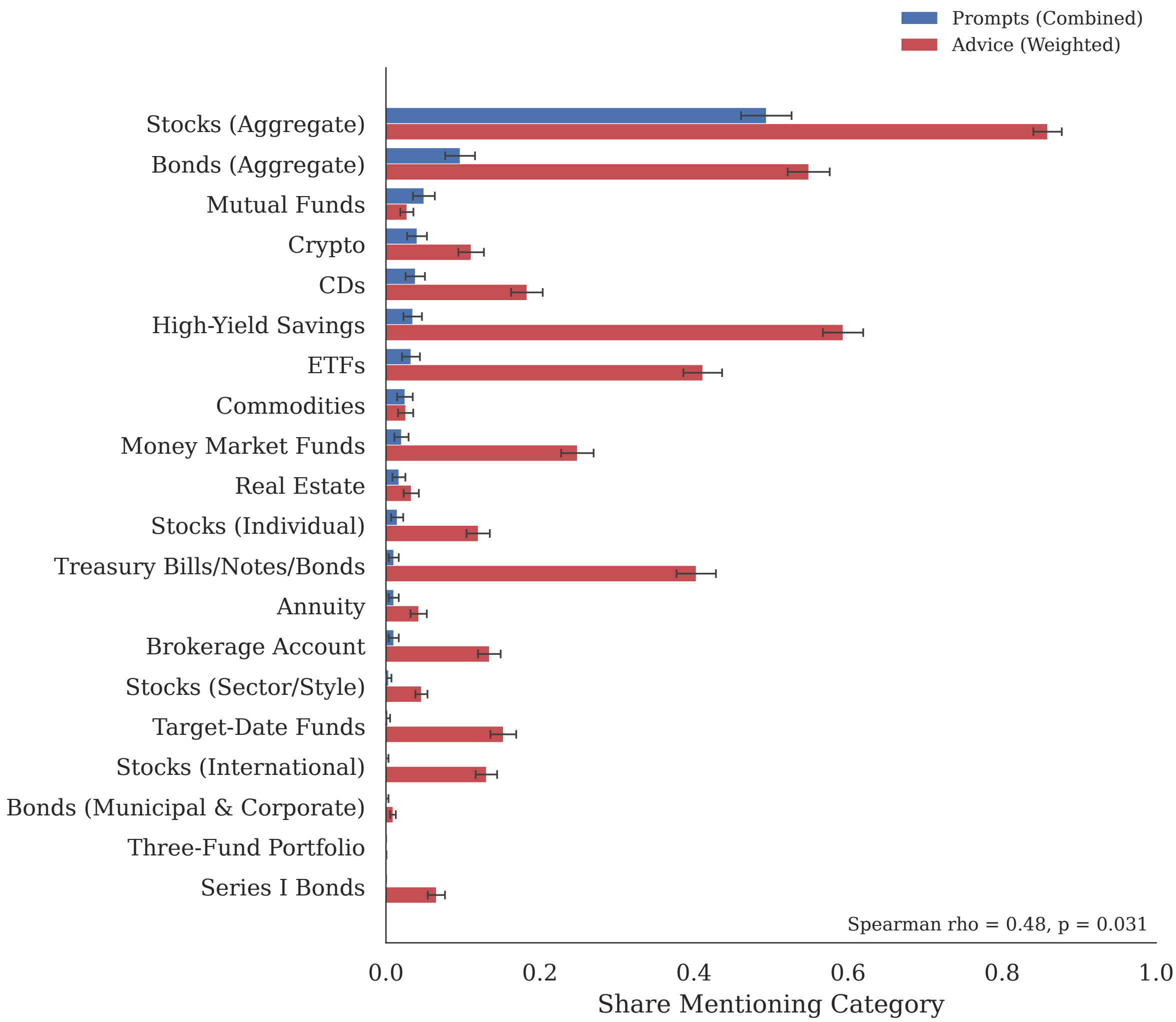
Notes: This figure repeats the demand–supply decomposition for age and income in Figure 14 using the one-step direct-to-JSON pipeline (Figure E12). In this pipeline, the simulated individual's state variables are appended to the prompt, and the LLM converts the combined input directly into a structured JSON output. This allows us to use the full sample of prompts rather than only the 23% (age) and 38% (income) that explicitly mention each variable. We estimate $Y_{i,t} = \beta_D$ AuthorVar$_i$ + β_S InsertedVar$_{i,t}$ + $\delta_{i,t}^{\text{bucket}} + \varepsilon_{i,t}$ separately for age and income, where AuthorVar$_i$ is the prompt writer's value of the variable, InsertedVar$_{i,t}$ is the value of the simulated agent inserted into the prompt, and $\delta_{i,t}^{\text{bucket}}$ are prompt bucket fixed effects. The top row uses age (in years) as the regressor; the bottom row uses income (in \$10,000 units). The left column uses the net saving rate as the dependent variable; the right column uses the change in the risky share. Within each panel, the left bar shows the combined effect $\hat{\beta}_D + \hat{\beta}_S$, the middle bar shows the demand effect $\hat{\beta}_D$, and the right bar shows the supply effect $\hat{\beta}_S$. We restrict the sample of analysis to working life (age < 65). Standard errors in parentheses.

Figure E26. Combined Dictionary Category Prevalence Across All Prompts



Notes: Combined respondent-level prevalence for all 27 dictionary categories (952 respondents). Each bar reports the share of respondents whose text contains at least one keyword from the category in any of the three prompts. Error bars show 95% confidence intervals based on the normal approximation for binomial proportions. Categories are ordered by descending prevalence.

Figure E27. All Investment Asset Groups: Prompts vs. Advice



Notes: Grouped bar chart comparing prompt and advice prevalence for all investment asset keyword groups (n = 944 matched pairs). Blue bars show the share of respondents mentioning the asset type in any of the three prompts. Red bars show the respondent-weighted mean mention rate in AI-generated advice. Categories are ordered by descending advice prevalence. Error bars show 95% confidence intervals. The main text (Figure 3, Panel A) shows the subset of categories with at least 5% prevalence in either series.